



सेंद्रल ट्रान्समिशन यूलिलिटी ऑफ इंडिया लिमिटेड

(पावर ग्रिड कॉर्पोरेशन ऑफ इंडिया लिमिटेड के स्वामित्व में)

(भारत सरकार का उद्यम)

CENTRAL TRANSMISSION UTILITY OF INDIA LTD.

(A wholly owned subsidiary of Power Grid Corporation of India Limited)

(A Government of India Enterprise)

Ref: CTU/N/00/CMETS_NR/27

Date: 28-01-2024

As per distribution list

Subject: 27th Consultation Meeting for Evolving Transmission Schemes in Northern Region-Minutes of Meeting

Dear Sir/Ma'am,

Please find enclosed the minutes of the 27th Consultation Meeting for Evolving Transmission Schemes in Northern Region held on 10th January, 2024 (Wednesday) through virtual mode.

The minutes are also available at CTU website (www.ctuil.in)

Thanking you,

Yours faithfully,

(Kashish Bhambhani)
General Manager (CTU)

Distribution List:

Chief Engineer (PSP&A – I) Central Electricity Authority Sewa Bhawan, R.K.Puram, New Delhi-110 066	Member Secretary Northern Regional Power Committee 18A, Shaheed Jeet Singh Sansanwal Marg, Katwaria Sarai, New Delhi – 110 016
Director (Power System) Solar Energy Corporation of India Ltd. D-3, 1 st Floor, A wing, Religare Building, District Centre, Saket, New Delhi-110017	Director Ministry of New and Renewable Energy, Block 14, CGO Complex, Lodhi Road, New Delhi-110003
Director (SO) Grid Controller of India Limited (erstwhile Power System Operation Corporation Ltd.) 9 th Floor, IFCI Towers, 61, Nehru Place, New Delhi-110 016	Executive Director Northern Regional Load Despatch Centre 18-A, Qutab Institutional Area, Shaheed Jeet Singh Sansanwal Marg, Katwaria Sarai, New Delhi– 110 016
Director (P&C), HPPTCL, Headoffice, Himfed Bhawan,Panjari, Shimla-171005, Himachal Pradesh	Director(W&P) UP Power Transmission Company Ltd. Shakti Bhawan Extn, 3rd floor, 14, Ashok Marg, Lucknow-226 001
Director (Technical) Punjab State Transmission Corporation Ltd. Head Office, The Mall, Patiala 147001, Punjab	Director (Projects) Power Transmission Corporation of Uttarakhand Ltd. Vidyut Bhawan, Near ISBT Crossing, Saharanpur Road, Majra, Dehradun.
Development Commissioner (Power) Power Development Department Grid Substation Complex, Janipur, Jammu	Director (Technical) Rajasthan Rajya Vidyut Prasaran Nigam Ltd. Vidyut Bhawan, Jaipur, Rajasthan-302005.
Member (Power) Bhakra Beas Management Board Sector-19 B, Madhya Marg, Chandigarh - 160019	Superintending Engineer (Operation) Electricity Circle, 5 th Floor, UT Secretariat, Sector-9 D, Chandigarh - 161009
Director (Operations) Delhi Transco Ltd. Shakti Sadan, Kotla Road, New Delhi-110 002	Director (Technical) Haryana Vidyut Prasaran Nigam Ltd. Shakti Bhawan, Sector-6, Panchkula-134109, Haryana
Executive Director (AM) POWERGRID Saudamini, Plot No.2, Power Grid Corporation Of India Limited, Near, August Kranti Marg, Sector 29, Gurugram, Haryana 122001	Executive Director (Engg.) POWERGRID Saudamini, Plot No.2, Power Grid Corporation Of India Limited, Near, August Kranti Marg, Sector 29, Gurugram, Haryana 122001

Distribution List:

Shri Angshuman Rudra Deputy General Manager Avaada RJBikaner Private Limited Avaada Energy Private Limited C-11, Sector-65, Gautam Buddha Nagar, Noida, Uttar Pradesh Ph.: 7835004673, 7835099717 Email: angshuman.rudra@avaada.com ; vivek.jain@avaada.com ;	Shri Amit Kumar Authorised Signatory Renew Solar (Shakti Six) Private Limited Auxo Sunlight Private Limited Renew Sun Power Private Limited Commercial Block-1, Zone-6, Golf Course Road, DLF City Phase-V, Gurugram, Haryana-122009 Ph.: 9717196796, 9873462717, 9871833188 Email: amit.kumar1@renewpower.in ; solarbidding.gm@renew.com ; mohit.jain@renew.com
Shri Pradeep Chauhan Director Anboto Solar Private Limited Level 1, Southern Park Building, D-2 District Centre Saket, New Delhi -110017 Ph.: 9870195134, 9818995610 Email: pradeep.chauhan@solarpack.es ; rohit.ahuja@solarpack.es ;	Shri Lalit Kumar Mehta Head Liasoning Malur Renewables Private Limited D-43, Janpath, Shyam nagar, Jaipur Rajasthan Ph.: 9587101222, 9588243677 Email: lalit@raysventures.com ; bhagwat.singh@rayspowerinfra.com ;
Shri Rajeev Thanvi Manager Solarcraft Power India 4 Private Limited 109, First Floor, Rishabh IPEX Mall, IP Extension Patparganj, Delhi- 110092 Ph.: 9462617997, 9999885815 Email: rajeev.thanvi@blupineenergy.com ; ashish.agarwal@blupineenergy.com ;	Shri Pavan Kumar Gupta Authorised Signatory Juniper Green Energy Private Limited Plot No. 18, 1st Floor, Institutional Area, Sector - 32, Gurgaon, Haryana - 122001 Ph.: 9560540654, 9999246072 Email: pavan.gupta@junipergreenenergy.com ; bd@junipergreenenergy.com ;
Mr. Stephen Lobo Director AK Renewable Infra Private Limited Flat No. G2, Satyam Housing, Plot No 64 Model Town Jaipur, Rajasthan -302017 Ph.: 9829059727, 9610608999 Email: akrenewableinfra@gmail.com ; karunain2000@yahoo.com ;	Shri Saurabh Mehta Authorised Signatory Hazel Hybren Private Limited Furies Solren Private Limited B-Block, 6th Floor, Embassy 247, Vikhroli West, Mumbai, Maharashtra- 400083 Ph.: 9930674683, 7838568948 Email: mehta.saurabh2@mahindra.com ; pathak.ankur@mahindra.com ;
Shri Karan Kishore Director Shudh Solar Power Private Limited Shop No. 303, City Pearl, Opp. Ganguar Hotel, M.I. Road, Jaipur, Rajasthan- 302001 Ph.: 9829059727, 7357879959 Email: shudhsolar@gmail.com ; karunain2000@yahoo.com ;	Shri Animesh Manna DGM NTPC Renewable Energy Limited Netra Building, E-3, Ecotech-II, Udyog Vihar, Greater Noida, Uttar Pradesh – 201306 Ph.: 9650997835, 9650991445 Email: amanna@ntpc.co.in ; rajivgupta@ntpc.co.in ;
Shri Sameer Mathur Business Head AM Green Energy Private Limited	Shri Vikram Malkotia Director Gamma Renewables India Project One Private Limited 208, Tower B, Pioneer Urban Square, Sector 62,

MyHome Twitza, 5 th Floor, Plot No. 30/A, Survey No. 83/1 APIIC Knowledge, city of Rai Durg, Rangareddy district, Hyderabad, Telangana -500081 Ph.: 9920324550, 8971081430 Email: approvals.amgepl@arcelormittal.com ; vishal.soni@arcelormittal.com ;	Gurugram, Haryana -122005 Ph.: 8800917799, 9717855521 Email: vikram.malkotia@upcrenewables.com ; alok.nigam@upcrenewables.com ;
Shri Ravindra Shekhawat Senior Manager HHP Two Private Limited 3rd Floor, Jmd Regent Arcade Mall, Haryana Ph.: 9784964606, 9871544211 Email: maharashtra@hygenco.in ; shashank@hygenco.in ;	Shri Rajesh Sodhi Head Secretarial Acme Cleantech Solutions Private Limited Plot No. 152, Sector-44, Gurugram, Haryana 122002 Ph.: 9811633237, 8744060601 Email: rajesh.sodhi@acme.in ; yogesh@acme.in ;
Shri Ravi Damaraju CEO First Energy Private Limited Unit No. 601, 6th Floor, Cello Platina, Fergusson College Road, Shivajinagar, Pune Maharashtra- 411 005 Ph.: 9971900659, 9604035720 Email: ravi.damaraju@feplglobal.com ; Vasundhara.sen@feplglobal.com ;	Shri Ashu Gupta Authorized Signatory Clean Max Gamma Private Limited First Floor, Clean Max Enviro Energy Solutions Private Limited The Peach Tree Complex, Sushant Lok Phase I, Gurugram, Haryana- 122009 Ph.: 9644400066, 7200615824 Email: ashu.gupta@cleanmax.com ; vidisha.dubey@cleanmax.com ;
Shri Aniket Vyas Director Iraax International (Opc) Private Limited R-3, Tilak Marg, civil Lines, Jaipur, Rajasthan-302005 Ph.: 9571724543, 9413333984 Email: iraaxinfo@gmail.com ; law.uni.aniketvyas@gmail.com ;	Shri Ajay Kumar Atrea Authorised Signatory Prerak Solar Urja Private Limited L-11, Green Park Extension, New Delhi 110016 Ph.: 9650999982, 8130696120 Email: ajayatrea@arightgreentech.com ; sandeep@arightgreentech.com ;
Shri Rajesh Kumar Chief Executive Officer Bundelkhand Saur Urja Limited TC-43/V, Vibhuti Khand, Gomti Nagar, Lucknow, Uttar Pradesh 226010 Ph.: 9350270238, 8130203048 Email: ceobsul@gmail.com ; sksbalyan@nhpc.nic.in ;	Shri Kuldeep Singh Director Deshraj Solar Energy Private Limited Plot no 51 & 51, M-Powered Building, PhaseIV, Udyog Vihar, Gurgaon Ph.: 9896052767, 8158880993 Email: kuldeepssindhu@gmail.com shubham.roy@dattainfra.com
Shri Ranjan Chhibba Additional Director Sunflag Power Limited D-47, 2nd Floor Defence Colony, New Delhi- 110024 Tel 011- 49576030/40/50 Ph.: 9810172113, 6398668185 Email: rrc@sunflagsteel.com ; rajeev_pandey@sunflagsteel.com ;	Shri Kumar Vipul Director Radiant Star Solar Park Private Limited D59, Sector 63, Noida, Uttar Pradesh Ph.: 9873122743, 9599709689 Email: vipul@provoltus.com ; info@provoltus.com ; nikhil@provoltus.com ;
Shri Aditya Pyasi Head – Regulatory and License & Permits	

Lanco Mandakini Hydro Energy Private Limited 401, 4th Floor, Salcon Rasvilas Building, Saket District Centre New Delhi - 110017 Ph.: 9313553021, 9818887963, 8467084956 Email: operations.india@statkraft.com rajeshkumar.srivastav@statkraft.com anurag.sharma@statkraft.com aditya.pyasi@statkraft.com rohit.gera@statkraft.com	
--	--

Minutes of 27th Consultation Meeting for Evolving Transmission Schemes in Northern Region held on 10.01.2024

The 27th Consultation Meeting for Evolving Transmission Schemes in the Northern Region (CMETS-NR) was held through VC on 10.01.24. **List of participants is enclosed at Annexure-A.**

A. Confirmation on the minutes of CMETS-NR meeting

It was informed that the Minutes of the 26th CMETS-NR meeting held on 01/12/2023 were issued vide letter dated 20/12/2023. No comments were received on the minutes, accordingly, minutes were confirmed as circulated.

However, regarding the minutes of 23rd & 25th CMETS NR meetings held on 29/08/23 and 31/10/23 respectively, following observations were received from applicants/observed by CTUIL:

1. For grant of Connectivity to M/s Shikhar Surya (One) Pvt. Ltd. (Appl. Nos. 1200003115, 1200003772 & 0312100003), 'High Capacity' was mentioned inadvertently in the DTL configuration, same may be considered deleted.
2. For grant of Connectivity to M/s Cannice (Appl. Nos. 0212100004, SW...16716 00589106 & 312100019), 'S/c on D/c tower' to be included in the DTL configuration.
3. For Connectivity grant to SJVNL for Sunni Dam HEP, the DTL capacity may be revised as High Capacity line along with associated bays at both ends." as the plant capacity is 382 MW. Further interconnection point may be considered as Nange PS instead of Ogli PS.
4. During, the 25th NR-CMETS meeting, it was mentioned that Budhil HEP is an existing project connected to ISTS at Chamera-III through 220 kV D/c line, however, it may be corrected as '220 kV S/c line.'

Accordingly, the above corrections to be considered in respective Minutes.

B. Network Expansion Schemes

1. Transmission system strengthening to facilitate evacuation of power from Bhadla/Bikaner complex

It was deliberated that comprehensive Transmission scheme for evacuation of power from Rajasthan REZ Ph-IV (Part-1) (Bikaner Complex) is under implementation for power transfer of 7.7GW incl. 4GW from Bikaner-III PS. The scheme comprises 765kV EHVAC corridor from Bikaner-III PS towards load centers of Delhi/UP. As part of the above scheme, 765 kV Bikaner-III -Neemrana 2xD/c and Neemrana-II – Bareilly (PG) D/c lines are being implemented.

Further, as part of Rajasthan REZ Ph-III (20GW) Transmission scheme, Bhadla-III & Ramgarh PS are being established for integration of 6.5GW & 2.9 GW RE potential respectively. Ramgarh PS is also being inter-connected with Bhadla-III for evacuation of RE power. In order to facilitate evacuation of 9.4GW RE power from Ramgarh/Bhadla-III PS (6.5GW+2.9GW) from Bhadla-III onwards, 765kV Bhadla-III - Sikar-II D/c line with implementation schedule of Mar'25 [for about 2.9GW power transfer requirement] as well as 6GW HVDC corridor (± 800 kV Bhadla (HVDC) -Fatehpur (HVDC)) with implementation schedule of Feb'28 (Pole-1) & Aug'28 (Pole-2) is being implemented as part of Ph-III scheme

Accordingly, in 19th CMETS-NR meeting, 765kV Bhadla-III - Bikaner-III D/c line was agreed to meet evacuation requirement from Bhadla-III PS onwards for some RE generators coming up in 2025-26 as well as Optimal utilization of EHVAC transmission system beyond Bikaner-III PS while providing flexibility of power transfer from Bhadla/Bikaner RE clusters. The scheme is under bidding.

To facilitate above evacuation of power as well as to meet N-1 criteria (beyond 765kV Bareilly S/s), additional 400 kV corridor i.e. 400kV Bareilly(765/400kV) – Bareilly(PG) D/c line (Quad) (2nd) along with augmentation with 1x1500 MVA, 765/400 kV ICT at Bareilly (765/400kV) S/s (3rd) is proposed. The above strengthening scheme is also included as part of connectivity system of various RE applications granted at Bikaner-III.

CEA agreed for the proposal, however they enquired about short circuit of 400kV Bareilly(765/400kV) & Bareilly(PG) S/s. CTU stated that short circuit level of 400kV Bareilly(765/400kV) is about 52kA (designed capacity:50 kA) & 400kV Bareilly(PG) S/s is about 51kA (designed capacity: 40 kA), however with above proposed scheme short circuit level increase marginally (1-2kA) on both the substations. CTU stated that to reduce short circuit level of various substation in NR, wherein short circuit level is already high (from designed capacity), measures will be taken as part of comprehensive scheme in a phased manner. Grid India stated that scheme is agreeable and they do not have any comment on above scheme.

It was stated that POWERGRID vide mail 21.01.23 confirmed the space availability for 765/400kV ICT (3rd) at Bareilly (PG) (765/400kV) as well as 2 nos. of 400kV line bays each at Bareilly (PG) (765/400kV) and Bareilly (PG) (400kV). In view of that following transmission system strengthening was agreed to facilitate evacuation of power as well as to meet N-1 criteria (beyond 765kV Bareilly) in ISTS

- 400 kV Bareilly(765/400kV) – Bareilly(PG) D/c line (Quad) (2nd) (~2 km)
- Augmentation with 1x1500 MVA, 765/400 kV ICT(3rd) at Bareilly (765/400kV) S/s

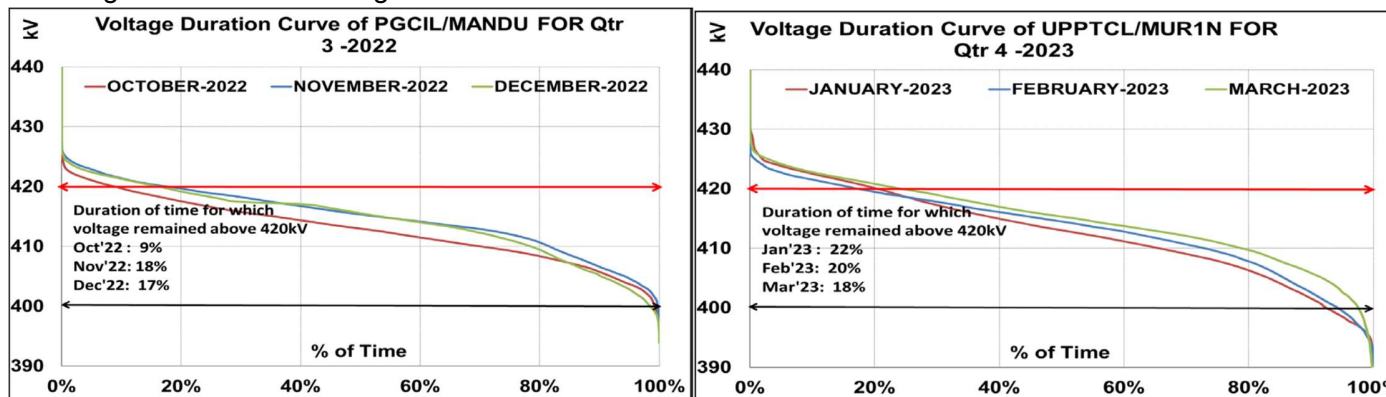
Implementation Schedule: 18 months from allocation

2. Replacement of existing 420kV 50MVAR Bus Reactors at Mandola & Muradnagar substation with 125MVAR bus reactor

It was stated that POWERGRID vide letter dated 30.10.23 informed that 400/220kV Mandola substation in NR-I was commissioned in 1990 and 50MVAR Bus reactor (BHEL make, 3-phase) on this substation was commissioned in 1991. Due to the ageing of reactor, frequent leakages have been observed despite all preventive measures/overhauling of the reactor. The furan value of the reactor is also in increasing trend and present value is 2.4PPM. As reactor has already completed more than 32 years of service life and any major maintenance may not be enough to ensure further reliable & longer service period

In the above letter, it was also mentioned that apart from the above, 01 no. 420kV, 50MVAR Bus reactor (3x16.67 MVAR, CGL make 1 phase unit, DOCO - 1988) was commissioned at Muradnagar substation of UPPTCL (owned by POWERGRID). Frequent oil leakages and deteriorated oil parameters along with high vibrations have been observed in these units. This reactor is also not reliable and is vulnerable to fail due to its ageing. Further any major maintenance may not be enough to ensure further reliable & longer service period. As the system fault level at both the substations have increased over a period of time (more than 30 years), therefore bus reactors of 50 MVAR may not be adequate at both stations (Mandola & Muradnagar). POWERGRID requested to review the augmentation of above said bus reactors for better voltage regulation.

In Grid India operational feedback report FY 2022-23(Q3) also, it was mentioned that voltage at 400kV Mandola remains more than 420kV for 9-18% of time and 415kV and above for about 40% of time. Similarly from operational feedback report FY 2022-23(Q4) it emerged that voltage at 400kV Muradnagar remains more than 420kV for 18-22% of time and 415kV and above for about 40% of time.



Voltage duration curve of 400kV Mandola & Muradnagar for 2022-23 (Source: Grid-India)

Grid-India stated that voltage on 400kV Mandola and Muradnagar S/s remains high in various operational scenarios and already highlighted in Grid-India operational feedback report, so proposal is agreeable to them. CEA also agreed for the proposal.

POWERGRID suggested that proposal will be considered in ADD CAP in taken up in NRPC meeting. CTU stated that POWERGRID may send proposal to NRPC for its approval in ensuing NRPC meeting.

From Grid-India operational feedback as well as inputs received from POWERGRID, it was decided in the meeting that it is technically recommended to replace 50MVA bus reactor each at Mandola(PG) & Muradnagar (UPPTCL) with 125MVA (420kV) bus reactor.

3. Additional Transmission system for evacuation of power from Bhadla-III PS as part of Rajasthan REZ Phase-III scheme (20GW)

It was stated that, Transmission System requirement for additional 20 GW REZ in Northern Region (Phase-III) was approved in 5th NCT meeting held on 25.08.21 & 02.09.21. As part of the above scheme, establishment of 765/400/220kV Bhadla-III PS (765/400kV : 2x1500MVA, 400/220kV : 3x500MVA) was approved in Phase-III Part B1 package, which is under implementation. Further, 7x500MVA ICTs (incl. 10 nos. of 220kV line bays & 220kV sectionalizer was also approved) under RTM in Phase-III Part B1 package with following condition.

- The implementation of number of 220 kV bays and 400/220 kV transformers to be taken up based on receipt of stage-II connectivity and commensurate LTA respectively (beyond 1500 MW at Bhadla-III).
- 220 kV line bays and Transformer augmentations shall be reviewed based on stage-II connectivity at 220 kV voltage level and LTA applications respectively.

Further, connectivity application of about 7GW (against potential of 6.5GW) is already received at Bhadla-III PS, out of which 6.5GW is already granted/agreed for grant. Out of above RE application of about 3GW (400kV - 1GW, 220kV- 2GW) is granted through EHVAC system (Sch. Progressively from Mar'25) and balance 3.5GW is being granted on HVDC system (Expected Sch.: Feb'28) at Bhadla-III PS.

It was stated that with fulfilment of connectivity grant process (Intimation, BGs etc.), total 5 nos. of 400/220kV ICTs are required, out of which 3 nos. ICTs already under implementation (Ph-III Part B1). Therefore balance 2 nos. ICTs (4th & 5th) are required with 18 months schedule considering RE generation schedule. Further total 4 nos. of 765/400kV ICTs are also required, out of which 2 nos. 765/400kV ICTs (Ph-III Part B1) are already under implementation. Additionally, 2 nos. of 765/400kV ICTs (3rd & 4th) are required (ICT-3:18 months (best effort Jun'25) & ICT-4: Dec'25) considering about 3-3.5GW power evacuation requirement from envisaged Bhadla-III PS

as well as power flow through 400kV Fatehgarh-III- Bhadla-III PS D/c line. The above ICTs (765/400kV & 400/220kV) at Bhadla-III are also considered as part of connectivity transmission system for RE projects. Additionally, total 5 nos. of 220kV line bays are already under implementation in ISTS (Ph-III Part B1) which is sufficient at present.

CEA enquired about utilization of 765/400kV ICTs (3rd & 4th) after commissioning of Bhadla-Fatehpur HVDC System. CTU stated that due to time gap (about 3 years) in implementation time of EHVAC and HVDC system, ICTs would be required. Optimal utilization of 765/400kV ICTs may be planned with other future RE schemes from Bhadla-IV PS or other system strengthening schemes. CEA enquired that whether more power can be evacuated from Bhadla-III PS beyond 6.5GW identified potential. CTU stated that beyond 6.5GW, connectivity will be granted at Bhadla-IV PS on HVDC system for which system is under planning. It is not recommended to inject more generation at Bhadla-III PS, due to very less margin availability in Bhadla-III onwards EHVAC system. CEA & Grid-India agreed for proposal.

SECI enquired that balance potential of Ramgarh PS (2.2GW) may be evacuated to EHVAC system in place of HVDC. CTU stated that Common transmission corridors were planned for Ramgarh (2.9GW) and Bhadla-III (6.5GW) RE potential as part of integrated transmission planning. In order to facilitate evacuation of RE power from Ramgarh/Bhadla-III PS, EHVAC system towards Sikar-II S/s & and 6GW HVDC corridor (Bhadla-Fatehpur) is being implemented as part of Ph-III scheme. The common transmission system for evacuation of power to various RE developers were allotted based on application priority combinedly to Ramgarh & Bhadla-III PS applicants. SECI requested that additional EHVAC corridor may be planned for evacuation of 2.2GW RE potential. In view of deliberations held in the meeting, following transmission system is agreed in ISTS

1. Augmentation of 2x500 MVA (4th & 5th) 400/220 kV ICTs at Bhadla-III PS
2. 220 kV bus sectionalizer (1 set) along with 220kV BC (1 no.) bay and 220kV TBC (1 no.) bay at Bhadla-III PS
3. Augmentation of 2x1500 MVA 765/400kV (3rd & 4th) ICTs at Bhadla-III PS

Implementation Schedule: 18 months from allocation (for S.No 1 & 2)

For S.No. 3: ICT-3 : 18 months from allocation (Best effort: Jun'25), ICT-4 : Dec'25 (30.12.25)

4. Transmission system for evacuation of power from Bikaner & Fatehgarh/Barmer Complex as part of Rajasthan REZ Ph-IV (Part-3a :2.4GW) and REZ Ph-IV (Part-4 :3.5GW) respectively

It was deliberated that Renewable Energy Zones (REZs) were identified by MNRE/SECI with a total capacity of 181.5 GW for likely benefits by the year 2030 in eight states, which includes 75 GW REZ potential in Rajasthan comprising of 15 GW Wind and 60 GW

Solar. In this regard a Committee on Transmission Planning for RE was constituted by MOP for planning of the requisite Inter State Transmission System required for the targeted RE capacity by 2030 for which a Comprehensive transmission plan for evacuation of 75GW RE potential from Rajasthan was evolved. Details of schemes approved/Under Planning scheme as part of above is as under:

S.No	Transmission Scheme	RE Potential	Status
A	Under Bidding/ Approved		
1	Rajasthan REZ Ph-IV (Part-1 :7.7GW) (Bikaner Complex)	14 GW (Solar 14GW, BESS:6GW) Bikaner-II : 3.7GW Bikaner-III: 4GW	Awarded
2	Rajasthan REZ Ph-IV (Part-2 :5.5GW) (Jaisalmer/Barmer Complex)	5.5GW (Solar) Fatehgarh-IV: 4 GW Barmer-I: 1.5 GW	Under Bidding
3	Rajasthan REZ Ph-IV (Part-3 :3.6GW) (Bikaner Complex)	3.6 GW (Solar) Bikaner-IV:3.6GW	Under approval with NCT
B	Planned/Under Planning		
1	Rajasthan REZ Ph-IV (Part-3a :2.4GW) (Bikaner Complex)	2.4 GW (Solar) Bikaner-IV : 2.4GW	Scheme planned for 2026-27 time frame (EHVAC)- Present proposal
2	Rajasthan REZ Ph-IV (Part-4 :3.5GW) (Jaisalmer/Barmer Complex)	3.5 GW (Solar) Fatehgarh-IV: 1 GW Barmer-I: 2.5 GW	
3	Rajasthan REZ Ph-IV (Part-5 : 6GW) (Barmer Complex)	6 GW (Solar) Barmer-II : 6GW	Scheme planned for 2029 time frame (HVDC)
4	Rajasthan REZ Ph-IV (Part-6 : 6GW) (Bhadla/Bikaner Complex)	6 GW (Solar) Bhadla-IV: 2 GW Bikaner-V: 4 GW*	

*SECI to confirm total RE potential at Bikaner-V

1. Rajasthan REZ Ph-IV (Part-3a :2.4GW) (Bikaner Complex)

It was deliberated that Transmission scheme is evolved for about 7.7GW (Solar) in Bikaner complex (14 GW potential along with 6 GW BESS) in Rajasthan for RE potential identified at Bikaner complex as part of committee report. However, no application of BESS (linked with RE) against envisaged 6GW was received. Accordingly, RE potential of about 7.7GW (in place of 14GW) can be evacuated from planned system (Ph-IV scheme) from Bikaner complex (Bikaner-II(3.7 GW) & Bikaner-III(4 GW)).

Further, In the Joint study meetings held on 18.10.23 & 23.10.23, Transmission system for evacuation of power from Bikaner Complex as part of Rajasthan REZ Ph-IV (Part-3 :3.6GW (Total 6 GW)) was discussed. In the meetings, two alternatives were deliberated

- Bikaner-IV- Siwani 765kV D/c line & onwards 400kV transmission system from Siwani S/s
- Bikaner-IV- Churu- Siwani 765kV D/c line & onwards 400kV transmission system from Siwani S/s along with LILO of one ckt of 765 kV Sikar-II (PG) -Khetri (PG) D/c line at Churu S/s and Churu- Fatehabad (PG) 400 kV D/c line.

In above meetings, Grid-India opined that in case of direct interconnection of Bikaner-IV to Siwani i.e. not considering 765/400kV Churu in between and onward transmission system, the Transmission system is kind of radial system connected with RE generation pocket and poses stability issues in various operational scenarios in future. Grid-India also emphasized that some margin should be kept in planning studies for operational scenarios. Further, the angular difference (>30 degrees) in N-1-1 /N-2 may cause problems in synchronization of lines after corridor outage. The proposed system is stable and well interconnected with Grid provides reliable power evacuation under various operational scenarios

CEA stated that N-1-1 or N-2 is a rare contingency and deliberations are required on N-1-1/N-2 compliance in planning studies as it may incur additional investment for strengthening of transmission system.

The scheme was further deliberated in 25thCMETS-NR meeting held on 31.10.23. In the meeting, CEA opined that Bikaner-IV S/s is planned for 6GW evacuation and at present direct interconnection between Bikaner-IV and Siwani is to be planned in first phase (3GW) and 765/400kV Churu S/s along with associated transmission scheme may be taken up in later stage while planning of balance 3 GW transmission scheme. Grid-India further emphasized that in case Churu S/s to be taken up at later stage of planning, 765kV Bikaner-IV-Siwani D/c line may not able to synchronize after corridor outage (765kV Bikaner-IV-Siwani D/c line) due to higher angular difference (>30 degrees) in N-1-1/N-2 in peak solar hours. Grid-India stated that due to potential delays in the restoration of the 765 kV Bikaner-IV to Siwani D/c line, synchronization of lines might not be feasible during peak solar hours. Instead, this process might need to be deferred to non-solar hours, which would entail running the system in a depleted condition for several hours.

In view of deliberations held in 25th CMETS-NR meeting, following transmission scheme was agreed in CMETS-NR meeting for evacuation of power from Rajasthan REZ Ph-IV (Part-3) [Bikaner IV :3.6GW]

Bikaner-IV: 3.6GW (Solar)

- Establishment of 4x1500 MVA, 765/400 kV & 4x500 MVA, 400/220 kV Bikaner-IV Pooling Station along with 2x240 MVar (765kV) & 2x125 MVar (420kV) Bus Reactors at a suitable location near Bikaner
- STATCOM (2x±300MVar) along with MSC (4x125 MVar) & MSR (2x125 MVar) along with 2 nos. 400kV line bays at Bikaner-IV PS
- 220kV line bays (5 nos.) for RE connectivity at Bikaner-IV PS
- 400kV line bays (1 no.) for RE connectivity at Bikaner-IV PS
- 220kV Sectionalization bay (1 set) along with BC (2 nos.) and 220 kV TBC (2 nos.) at Bikaner-IV PS
- 400kV Sectionalization bay (1 set) at Bikaner-IV PS
- LILO of one ckt of 765 kV Bikaner-III -Neemrana-II D/c line (2nd) at Bikaner-IV PS (~20 km) along with 330 MVAR switchable line reactor at Bikaner-IV PS end of 765 kV Bikaner-IV -Neemrana-II line (formed after LILO)
- Establishment of 765/400 kV, 2x1500 MVA S/s at suitable location near Churu along with 2x240 MVar (765kV) Bus Reactor & 2x125 MVar (420kV) Bus Reactor
- Bikaner-IV PS – Churu 765 kV D/c line along with 240 MVar switchable line reactor for each circuit at Bikaner-IV PS end (~180 km)
- LILO of one ckt of 765 kV Sikar-II (PG) -Khetri (PG) D/c line at Churu S/s (~90 km)
- Churu – Fatehabad (PG) 400 kV D/c line (Quad) along with 63 MVar switchable line reactor for each circuit at Churu S/s end (~150 km)
- Establishment of 765/400 kV, 3x1500 MVA S/s at suitable location near Siwani (Distt. Bhiwani) along with 2x240 MVar (765kV) Bus Reactor & 2x125 MVar (420kV) Bus Reactor*
- Siwani – Churu 765 kV D/c line (~95 km)
- Siwani – Jind (PG) 400 kV D/c line (Quad Moose) (~110 km)
- Siwani – Patran (Indi Grid) 400 kV D/c line (Quad) (~160 km) (400kV GIS duct :700m) along with 80 MVar switchable line reactor for each circuit at Siwani S/s end

Subsequently, Joint Study Meeting(s) were held in virtual mode on 28.12.23 with SECI, CEA, GRID-INDIA, RVPN, UPPTCL and other STUs of Northern region to finalize the Transmission system for evacuation of power from Bikaner & Fatehgarh/Barmer Complex as part of Rajasthan REZ Ph-IV (Part-3a :2.4GW) and REZ Ph-IV (Part-4 :3.5GW) respectively (**Minutes of Meeting is enclosed in Annexure-1**).

Gist of discussion in Joint study meeting held on 28.12.23

As discussed earlier, Transmission scheme was evolved and approved in 25th CMETS-NR meeting for about 3.6 GW (Solar) as part of Rajasthan REZ Ph-IV (Part-3 :3.6GW) (Bikaner Complex) for injection at Bikaner-IV in Rajasthan. At present connectivity of about 6 GW is already agreed/received against its total potential (6GW) and more applications are expected due to land availability and being outside of GIB area.

It was stated that for balance 2.4GW capacity (thus makes total capacity of 6 GW at Bikaner-IV PS), EHVAC evacuation system is planned for comprising Bikaner-IV PS – Churu 765 kV D/c line (2nd) line and LILO of 2nd ckt of 765 kV Sikar-II (PG) -Khetri (PG) D/c line at Churu S/s. For onward dispersal of power, Siwani – Sonipat (PG) 400 kV D/c line (Quad) is also planned.

For EHVAC system, studies were carried out in 2026-27 time frame in solar maximized scenario and Study files for solar maximized scenario was shared on 20.12.23 with all constituents. Considering grant of connectivity to new RE generators in Bikaner complex as well as for evacuation of power beyond Bikaner complex, following transmission scheme is proposed for evacuation of power from Rajasthan REZ Ph-IV (Part-3a :2.4GW) [Bikaner complex]. Details of proposed EHVAC scheme is as under

A. Transmission system for evacuation of power from Bikaner Complex as part of Rajasthan REZ Ph-IV (Part-3a :2.4GW)

In the above meeting following transmission system was proposed for evacuation of power from Rajasthan REZ Ph-IV (Part-3a :2.4GW) for evacuation of power from Bikaner complex

Bikaner-IV: 2.4GW (Solar)

- Augmentation of 1x1500 MVA(5th), 765/400 kV ICT at Bikaner-IV Pooling Station
- Augmentation of 3x500 MVA (5th to 7th), 400/220 kV ICT at Bikaner-IV Pooling Station[#]
- 220kV line bays (3 nos.) for RE connectivity at Bikaner-IV PS[#]
- Bikaner-IV PS – Churu 765 kV D/c line (2nd) along with 240 MVAR switchable line reactor for each circuit at Bikaner-IV PS end (~175 km)
- LILO of other ckt of 765 kV Sikar-II (PG) -Khetri (PG) D/c line at Churu S/s* (~80 km)
- Augmentation of 765/400 kV, 1x1500 MVA ICT (4th) at Siwani S/s
- Siwani – Sonipat (PG) 400 kV D/c line (Quad Moose) (~150 km) along with 63 MVAR switchable line reactor for each circuit at Siwani S/s end

**one ckt is already LILOed as part of Rajasthan REZ Ph-IV (Part-3 :3.6GW) scheme)*

To be finalized as per application received

In the meeting, CEA stated that with Phase-IV Part 3a (2.4GW) proposed scheme, there will be no requirement of 765/400kV Churu S/s & associated interconnections, which were earlier agreed in 25th CMETS-NR meeting as part of Phase-IV Part 3 (3.6GW) scheme. CEA stated that in the earlier agreed Phase-IV Part 3 (3.6 GW) scheme, Churu S/s & associated transmission scheme (LILO of one ckt of 765 kV Sikar-II (PG) -Khetri (PG) D/c line at Churu S/s and Churu- Fatehabad (PG) 400 kV D/c line) was considered to address issue on synchronization of line after corridor outage (765kV Bikaner-IV-Siwani D/c line) in N-1-1/N-2 in peak solar hours as emphasized by Grid-India. CEA stated that in place of Bikaner-IV – Churu D/c line, a parallel corridor towards Siwani S/s i.e. 765kV Bikaner-IV- Siwani D/c line may be considered for additional 2.4GW evacuation of power as part of scheme. With the above parallel 765kV corridor i.e. 765kV Bikaner-IV – Siwani D/c line, angular stability issue under N-1-1 contingency will be resolved as 4 nos. of

765kV circuits are available towards Siwani S/s. Further in future, if there is requirement of establishment of Churu S/s, Bikaner-IV – Siwani D/c line may be LILOed at Churu S/s.

It was deliberated that Transmission system for evacuation of power from Bikaner Complex as part of Rajasthan REZ Ph-IV (Part-3 :3.6GW) is being taken up for approval in next NCT. In view of that both the schemes viz Ph-IV: Part 3 (3.6GW) & Ph-IV: Part 3a (2.4GW)) may be considered comprehensively for 6GW evacuation.

CTU stated that Rajasthan REZ Ph-IV (Part-3 :3.6GW) is already approved in 25th CMETS-NR meeting held in Oct'23 and change in earlier agreed scheme (3.6GW) will delay the implementation timeline for 3-4 months and may impact the evacuation of RE generation from Bikaner-IV PS. CTU also stated that in 26th CMETS-NR meeting, SECI vide mail 07.12.23 informed that South West Churu has solar potential & therefore considering solar potential at Churu as indicated by SECI, 220kV future scope was retained at Churu S/s as earlier agreed in 25th CMETS-NR meeting. CTU stated that they will review the proposal based on CEA observations and deliberations in meeting and evaluate various alternatives (with and without Churu S/s) in a comprehensive manner. Accordingly, the scheme will be further deliberated and finalized in next Joint study meeting.

Gist of discussion in Joint study meeting held on 09.01.24

In view of the above deliberations, CTU re-examined the proposals and shared the study files along with two alternatives ;

- **Alternative-1** : Bikaner-IV- Churu- Siwani 765kV D/c & Bikaner-IV- Siwani 765kV D/c & associated transmission interconnection at Churu S/s i.e. LILO of one ckt of 765 kV Sikar-II (PG) -Khetri (PG) D/c line at Churu S/s and Churu- Fatehabad (PG) 400 kV D/c line & onwards 400kV transmission system from Siwani S/s (Towards Jind, Patran, Sonipat)
- **Alternative-2:** Bikaner-IV- Siwani 765kV 2xD/c line & onwards 400kV transmission system from Siwani S/s (Towards Jind, Patran, Fatehabad, Sonipat)

It was stated that recently additional margin of about 600MW (over 4GW planned capacity) was released at Bikaner-III PS, with Bikaner-IV transmission system. In the earlier agreed system (LILO of one ckt of 765 kV Bikaner-III -Neemrana-II D/c line at Bikaner-IV PS), 765/400kV ICTs at Bikaner-III PS and Bikaner (PG) is higher in N-1 contingency. Space is not available for augmentation of new at 765/400kV ICT (5th) at Bikaner (PG) and 765/400kV ICT (7th) at Bikaner-III PS. Therefore, in place of LILO of one ckt of 765 kV Bikaner-III -Neemrana-II D/c line at Bikaner-IV PS, LILO of both ckts of 400kV Bikaner II- Bikaner-III D/c line (Quad) at Bikaner-IV is proposed which will relieve the above ICT loadings and increase the operational margin on above ICTs. Additionally STATCOM at Siwani (± 300 MVar) is proposed in Alternative-2 to mitigate low voltage issue.

Further, Observations were received on 04.01.24 from Grid-India and CEA on both the alternatives for transmission system for evacuation of power from Bikaner complex as part of Rajasthan REZ Ph-IV (Part-3). Major observations are as under

A. Grid-India Observations/Comments

- During peak solar period, voltages at Siwani (730kV) and Bikaner-IV (740kV) are very low even after taking B/Rs out. If line reactors are to be taken out of service for maintaining voltages, the same needs to be mentioned along with the scheme.
- Considering the identified solar potential at Churu by SECI, establishment of Churu substation will help in tapping the solar generation in that complex. CTUIL may clarify its plan and the timelines for establishing the ISTS Churu substation, if the same is not being taken up right now.

Additionally, Grid-India in 25th CMETS-NR meeting opined that in case of direct interconnection of Bikaner-IV to Siwani i.e. not considering 765/400kV Churu in between and onward transmission system, the Transmission system is kind of radial system connected with RE generation pocket and poses stability issues in various operational scenarios in future. Further Grid-India also requested to share Study cases for off peak scenarios. Other study related comments received from Grid-India was also deliberated in above Joint study meeting

B. CEA Observations/Comments

- Power flow in both the alternatives seems to be in order. Transmission system requirement is less in Alt 2. This would result in saving of hundreds of crore.
- In Alt 2, with Bikaner-IV- Siwani 765 kV 2xD/c line, issue of synchronization of line after corridor outage (765 kV Bikaner-IV-Siwani D/c line) in N-1-1/N-2 in peak solar hours as highlighted by Grid-India gets resolved.
- As intimated by CTUIL, connectivity applications of 5.4 GW RE has already been received against potential of 6 GW. Therefore, a comprehensive scheme for 6 GW RE evacuation could be implemented.
- In view of the above, Alt 2 seems to be preferable at present. Churu substation could be implemented in future, if required, with LILO of Bikaner-IV- Siwani 765 kV line to cater to the potential in Churu district.

CTU analyzed both the alternatives with above changes and observations are as under :

- Estimated Cost of scheme is about 12% lower (about Rs 1200 Cr) in Alt-2 than Alt-1
- In Alt-2, due to longer line length, under N-1 contingency of 765kV Bikaner-IV-Siwani, voltage dipping will be higher (about 10kV than Alt-1) and in the range of 730-740kV considering STATCOM at Bikaner-IV under various operating scenario. In above scenario STATCOM (± 300 MVar) at Siwani may also be required (in case of operational difficulty of line reactors switching on EHVAC lines in RE pockets).
- With Alt-2, Churu Substation may be planned with future transmission proposals i.e. Bikaner-V, other strengthening schemes based on requirement. Therefore, RE evacuation from Churu S/s will only be possible at later stage (beyond 2027) in Alt-2.
- Comprehensive scheme for 6GW will be implemented with tentative timeline of Sep'26 in Alt-2, however in Alt-1 tentative timeline for part scheme (3.6GW) was Jul'26

Revised study files with proposed changes (LILO of both ckts of 400kV Bikaner II- Bikaner-III D/c line at Bikaner-IV PS in place of LILO of one ckt of 765 kV Bikaner-III -Neemrana-II D/c line at Bikaner-IV PS & Siwani STATCOM (in Alt-2)) was circulated on 06.05.24 along with study plots. SECI stated that Churu complex was not identified in solar potential zone as part of 500GW report and thereafter potential was indicated only because the substation was planned as part of agreed Bikaner-IV transmission scheme (3.6GW), however no response from RE developers is yet received to assess & harness RE potential at Churu complex. CTU stated that in above case till no confirmation received from SECI, RE potential at Churu will not be considered and may be evolved based on SECI inputs at later stage.

Grid-India stated that in June solar scenario studies, Punjab/Haryana load pf is considered near to 0.99 which is very optimistic. However, in real time they have faced voltage dipping issues in Punjab and Haryana intra state network and interstate network (incident to intra state) due to high reactive power drawl in paddy season. This Intra state reactive support may further reduce in paddy season due to envisaged hydro generation from J&K/HP and solar generation from Rajasthan.

CTU enquired to Grid-India about capacitive support by STUs/DISCOMs in real time to improve load pf and mitigate voltage dipping issues in peak loading season. Grid-India stated that in real time capacitor bank support is available in Punjab, however in Haryana low voltage issue still persists in June-July peak loading season. CTU stated that in the integrated planning of transmission system, reactive power (MVar) drawl/absorption requirement at load end to be compensated at intra state network level only and ISTS system is not supposed to be planned to compensate MVar injection/ drawl from intra state network under various operating scenario as it will also increase losses due to MVar flow to downstream network from upstream network which is not desirable. In view of that load pf ($>0.98\text{pf}$) may be maintained by STUs/DISCOMs specially in Punjab & Haryana through new capacitor banks, if required and also provide progress of installing new capacitors in monthly OCC/NRPC meetings.

Considering above, it was decided that STUs specially Haryana may assess the reactive compensation requirement for peak load season in next 2-3 years and may take necessary actions to maintain pf and mitigating low voltage issues in intra state network. HVPN told that they have planned capacitor banks installation at downstream network.

Grid-India highlighted the voltage dipping issue at Siwani substation ($\sim 730\text{kV}$) in N-1 contingency. The voltage will further decreases to 725kV , with no STATCOM reactive support (STATCOM support reserved for dynamic support only). CEA stated that line reactors on EHVAC lines can be switched off during low voltage as it will give direct relief to improve voltage to $15\text{-}20\text{kV}$, however operational difficulties in switching on/off of line reactors may be deliberated. Further $\pm 600\text{MVar}$ STATCOM at Siwani S/s may be considered (earlier $\pm 300\text{MVar}$ proposed in scheme) along with MSC/MSR to provide voltage support. Grid-India stated that earlier, voltage in Fatehgarh complex was observed up to 720kV in peak solar hours and under such condition, sequential line reactor switching was carried out in for volage control, however they have faced resistance from TSPs regarding line reactor switching and line opening for voltage control. In view of that provision of switching of line reactors may be clearly stipulated in the RfP of proposed scheme.

CEA stated that line reactor switching in RE pockets is in general operational requirement and need not to be specific for one scheme and should be carried out in real time as per Grid requirement. CTU stated that presently as per IEC standard followed for breaker duty cycle, class M2 circuit breakers i.e frequently operated CBs for special service requirements and designed so as to meet only limited maintenance is mechanical type tested for 10,000 operations, however whenever such line reactor switching issue is raised by TSP in future, it should come with proper reasoning & justification from TSP side.

Grid-India stated that in steady state study file, STATCOMs along with MSC/MSR may be kept off for worst scenario (reserved for dynamic range). To mitigate low voltage issues without STATCOM support, line reactors to be switched off as per requirement. CTU stated that in base case voltage at Siwani S/s is about 733kV and in N-1 contingency of one ckt of Bikaner-IV-Siwani 2xD/c lines, voltage may further reduce to 728kV. In above scenario, voltage will be improved to 740kV considering STATCOM (at Bikaner-IV and Siwani) and 748kV with STATCOM+MSC at Bikaner-IV & Siwani S/s. However, in case STATCOM along with MSCs is kept off as suggested by Grid-India and line reactor switching is carried out, it will further improve the voltage of Siwani S/s by 750kV (line reactors on other 3 ckts of 765kV Bikaner-IV-Siwani kept off). In view of that it is recommended that steady state voltage to be maintained over 750kV in planning files considering line reactor switching (760kV at Siwani S/s in present case with line reactor on 765kV Bikaner-IV - Siwani kept off) . Accordingly, it was decided by CEA, CTU & Grid-India that line reactor switching may be carried out in a sequential manner in RE pockets for better voltage control.

CTU stated that in Feb solar maximized scenario, loading of 400kV RAPP- Shujalpur D/c line is critical (about 950MW) in N-1 contingency. To relieve loading of above line, EHVAC corridor from RAPP generating station or other transmission corridor in parallel to above line is to be planned. In 500GW report also, 765kV Ajmer-Kota-Shujalpur transmission corridor was planned to integrate RE complexes of Ajmer. In view of that, it was decided that loading of above line will be reviewed with progress of RE generation projects at Rajasthan and planning to be carried out in next phase to relieve loading on above line. Grid-India & CEA agreed for the same

Grid-India enquired that timeline for implementation of two nos. of Bikaner-IV-Siwani D/c lines will be same or different. CEA stated that scheme will be segregated in various packages and efforts to be made that all the packages are implemented simultaneously with implementation timeline decided in NCT meeting. Grid-India stated that as per past experiences, there will be always some mismatch (up to 6 months) in implementation time of two parallel double ckt corridors from same RE pooling stations. In such situation, in the event of only one corridor i.e. 765kV Bikaner-IV-Siwani D/c is commissioned and 2nd 765kV D/c gets delayed, RE generation of 2-3GW may only be evacuated in base case with one D/c and in N-1 contingency, however in N-1-1 contingency, restoration of line becomes delayed in solar peak hours and shall impact the RE generation.

CTU stated that considering HVPNL existing & proposed interconnections, interconnections as well as GNA granted to bulk consumer for 100MW GNA at 220kV Jind(PG), 400/220kV ICTs (3x500 MVA) at Jind(PG) are critically loaded in N-1 contingency. Grid-India stated that at present 400/220kV Jind(PG) S/s has 2x500MVA ICTs & 4 nos. of 220kV interconnections whereas 3rd ICT (500MVA) in ISTS by POWERGRID and 4 nos. of STU feeders are under implementation by HVPN. At present loading on 2x500MVA ICT is about 500MW (peak) and in view of that ICT augmentation at Jind may considered at later stage with commissioning of HVPNL

interconnections & 400/220kV ICT (3rd) at Jind(PG). HVPNL stated that 400kV Munakh S/s for power evacuation of YTPP (800MW) is under planning. Additionally, 640MVA load will be envisaged progressively from 2027 by IOCL. Accordingly, requirements will be assessed based on Grid-India operational feedback report and loading diversion in planning studies considering HVPNL proposal for evacuation of power from YTPP generation (800MW). CEA agreed for the same.

Further in 27th CMETS-NR meeting, RVPN stated that they do not have any drawl requirement in future from proposed Merta-2 S/s, however as informed earlier in joint study meeting, in view of envisaged RE potential in Nagaur distt., space provision to be kept for 220kV scope i.e. 400/220kV ICTs., 220 kV line bays for drawl & 220kV Sectionalization bay for RE injection.

Subsequently, off peak file was circulated on 12.01.24 to all stakeholders. Result of system studies incorporating observations received from stakeholders is enclosed in **Exhibit-1**. Considering grant of connectivity to RE generators in Bikaner complex as well as for evacuation of power beyond above complex, following transmission scheme was agreed in Joint study meetings (28.12.23 & 09.01.24) as well as in 27th CMETS-NR meeting held on 10.01.24 as part of Rajasthan REZ Ph-IV (Part-3 : 6 GW) system

Transmission system for evacuation of power from Rajasthan REZ Ph-IV (Part-3 : 6GW) Bikaner-IV: 6 GW (Solar)

- Establishment of 6x1500 MVA, 765/400 kV & 6x500 MVA, 400/220 kV Bikaner-IV Pooling Station along with 2x240 MVA (765kV) & 2x125 MVA (420kV) Bus Reactors at a suitable location near Bikaner *

Future provisions at Bikaner-IV PS (excl. scope for present scheme): Space for

- 765 kV line bays along with switchable line reactors – 6 nos.
- 765kV Bus Reactor along with bay: 1 no.
- 400 kV line bays along with switchable line reactor –4 nos.
- 400 kV line bays–2 nos.
- 400/220kV ICT along with bays -4 nos.
- 400 kV Bus Reactor along with bay: 1 no.
- 400kV Sectionalizer bay: 2 sets
- 220 kV line bays for connectivity of RE Applications -5 nos.
- 220kV Sectionalizer bay: 2 sets
- 220 kV BC (2 nos.) bays and 220 kV TBC (2 nos.) bays

****along with provision of 80MVA spare reactor (Single phase) & 500MVA spare transformer unit (Single phase)***

- 220kV line bays (6 nos.) for RE connectivity at Bikaner-IV PS
- 400kV line bays (3 no.) for RE connectivity at Bikaner-IV PS
- 220kV Sectionalizer bay (1 set) along with BC (2 nos.) bays and 220 kV TBC (2 nos.) bays at Bikaner-IV PS
- 400kV Sectionalizer bay (1 set) at Bikaner-IV PS

- STATCOM (2x±300MVar) along with MSC (4x125 MVar) & MSR (2x125 MVar) along with 2 nos. 400kV bays at Bikaner-IV PS
- LILO of both ckts of 400kV Bikaner II PS- Bikaner III PS (Quad) direct line at Bikaner-IV PS (~20kms)
- Establishment of 765/400 kV, 6x1500 MVA S/s at suitable location near Siwani (Distt. Bhiwani) along with 2x240 MVar (765kV) Bus Reactor & 2x125 MVar (420kV) Bus Reactor*

Future provisions at Siwani S/s (excl. scope for present scheme): Space for

- 765 kV line bays along with switchable line reactors – 6
- 765kV Bus Reactor along with bay: 1 nos.
- 400 kV line bays along with switchable line reactor –4
- 400 kV Bus Reactor along with bays: 1 no.
- 400kV Sectionalizer bay: 1 set

***along with provision of 80MVar spare reactor (Single phase) & 500MVA spare transformer unit (Single phase)**

- Bikaner-IV PS – Siwani 765 kV 2xD/c line along with 240 MVar switchable line reactor for each circuit at each end (~260 km)
- Siwani – Jind (PG) 400 kV D/c line (Quad) (~110 km)
- Siwani – Patran (Indi Grid) 400 kV D/c line (Quad) (~160 km) (400kV GIS duct :700m) along with 80 MVar switchable line reactor for each circuit at Siwani S/s end
- Siwani – Fatehabad (PG) 400 kV D/c line (Quad) (~80 km)
- Siwani – Sonipat (PG) 400 kV D/c line (Quad) (~150 km) along with 63 MVar switchable line reactor for each circuit at Siwani S/s end
- STATCOM (2x±300MVar) along with MSC (4x125 MVar) & MSR (2x125 MVar) along with 2 nos. 400kV bays at Siwani S/s
- 400kV Sectionalizer bay (1 set) at at Siwani S/s

It was stated that for evacuation of power beyond 6GW from Bikaner-IV PS, HVDC from Bikaner complex is planned. For additional power transfer from Bikaner complex, a new substation at suitable location in Bikaner i.e. Bikaner-V is planned. In meeting, CTU enquired **SECI to conform RE potential for Bikaner-V PS** as 14 GW evacuation system is already planned from Bikaner complex (Bikaner-II/III:8GW, Bikaner-IV: 6GW) against 14 GW potential (along with 6 GW BESS) in Rajasthan for RE potential identified at Bikaner complex as part of committee report.

SECI stated that potential identified in Bikaner complex (incl. 6GW BESS which has not yet come up) as part of 500GW report is already considered for planning (Bikaner-II/III/IV). CTU stated that transmission system requirements (with ATS/Without ATS) for connectivity to various RE developers granted at Bikaner-V PS depends on RE potential qualification which impact BGs applicability to RE developers under GNA regulation. Further, intimation to be issued in a timebound manner as per GNA regulation timelines for RE application granted connectivity at Bikaner-V PS. Therefore RE capacity along with potential qualification at Bikaner-V or any other such locations may be expedited by SECI in consultation with MNRE

SECI stated that Bikaner-V station cannot be considered alone for decision of RE potential and it may be considered in totality for all such new potential locations. SECI stated that they have already communicated to RE developers for such information and once data is received from RE developers, SECI will discuss it with MNRE. Subsequently, MNRE will consult all the stakeholders and requirement may be firmed up based on views of MNRE to take up part system for one or two locations immediately or in totality for all such potential locations. CTU suggested that total RE capacity along with its potential qualification at Bikaner-V may be finalized within 2-3 weeks. SECI stated that they will expedite the process and discuss the matter with MNRE and revert on the same.

Further as part of 75GW REZ in Rajasthan (as part of 500GW report), 5 GW potential (3GW solar, 2 GW wind) along with net dispatch of 2GW considering 2GW BESS was considered at Bhadla complex (Bhadla-IV: 2GW). A comprehensive HVDC scheme is planned for 6GW evacuation from Bikaner-V (4GW) and Bhadla-IV (2GW) PS towards Baripada in Orissa (ER) through HVDC system, for which studies is undergoing for 2029 timeframe and will be taken up in Joint study meeting. However grant of connectivity shall depend on application received at Bikaner-V & Bhadla-IV which may differ than Bikaner-V (4GW) & Bhadla-IV (2GW). Details of broad scheme is as under

**Proposed Transmission system for evacuation of power from Rajasthan REZ Ph-IV (Part-6 :6 GW) [Bhadla/Bikaner Complex]
Bhadla-IV: 2 GW (Solar), Bikaner-V : 4GW (Solar)***

- Establishment of 6000 MW, $\pm$ 800 kV Merta (HVDC) [LCC] terminal station (4x1500 MW) in Merta-II substation
 - 400/33 kV, 2x50 MVA transformers for exclusively supplying auxiliary power to HVDC terminal.
 - 400kV bus sectionaliser -2 nos (1 Set) at Merta-II (HVDC) station
- Establishment of 765/400kV, 4x1500 MVA S/s & 400/220kV 4x500MVA pooling station at suitable location near Bikaner (Bikaner-V PS) along with 2x125 MVar & 2x240 MVar bus reactor

Future provisions

Space for

- 765/400kV ICTs along with bays- 2
- 765 kV line bays along with switchable line reactors – 10
- 765kV Bus Reactor along with bay: 1 nos.
- 400 kV line bays along with switchable line reactor –10
- 400kv line bays : 4 nos.
- 400 kV Bus Reactor along with bays: 1 no.
- 400kV Sectionalization bay: 2 sets
- 400/220kV ICT along with bays -6 nos.
- 220 kV line bays - 10 nos.
- 220kV Sectionalization bay: 3 sets

**along with provision of 80-MVAr spare reactor (Single phase) & 500MVA spare transformer unit (Single phase)*

- 220kV line bays (4 nos.) for RE connectivity at Bikaner-V PS
- Bikaner-V PS – Merta-2 765KV D/c line (~ 200kms) along with 240 MVAr switchable line reactor for each circuit at each end of Bikaner-V PS – Merta-2 765KV D/c line
- Establishment of 765/400kV, 3x1500 MVA S/s & 400/220kV 3x500MVA pooling station at suitable location near Bhadla (Bhadla-IV PS) along with 2x125 MVAr & 2x240 MVAr bus reactor

Future provisions

Space for

- 765/400kV ICTs along with bays- 3
 - 765 kV line bays along with switchable line reactors – 10
 - 765kV Bus Reactor along with bay: 1 nos.
 - 400 kV line bays along with switchable line reactor –10
 - 400kv line bays : 4 nos.
 - 400 kV Bus Reactor along with bays: 1 no.
 - 400kV Sectionalization bay: 2 sets
 - 400/220kV ICT along with bays -7 nos.
 - 220 kV line bays - 10 nos.
 - 220kV Sectionalization bay: 3 sets
- Bhadla-IV PS – Merta-II 765KV D/c line (~ 250kms) along with 240 MVAr switchable line reactor for each circuit at each end of Bhadla-IV PS – Merta-2 765KV D/c line
 - Augmentation of 765/400 kV, 3x1500 MVA ICTs at Merta-II S/s
 - 6000 MW, ± 800 kV Merta (HVDC) [LCC] terminal station (4x1500 MW)
 - Establishment of 6000 MW, ± 800 kV Baripada (HVDC) [LCC] terminal station (4x1500 MW) at suitable location near Baripada (Distt. Mayurbhanj)
 - ± 800 kV HVDC line (Hexa lapwing) between Merta-II (HVDC) & Baripada (HVDC) (with Dedicated Metallic Return) - 1600 km

EHVAC system beyond Baripada (HVDC) S/s is under planning.

** Evacuation capacity among Bikaner-V and Bhadla-IV may vary depend on application received at Bikaner-V and Bhadla-IV*

2. Rajasthan REZ Ph-IV (Part-4 :3.5GW) (Jaisalmer/Barmer Complex)

Joint Study Meeting(s) were held in virtual mode on 28.12.23 with SECI, CEA, GRID-INDIA, RVPN, UPPTCL and other STUs of Northern region to finalize the Transmission system for evacuation of power from Bikaner & Fatehgarh/Barmer Complex as part of

Rajasthan REZ Ph-IV (Part-3a :2.4GW) and REZ Ph-IV (Part-4 :3.5GW) respectively respectively (**Minutes of Meeting is enclosed in Annexure-1**). Details of deliberations for Transmission system for evacuation of power from Bikaner & Fatehgarh/Barmer Complex as part of Rajasthan REZ Ph-IV (Part-4 :3.5GW) is as under

Gist of discussion in 1st Joint study meeting held on 28.12.23

It was informed that as part of Renewable Energy Zones (REZs) identified by MNRE/SECI with a total capacity of 75 GW REZs in Rajasthan, evacuation system for 13GW was planned at Fatehgarh (5GW) & Barmer (8GW) complex. Transmission scheme for Rajasthan REZ Ph-IV (Part-2:5.5GW) (Jaisalmer/Barmer Complex) was approved in 14th NCT meeting for injection at Fatehgarh-IV PS (4GW) & Barmer-I PS (1.5GW) in Rajasthan and under bidding.

At present connectivity of about 9 GW capacity is already granted/received. For evacuation of power beyond 5.5GW from Fatehgarh-IV PS/Barmer-I PS, hybrid transmission system (EHVAC(3.5GW)+HVDC(6GW)) from Fatehgarh/Barmer complex is planned. Studies were carried out in 2026-27 time frame in solar maximized scenario for EHVAC system and Study files for solar maximized scenario was shared on 20.12.23 with all constituents.

It was stated in meeting that the EHVAC evacuation system comprises 3.5 GW capacity transmission system from Fatehgarh-IV/Barmer-I (thus making total planned capacity of 9GW (5.5GW+3.5GW)) and 6GW HVDC system from Barmer-II PS (new pooling station is being established as part of HVDC scheme). The 6GW HVDC scheme comprises Barmer-II to Suitable locations in WR/SR through HVDC system.

In the joint study meeting, CTU stated that as per committee report Barmer-II PS was planned for 6GW RE potential (2GW BESS capacity) with 4GW evacuation capacity. In 25th CMETS-NR meeting, SECI informed that at present there is no clear visibility for RE projects with BESS before 2027 as award process will take time (1-2 years). In view of that 3.5GW RE potential (solar) remains untapped due to non-materialisation of BESS capacity (Barmer-I : 1.5GW+Barmer-II: 2GW) in Barmer complex

Details of RE potential considered in Fatehgarh/Barmer complex is under:

S.No	Transmission Scheme	RE Potential	Cumulative RE Potential
1	Rajasthan REZ Ph-IV (Part-2:5.5GW) (Jaisalmer/Barmer Complex) EHVAC system -Under Bidding	5.5GW (Solar) Fatehgarh-IV: 4 GW Barmer-I: 1.5 GW	5.5GW (Solar) Fatehgarh-IV: 4 GW Barmer-I: 1.5 GW
2	Rajasthan REZ Ph-IV (Part-4:3.5GW) (Jaisalmer/Barmer Complex) EHVAC system-Present proposal	3.5 GW (Solar) Fatehgarh-IV: 1 GW Barmer-I: 2.5 GW	9GW (Solar) Fatehgarh-IV: 5 GW Barmer-I: 4 GW
3	Rajasthan REZ Ph-IV (Part-5 : 6GW) (Barmer Complex) HVDC system towards WR/WR-SR	6 GW (Solar) Barmer-II : 6 GW*	15 GW (Solar) Fatehgarh-IV: 5 GW Barmer-I: 4 GW Barmer-II: 6 GW*

**Subject to inputs from MNRE/SECI for enhancement of RE potential at Barmer-II PS*

In the joint study meeting as well as in present CMETS meeting, CTU stated that as per committee report Barmer-II PS was planned for 6GW RE potential with 4GW evacuation capacity considering 2GW BESS capacity. SECI informed that total RE potential identified at Barmer complex is 13GW considering BESS of 3.5GW capacity. SECI reiterated that at present there is no clear visibility for RE projects with BESS in Rajasthan and therefore transmission system to be planned for 13 GW RE potential (solar) from Barmer complex due to non-materialization of BESS capacity. CTU stated that 4GW EHVAC system is already planned from Barmer-I PS and with 6GW RE potential considered at Barmer-II PS, transmission system for balance 3GW RE potential will be planned at later stage from new pooling station envisaged in Barmer complex

Accordingly, based on inputs from SECI it was decided in the meeting that total 6GW RE potential may be considered for planning of transmission scheme from Barmer-II PS.

In the Joint study meeting held on 28.12.2023, following transmission system for evacuation of power from Fatehgarh/Barmer Complex as part of Rajasthan REZ Ph-IV (Part-4 :3.5GW) was proposed

Fatehgarh-IV: 1 GW (Solar), Barmer-I :2.5GW (Solar)

- Establishment of 765/400kV, 2x1500 MVA S/s at suitable location near Merta (Merta-II Substation) along with 2x125 MVar & 2x240 MVar bus reactor at Merta-II* S/s
- STATCOM (2x±300MVar) along with MSC (4x125 MVar) & MSR (2x125 MVar) along with 2 nos. 400kV line bays at Barmer-I PS
- Augmentation with 765/400 kV, 2x1500 MVA Transformer (4th & 5th) at Barmer-I PS
- Augmentation of 3x500 MVA (3rd to 5th), 400/220 kV ICT at Barmer-I PS
- 220kV line bays (4 nos.) for RE connectivity at Barmer-I PS
- 220kV Sectionalization bay (1 set) along with BC (1 nos.) and 220 kV TBC (1 nos.) at Barmer-I PS
- Fatehgarh-IV PS – Barmer-I PS 400kV D/c line (Quad) (~40km)
- Barmer-I PS – Merta-II 765 kV D/c line along with 240 MVar switchable line reactor for each circuit at each end of Barmer-I PS – Merta-II 765 kV D/c line (~320 km)
- Merta-II – Beawar 400 kV D/c line (Quad) (~55 km)
- Merta-II – Dausa 765 kV D/c line along with 240 MVar switchable line reactor for each circuit at each end of Merta-II – Dausa 765 kV D/c line (~250 km)
- Establishment of 765/400kV, 2x1500 MVA S/s at suitable location near Ghiror (Distt. Mainpuri) along with 2x125 MVar & 2x240 MVar bus reactor at Ghiror S/s (UP)
- Dausa - Ghiror 765 kV D/c line along with 240 MVar switchable line reactor for each circuit at each end of Dausa - Ghiror 765 kV D/c line (~285 km)
- LILO of both ckt of 765 kV Aligarh (PG) -Orai (PG) D/c line (~40 km) at Ghiror S/s along with 240 MVar switchable line reactor for each circuit at Ghiror S/s end of 765 kV Ghiror -Orai D/c line
- LILO of one ckt of 765kV Agra (PG) – Fatehpur(PG) 765kV D/c line at Ghiror along with 240 MVar switchable line reactor at Ghiror S/s end of 765 kV Ghiror -Fatehpur line (~40 kms)

In above meeting, Grid-India enquired about installation of STATCOMs at new pooling stations in Bikaner and Fatehgarh/Barmer complex. CTU stated that at present 3 nos. STATCOMs are already operational at Fatehgarh-II, Bhadla-II & Bikaner-II and 2 nos. STATCOMs at Ramgarh and Fatehgarh-III are under implementation. Additionally, 3 nos. STATCOMs at Bikaner-IV, Siwani and Barmer-I is also planned in proposed schemes (Ph-IV: Part 3 & 4). Additionally, space provision to be kept STATCOM at Merta-II, & Ghiror S/s as part of future scope.

Grid-India stated that proposal is in order, however 400/220kV ICTs (2x315MVA) at Agra (PG) is critically loaded in N-1 contingency. CTU stated that with proposed scheme there is marginally increment (~5MW) in 400/220kV ICT loading in Agra. CTU stated that space is not available for installation of 3rd ICT at Agra(PG) S/s, therefore replacement of ICTs may be considered in future based on real time loadings.

CTU stated that in the present proposal they have optimally utilized the Aligarh-Orai and Fatehpur -Agra high capacity 765kV corridors as the loadings of lines are less in solar maximized scenario. 765/400kV Orai Substation is well connected with WR Grid and Power from 765/400kV Orai shall be dispersed to load centres of WR through various high capacity lines. CTU stated that they are planning to interconnect proposed Ghiror S/s to nearby UPPTCL substation i.e. 400/220kV Firozabad at 400kV level as part of present scheme. CTU requested UPPTCL to provide space availability for 2 nos. of 400kv line bays at Firozabad S/s. This will also relieve the loading of Agra ICTs and increase resiliency of system. UPPTCL agreed for the same.

UPPTCL requested that studies may be performed with Bhadla-Fatehpur HVDC system to check loadings with proposed system. CTU stated that Bhadla-Fatehpur system is under bidding with scheduled implementation Feb'28 (Pole-1) and therefore not considered in studies, however they will carry out sensitivity case considering HVDC system and commensurate RE generation and will provide to UPPTCL. Same was provided to UPPTCL on 02.01.24.

CEA stated that proposal in order however, studies may be reviewed with UP intra state transmission system and associated RE generation in GEC-II. CTU agreed for the same. CTU enquired SECI and UPPTCL for envisaged RE potential in Bundelkhand region. UPPTCL stated that UPNEDA communicated 10GW RE potential in Bundelkhand region (UP), out of which UPPTCL has planned transmission scheme for evacuation of 4GW solar in intra state and transmission scheme is already approved in GEC-II. CTU stated that they have also received 1200MW solar application in Bundelkhand region for injection at Orai S/s which will be discussed in ensuing CMETS-NR meeting. It was stated that the above 4GW potential (out of 10GW potential) is part of 500GW report, however balance 6GW potential as indicated by UPNEDA is not part of 500GW report, therefore UPPTCL may provide communication to CEA/CTU in 2-3 days, so that same can be forwarded to SECI for evaluation of developable RE potential in this region. UPPTCL agreed for the same.

Further, UPPTCL informed in the meeting that space is available for 2 nos. of 400kV line bays at Firozabad S/s however, they will re-confirm the same after verification. Prima facie loading of 400kV Ghiror- Firozabad D/c line was observed to be in order. In view of that CTU included 400kV Ghiror- Firozabad as part of proposed Transmission system for evacuation of power from Rajasthan REZ Ph-IV (Part-4 :3.5 GW) [Fatehgarh/Barmer] Complex. This will also relieve the loading at Agra ICTs and increase resiliency of the system.

Revised study files with proposed changes (considering 400kV Ghiror- Firozabad D/c line & Barmer-I STATCOM as part of proposed scheme) and considering UP intra state solar generation and associated transmission scheme (GEC-II) was circulated on 01.05.24 along with study plots.

Gist of discussion in Joint study meeting held on 09.01.24

CTU stated that in Joint study meeting held on 28.12.23, Transmission system for evacuation of power from Fatehgarh/Barmer Complex as part of Rajasthan REZ Ph-IV (Part-4 :3.5GW) was in-principally agreed. Grid-India also stated that the surplus power in NR and WR during solar peak period is flowing towards ER and high loading is observed in 765 kV Ranchi - Dharamjaygarh line and

other EHV lines towards ER with higher angular separation after N-1. Due to high loading, low voltages observed in ER region on various EHVAC buses.

CTU stated that based on Grid-India observations, in revised studies, Guzuwaka HVDC is already reversed (1000MW) towards ER and Talchar-Kolar capacity is also kept minimum in forward direction (600-700MW). This will relieve loading of various WR-ER corridors incl. 765kV Ranchi – Dharamjaygarh D/c up to a certain extent. The files were already sent to other regional planning groups (CTU) to review loadings (ER-WR) commensurate to envisaged thermal generation in both the regions.

CTU stated the loading of some of WR-ER corridors are in general high and with proposed scheme (Rajasthan Ph-IV (Part-4 :3.5 GW)), there is marginal increment on above Inter regional links in winter solar maximized scenario. In the next phase of studies (2028-29) with proposed HVDC corridors from NR to ER and NR to WR, loadings will be reviewed, and measures will be taken if required. Grid-India & CEA agreed for the same

CTU enquired about RE potential at Ghiror complex for provision of 220kV scope at Ghiror S/s. SECI replied that solar irradiation level at Ghiror complex is less and at present they are not anticipating RE applications for injection at Ghiror. In view of that it was decided that 220kV future scope shall not be considered at Ghiror for RE injection.

UPPTCL informed that space is available at 400kV Firozabad S/s for interconnection of proposed 400kV Ghiror – Firozabad D/c line and power flow is in order on the above line. CTU requested UPPTCL to provide the above information in the mail at the earliest. Regarding RE potential in Bundelkhand region, UPPTCL stated that they have received inputs from UPNEDA however UPPTCL sought some clarifications as well as some additional inputs from UPNEDA. CTU requested UPPTCL to provide the same at the earliest. Further CTU enquired for drawl requirement at Merta-II and Ghiror S/s. UPPTCL stated that in view of growing load of UP, space provision may be kept for future drawl requirement at Ghiror S/s, however they will confirm the same on mail. RVPN stated that they will revert with the drawl requirement at Mera-II S/s. No other comments were received from any Stakeholders. Revised study files with proposed changes in Ph-IV (Part-3: 6GW) and incorporating stakeholder comments on all India LGB, study files were circulated on 06.05.24 along with study plots.

As discussed earlier, off peak file were circulated on 12.01.24 to all stakeholders and based on voltages specially in Ghiror complex, reactive compensation was reviewed. Result of system studies incorporating observations received from stakeholders is enclosed in **Exhibit-1**. Considering grant of connectivity to new RE generators in Fatehgarh/Barmer complex as well as for evacuation of power beyond Fatehgarh-IV PS/Barmer-I PS following transmission scheme is proposed as agreed in joint study meeting for evacuation of power from Rajasthan REZ Ph-IV (Part-4: 3.5GW) [Jaisalmer/Barmer Complex]

Transmission system for evacuation of power from Rajasthan REZ Ph-IV (Part-4 :3.5 GW) [Fatehgarh/Barmer] Complex

Fatehgarh-IV: 1 GW (Solar), Barmer-I :2.5GW (Solar)

- Establishment of 765/400kV, 2x1500 MVA S/s at suitable location near Merta (Merta-II Substation) along with 2x125 MVar & 2x240 MVar bus reactor at Merta-II* S/s

Future provisions at Merta-II S/s(excl. scope for present scheme): Space for

- 765/400kV ICTs along with bays- 4
- 765 kV line bays along with switchable line reactors – 8
- 765kV Bus Reactor along with bay: 1 nos.
- 400 kV line bays along with switchable line reactor –8
- 400 kV line bays – 2 nos.
- 400 kV Bus Reactor along with bays: 1 no.
- 400kV Sectionalizer bay: 2 sets
- 400/220kV ICT along with bays -6 nos.
- 220 kV line bays for RE injection -8 nos.
- 220kV Sectionalizer bay: 2 set
- 220 kV BC (3 nos.) bays and 220 kV TBC (3 nos.) bays
- 6000 MW, ± 800 kV Merta (HVDC) [LCC] terminal station (4x1500 MW) along with bays
- STATCOM (2x ± 300 MVar, 4x125MVar MSC, 2x125MVar MSR) along with 400kV bays (2 nos.)
**along with provision of 80-MVar & 110MVar spare reactors (Single phase) & 500MVA spare transformer unit (Single phase)*

- STATCOM (2x ± 300 MVar) along with MSC (4x125 MVar) & MSR (2x125 MVar) along with 2 nos. 400kV bays at Barmer-I PS
- Augmentation with 765/400 kV, 2x1500 MVA Transformer (4th & 5th) at Barmer-I PS
- Augmentation of 5x500 MVA (5th to 9th), 400/220 kV ICTs at Barmer-I PS
- 220kV line bays (6 nos.) for RE connectivity at Barmer-I PS
- 220kV Sectionalizer bay (1 set) along with 220 kV BC (1 nos.) bay and 220 kV TBC (1 nos.) bay at Barmer-I PS
- 400kV Sectionalizer bay (1 set) at Barmer-I S/s
- Fatehgarh-IV PS (Sec-2) – Barmer-I PS 400kV D/c line (Quad) (~25km)
- Barmer-I PS – Merta-II 765 kV D/c line along with 240 MVar switchable line reactor at Barmer-I PS end and 330MVar switchable line reactor at Merta-II PS end for each circuit of Barmer-I PS – Merta-II 765 kV D/c line (~320 km)
- Merta-II – Beawar 400 kV D/c line (Quad) (~55 km)
- Merta-II – Dausa 765 kV D/c line along with 240 MVar switchable line reactor for each circuit at each end of Merta-II – Dausa 765 kV D/c line line (~250 km)

- Establishment of 765/400kV, 2x1500 MVA S/s at suitable location near Ghiror (Distt. Mainpuri) along with 2x125 MVar & 2x240 MVar bus reactor at Ghiror S/s (UP)

Future provisions at Ghiror S/s (excl. scope for present scheme): Space for**

- 765/400kV ICTs along with bays- 4
 - 765 kV line bays along with switchable line reactors – 6
 - 765kV Bus Reactor along with bay: 1 nos.
 - 400 kV line bays along with switchable line reactor –6
 - 400 kV Bus Reactor along with bays: 1 no.
 - 400kV Sectionalizer bay: 1 set
 - 400/220kV ICT along with bays -4 nos.
 - 220 kV line bays for drawl -6 nos.
 - 220kV Sectionalizer bay: 1 set
 - 220 kV BC (2 nos.) bay and 220 kV TBC (2 nos.) bay
 - STATCOM (2x±300MVar, 4x125MVar MSC, 2x125MVar MSR) along with 400kV bays (2 nos.)
- *along with provision of 80MVar spare reactor (Single phase) & 500MVA spare transformer unit (Single phase)*
***Drawl requirement at Ghiror S/s to be confirmed by UPPTCL, accordingly future scope will be finalized*

- Dausa - Ghiror 765 kV D/c line along with 240 MVar switchable line reactor for each circuit at each end of Dausa - Ghiror 765 kV D/c line (~285 km)
- LILO of both ckt of 765 kV Aligarh (PG) -Orai (PG) D/c line (~40 km) at Ghiror S/s along with 240 MVar switchable line reactor for each circuit at Ghiror S/s end of 765 kV Ghiror -Orai D/c line
- LILO of one ckt of 765kV Agra (PG) – Fatehpur(PG) 765kV D/c line at Ghiror along with 240 MVar switchable line reactor at Ghiror S/s end of 765 kV Ghiror -Fatehpur line (~40 kms)
- 400kV Ghiror-Firozabad (UPPTCL) D/c line (Quad) (~40 kms)

As deliberated earlier, HVDC system is to be planned for additional 6GW at Barmer complex with 2029 implementation schedule. Studies for HVDC scheme is undergoing and will be taken up in next Joint study meeting. Details of broad HVDC scheme is as under:

Proposed Transmission system for evacuation of power from Rajasthan REZ Ph-IV (Part-5 :6 GW) [Barmer Complex] Barmer-II : 6GW (Solar)

- Establishment of 400/220kV, 4x500MVA S/s at suitable location near Barmer (Barmer-II Substation) along with 2x125 MVar bus reactor

Future provisions at Barmer-II S/s:

Space for

- 400 kV line bays along with switchable line reactor –6
- 400 kV line bays –4

- 400 kV Bus Reactor along with bays: 1 no.
- 400/220 kV ICT along with bays -10 Nos.
- 400 kV Sectionalization bays: 2 sets
- 220 kV line bays for connectivity of RE Applications -15 Nos.
- 220kV Sectionalization bay: 3 sets
- 220 kV BC (4 Nos.) & TBC (4 Nos.)
- 220kV line bays (4 nos.) for RE connectivity at Barmer-II PS
- Barmer-I PS – Barmer-II PS 400kV D/c line (Quad)
- Establishment of 6000 MW, $\pm$ 800 kV Barmer-II (HVDC) [LCC] terminal station (4x1500 MW) at a suitable location near Barmer-II substation
 - 400/33 kV, 2x50 MVA transformers for exclusively supplying auxiliary power to HVDC terminal.
 - 400kV bus sectionaliser -2 nos (1 Set) at Barmer-II (HVDC) station
- Establishment of 6000 MW, $\pm$ 800 kV (HVDC) [LCC] terminal station (4x1500 MW) at suitable location in WR/SR or 2 nos. 3000MW, $\pm$ 800 kV (HVDC) [LCC] terminal stations (Multi terminal) at suitable location(s) in WR/SR
- $\pm$ 800 kV HVDC line (Hexa lapwing) between Barmer-II (HVDC) & WR /SR (HVDC) (with Dedicated Metallic Return)
- Associated EHVAC system strengthening in WR/SR

Exact HVDC terminal points in WR/WR-SR boundary are proposed to be identified based on detailed system studies involving stakeholders from NR, WR & SR. Decision on HVDC technology i.e. LCC/ VSC will be carried out at later stage during joint study meeting

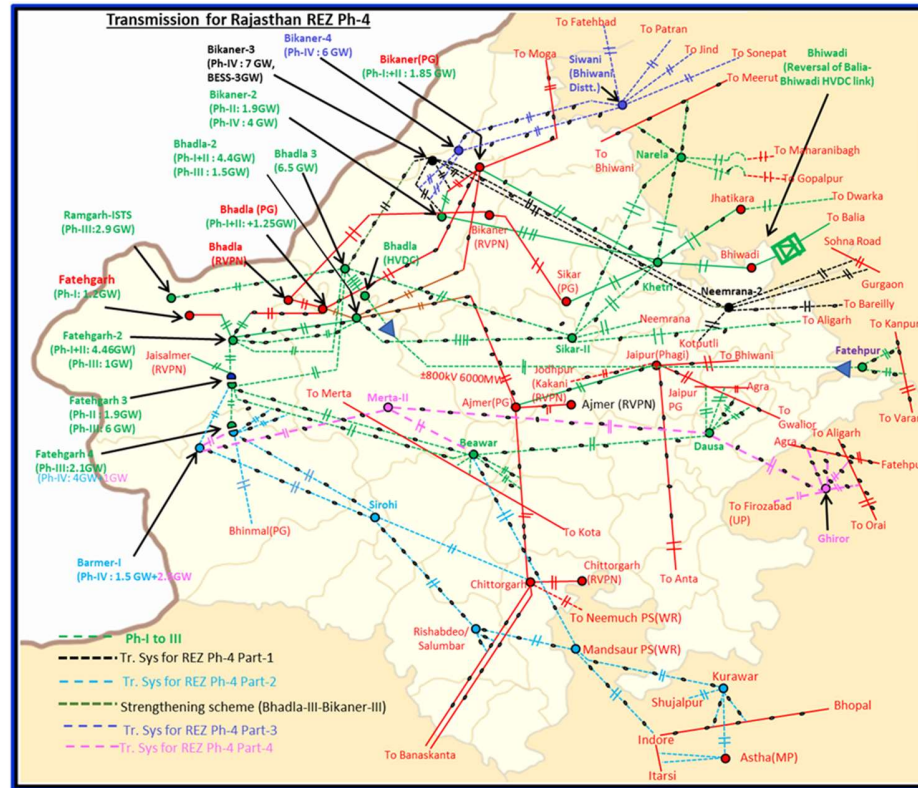


Fig: Transmission system for Rajasthan REZ Ph-IV (Part-3: 6GW) [Bikaner Complex] and Rajasthan REZ Ph-IV (Part-4 :3.5GW) [Fatehgarh/Barmer complex]

5. Transmission system scheme for Ratle HEP (850MW)

It was stated that OPGW installation on 2nd E/W peak of 400kV Kishenpur-Dulhasti existing line (120 kms.) (Circuit-1) along with LILO portion (2nd LILO at Kishtwar) was deliberated in the 26th CMTES-NR meeting. OPGW on 1st E/W peak is already available. This arrangement shall provide redundant communication to Dulhasti & Kishtwar stations which are presently connected / to be connected on single path only which are critical in view of Pakaldul & Ratle Generation to be evacuated from Dulhasti & Kishtwar S/s. Proposed OPGW arrangement is demonstrated at Fig-1.

Feasibility of OPGW installation on 400kV S/c line (Circuit-1) strung over modified 220kV towers for providing redundant communication to Dulhasti Station was also deliberated in the 2nd / 3rd ISTS Communication Planning Meeting (CPM) of NR, 22nd & 23rd NRPC TeST sub-committee meeting and in 57th NRPC. In the 23rd NRPC TeST meeting POWERGRID stated that OPGW installation is not feasible on this line due to tower strengthening not feasible.

During the meeting, all members agreed to the proposal and NRLDC/Grid-India provide their confirmation through email, which is attached at **Annexure-II**.

In 27th CMETS-NR meeting POWERGRID representatives informed that reconductoring scope is also involved in the scheme on the 400kV Kishenpur-Dulhasti existing line (120 kms) Ckt-1. During reconductoring work OPGW can be installed on this line as sag can be relaxed during installation which shall remove hindrance of tower strengthening due to OPGW installation. Further this will also create diverse physical path in line with CEA Manual of Communication Planning. The modified scheme is presented in the Fig-2. After the deliberation, all members agreed to the proposal.

This proposal shall be made part of Transmission scheme and to be put up for approval in NCT along with LILO of 400 kV Kishenpur-Dulhasti line at Kishtwar S/s as per MoP "Guidelines on Planning of Communication System for Inter-State Transmission System (ISTS)" under Category (A). Proposed OPGW Diagrams are as under:

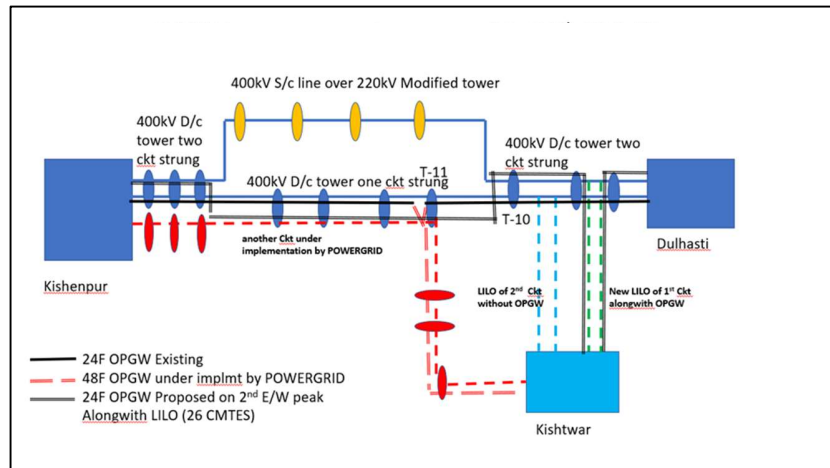


Fig-1 : OPGW arrangement proposed in 26th CMETS-NR meeting

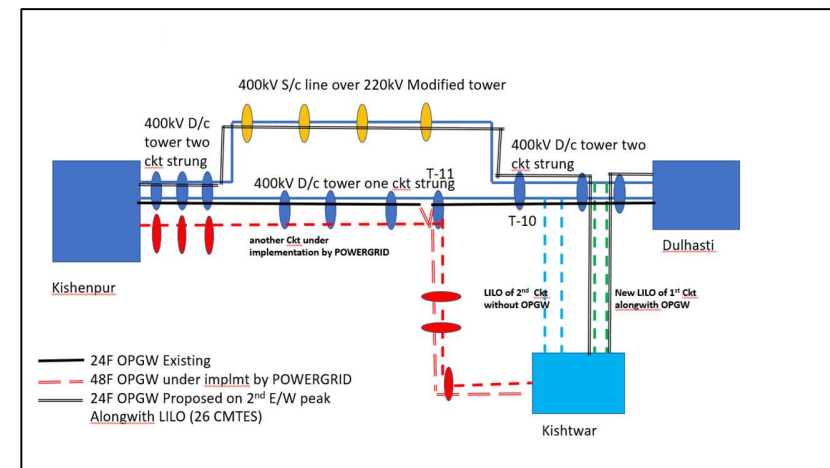


Fig-2 : Final OPGW arrangement proposed in 27th CMETS-NR meeting

C. Application related matters in Northern Region

C1. Connectivity Application for 76 MW Phatabyung HEP from M/s Lanco Mandakini Hydro Energy Pvt. Ltd. received under GNA Regulations:

S. No.	Application ID	Applicant	Project Location	Date of Application	Connectivity Sought (MW)	Nature of Applicant	Start Date Requested	Connectivity location (requested)
1.	2200000146	Lanco Mandakini Hydro Energy Pvt. Ltd.	Rudraprayag Uttarakhand	11.07.2023	76	Generator (Hydro)	31.12.2027	Rudrapur (Brahmwari)

It was mentioned that Connectivity application from M/s Lanco Mandakini Hydro Energy Pvt. Ltd. (LMHEPL) was discussed in the 23rd NR-CMETS meeting held on 29/08/2023 for grant of Connectivity of its Phatabyung HEP in Uttarakhand. During the meeting, CEA confirmed that Phatabyung HEP falls under hydro potential area of Alaknanda basin in Uttarakhand. Considering this, the transmission system for Connectivity under GNA was agreed in this meeting as Augmentation without Associated Transmission System (ATS) as given below:

Details of Transmission System for Connectivity under GNA (Augmentation Without ATS):

A. Associated Transmission System (ATS): NIL

B. Transmission system for Connectivity (under scope of Applicant):

- Lanco Mandakini switchyard – Baramwari S/s (PTCUL) 220 kV D/c line along with 220 kV bays at both ends

C. Common Transmission system for Connectivity under GNA

I. Under ISTS:

- Khandukhal - Rampura 400 kV D/c Twin HTLS line along with associated bays at both ends

II. Under scope of PTCUL as deemed ISTS Licensee:

- Establishment of Baramwari 220 kV Pooling station
- Baramwari (PTCUL) – Till the Point of Interconnection of Khandukhal (PTCUL)- Baramwari 220 kV D/c line with Singoli Bhatwari HEP along with 220 kV bays at Baramwari end

Further, in the present meeting, PTCUL informed that earlier 220kV Baramwari S/s was to be implemented as part of deemed ISTS scheme (UITP) for providing Connectivity to Phata Byung HEP. However, since Phata Byung HEP was under NCLT from long time

and Intra State Small hydro Projects (SHP) were coming up for interconnection at Baramwari S/s, PTCUL planned it to be developed as an Intra-state scheme. On this, M/s LMHEPL (Statkraft) had confirmed that their project is out of NCLT now, and they would wish to get connected to Baramwari pooling station for Connectivity.

In view of this, PTCUL suggested in such case Baramwari substation status needs to get changed from Intra State to Deemed ISTS for which they would have to take up management approval. PTCUL was suggested to confirm the status of implementation of Baramwari PS in a time bound manner. Accordingly, it was decided that Connectivity arrangement for Phata Byung HEP shall be discussed in a separate meeting among CEA, CTUIL, Generation developer & PTCUL.

Subsequently, a meeting was taken up by CEA on 21/11/2023 through VC regarding status of implementation of Baramwari S/s by PTCUL. During this meeting, M/s LMHEPL (Statkraft) updated the status of project and submitted that the progress of Phata Byung HEP project is in full swing, CEA & State approvals are already received and the Project is scheduled for commissioning by December, 2027. Therefore, the Connectivity of HEP may be granted at the earliest. Regarding the present progress of Baramwari sub-station, PTCUL informed that the Substation land is under possession and sub-station works shall be awarded soon. Further, the 220 kV transmission line upto Baramwari S/s is about 90% complete. PTCUL stated that the Baramwari system could be completed in the matching timeframe of Phata Byung HEP.

Subsequently, PTCUL vide their letter dated 06/12/2023 informed CEA that 220 kV Baramwari switching station & 220 kV D/c line from Baramwari S/s to the point of interconnection of Singoli Bhatwari HEP may be considered as Deemed ISTS. PTCUL confirmed that the implementation shall be taken up under Deemed ISTS.

CTUIL enquired about the status of the Phata Byung generation. M/s LMHEPL stated that the progress of their project is in full swing and is scheduled for commissioning by December, 2027, therefore, the Connectivity of HEP may be granted at the earliest. Further, PTCUL stated that Baramwari PS may be considered as ATS for Connectivity grant to Phata Byung HEP as it has not been awarded by PTCUL as on date and accordingly, Bank Guarantee matched with the cost of ATS may be taken from M/s LMHEPL for Phatabyung generation. CTUIL clarified that during the present meeting, CEA had confirmed that Phatabyung HEP falls under hydro potential area in Alaknanda basin in Uttarakhand. Further, PTCUL is also integrating UJVNL's small HEPs at Baramwari S/s and considering the same, when any comprehensive transmission system is being planned for multiple generation projects, the transmission system for Connectivity under GNA is considered as augmentation without Associated Transmission System (ATS) as it shall not be used by any single generator, accordingly, no bank guarantee for ATS is applicable to M/s LMHEPL. PTCUL noted the same.

In view of above, after deliberations, it was agreed that Connectivity under GNA for Phatabyung HEP shall be granted through following Transmission system:

A. Transmission system for Connectivity (under the scope of Applicant):

- Lanco Mandakini HEP generation switchyard – Baramwari S/s (Deemed ISTS) 220 kV D/c line along with 220 kV bays at both ends

B. Common Transmission system for Connectivity under GNA

I. Under ISTS:

- Khandukhal - Rampura 400 kV D/c (Twin HTLS) line along with associated bays at both ends

II. Under scope of PTCUL as deemed ISTS Licensee:

- Establishment of Baramwari 220 kV Pooling station
- Baramwari (PTCUL) – LILO Point* of Baramwari - Khandukhal 220 kV D/c line at Singoli Bhatwari HEP along with 220 kV bays at Baramwari end

**Upon completion of Final arrangement for Connectivity of Singoli Bhatwari HEP (ie. a separate 220 kV D/c line from Singoli Bhatwari HEP to Baramwari HEP as a final arrangement for Connectivity to Singoli Bhatwari HEP [to be implemented by M/s Renew]), a direct connection from Baramwari S/s to Khandukhal S/s through 220 kV D/c line shall be under the scope of PTCUL.*

Transmission scheme is depicted at **Annexure-III**.

C2. Connectivity Application for 60 MW Hanol Tuni HEP from M/s Sunflag Power Limited received under GNA Regulations:

S. No.	Applicati on ID	Applicant	Project Location	Submissi on Date	Nature of Applicant	Criterion for applying	Injection at ISTS point (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)
1	2200000357	Sunflag Power Limited	Dehradun distt., Uttarakhand	15.11.2023	Generating station(s), including REGS(s), without ESS (Hydro)	LOA or PPA	60 MW	25.09.2028	Mori S/s

It was deliberated that Connectivity application from M/s Sunflag Power Limited (SPL) has been received for their 60 MW Hanol Tuni HEP in Yamuna basin of Uttarakhand. Connectivity point requested by the Applicant is Mori substation (earlier planned to be developed under ISTS).

Regarding Mori Substation, it was informed that for evacuation of power from various hydro projects including Naitwar Mori HEP in Yamuna Basin of Uttarakhand, Mori Substation along with 220 kV D/c Mori - Dehradun (Twin Zebra) line was planned under UITP scheme for implementation by PTCUL as deemed ISTS scheme. However, during the 3rd NRPC(TP) held on 19/02/2021, it was agreed that Naitwar Mori HEP would be connected to STU system for evacuation of power from their HEP, as no other hydro project in Yamuna basin is expected within next 2-3 years. Therefore, implementation of 220 kV D/c Mori-Dehradun line shall be considered in future under Central sector with materialization of other projects in Yamuna basin.

Status of Generation project was enquired from M/s Sunflag and they informed that the DPR has been approved for the project in 2013 and they have started pre-commissioning activities.

Further, it was deliberated that Connectivity application from M/s Sunflag is for the time frame Sep'2028 and at present, there is no other Connectivity application received from any hydro generators in Yamuna Basin. CEA informed that there are some other hydro generation projects of UJVNL in that basin which may be seeking Connectivity at Mori Substation in Future. Further, CEA also confirmed that Hanol Tuni HEP falls under hydro potential area in Yamuna basin in Uttarakhand. PTCUL also added that some other hydro projects (four generators with total 275 MW capacity by UJVN + THDC) in Yamuna basin in Uttarakhand are envisaged in future. CTU stated since they havenot received any application, PTCUL/CEA, may share hydro project status.

In view of this, it was discussed that the proposed Transmission system shall be considered as Common Augmentation without ATS (Associated Transmission System) as below:

Details of Transmission System required for Connectivity under GNA:

A. Under scope of Applicant:

- Hanol Tuni HEP- Mori Substation 220 kV D/c line along with 220 kV bays at both ends
- 220kV, 1x25 MVAr Bus reactor at Generation switchyard

B. Under ISTS:

- Establishment of 220 kV Switching Station at Mori
- Mori switching station-Dehradun (PG) 220 kV (Twin Zebra[#]) D/c line-approx. 160 km

[#]Line Capacity shall be based on envisaged power transfer requirement from hydro projects in Yamuna basin

However, CTUIL opined that though there are some other hydro generations envisaged in the Yamuna basin for Connectivity at Mori Substation, however, at present, there is no Connectivity application received under GNA and in view of this, for which substantial investment of a pooling station and 220 kV D/c line may not be justified that too for only 60 MW of generation, 220 kV D/c (considering N-1 scenario), shall be sub-optimally utilized (30 MW per circuit). Further, to mitigate the high voltage issues due to less line loadings, reactive power compensation shall need to be provided at generation switchyard also. It was enquired from PTCUL that whether there is any PTCUL substation nearby proposed Hanol Tuni HEP, so that Connectivity could be granted there, however, PTCUL informed that there is no nearby PTCUL substation for grant of Connectivity. PTCUL also informed that Govt. of Uttarakhand has cancelled the allocation of Hanol Tuni HEP to M/s Sunflag and has allotted it to UJVNL. On this, M/s Sunflag informed that, this matter is presently subjudice and they have applied for the Connectivity under ISTS before the matter went under litigation (in Dec'23).

Further, CTUIL suggested that M/s Sunflag could also approach HPPTCL for Connectivity grant using Transmission system implemented for evacuation of power from Naitwar Mori HEP. M/s Sunflag, informed that neither HPPTCL nor SJVNL gave them their consent for using their Transmission system for evacuation of power from Hanol Tuni HEP. Further, M/s Sunflag informed that they had also applied for LTOA to PTCUL in 2008, however, PTCUL replied that as per CERC Order, it was decided that all the previous Connectivity/LTA/LTOA applications will be closed at PTCUL's end and all applicants were advised to apply to CTU for grant of Connectivity and LTA. CEA also mentioned that as project DPR was approved in Jan'13, they need to check TEC clearance/approvals validity as well.

In view of above, it was agreed that in reference to PTCUL comments on allocation of Hanol Tuni project to UJVNL, M/s Sunflag shall submit status of litigation associated with this generation project in Uttarakhand. CEA & PTCUL shall also share status of other hydro projects in Yamuna basin. CEA will also revert on the pending clearances & approval validity for the project.

Subsequent to receipt of above information & resolution of matter under litigation, their Connectivity application shall be discussed again in the NR-CMETS meeting for further decision.

C3. Transmission system for reallocated entities of Bikaner-III/IV and Fatehgarh-IV/Barmer complex

It was informed that a meeting for reallocation of connectivity at Fatehgarh-IV PS & Bikaner-III PS was held on 02.01.2024 in which some entities granted connectivity were reallocated. Based on the reallocation at Bikaner-III PS & Fatehgarh-IV PS(Sec-II), revised transmission system for connectivity for the reallocated entities is as below :

Sl. No.	Application ID	Applicant	Project Location	Application Date	Nature of Applicant	Criterion for applying	Injection at ISTS point (MW)	Start Date of Connectivity (As per Application)	Connectivity location	Conn BG requirement as per earlier intimation	Revised Conn BG requirement
1.	2200000013	Avaada RJ Bikaner Private Limited	Bikaner distt., Rajasthan	27.04.2023	Generator (Solar)	Land Route	50	31.03.2025	Bikaner-IV PS	•Conn BG1: 50 lakh •Conn BG2: Nil(Bay in sharing) •Conn BG3: Rs. 1 Cr.	•Conn BG1: 50 lakh •Conn BG2: Nil(Bay in sharing) • Conn BG3: Rs. 1 Cr.
M/s Avaada vide letter dated 01.11.2023 informed that they have submitted application for 600 MW at same site and location and expansion of the same project (50 MW at BikanerIII). • M/s Avaada requested CTUIL that in case of margin constraint at Bikaner-III PS, total connectivity of 650 MW (including 50 MW being shared with M/s Proteus at Bikaner-III PS) may be granted together at Bikaner-IV PS. It was informed that in the reallocation meeting held on 02.01.2024, based on the request of M/s Avaada, it was proposed that connectivity of Avaada RJBikaner Pvt. Ltd. (50 MW) granted at Bikaner-III PS in sharing with M/s Proteus(217 MW) shall be shifted to Bikaner-IV PS and granted connectivity along with another application of M/s Avaada(600 MW) which is being discussed in the present CMETS meeting. M/s Avaada has agreed for the change in its connectivity system with Bikaner-IV transmission system. Accordingly, it is proposed to grant connectivity to M/s Avaada at 400 kV Bikaner-IV PS through 1 no. of bay for combined capacity of 650MW (50+600). M/s Avaada initially intended to obtain connectivity at 220 kV for 650 MW. However, subsequently in the same meeting, M/s Avaada requested to grant additional connectivity of 300 MW at Bikaner-IV PS for application No. 2200000289(300 MW) which was applied in Land BG route for connectivity at Bhadla-III PS. Considering the same, total connectivity at Bikaner-IV PS shall be 950 MW for M/s Avaada. Therefore, it was decided to grant connectivity at 400 kV through 1 no. of 400 kV line bay. M/s Avaada agreed for the same. M/s Avaada informed that 400 kV bay at Bikaner-IV end shall be under ISTS and tower configuration shall be S/c line on D/c tower. The same was noted & M/s Avaada was asked to submit necessary undertaking for S/c line on D/c tower configurations. M/s Avaada agreed for the same. Further, M/s Avaada was asked to confirm which application among these is to be considered as lead for bay allocation and Conn BG2 submission. M/s Avaada requested to consider App. No. 2200000254(600 MW) as lead. The same was agreed. Accordingly, details of revised transmission system for connectivity under GNA Avaada RJ Bikaner Private Limited (50 MW) and revised Start Date of Connectivity is as below: <u>Details of Transmission system for Connectivity under GNA:</u> A. <u>Associated Transmission System (ATS): NIL</u> B. <u>Transmission System under applicant scope</u> (i). Common Pooling station for Avaada RJ Bikaner Private Limited (App No. 220000013(50 MW), App. No. 2200000254(600 MW)) & Avaada Energy Private Limited (App No. 2200000289(300 MW)) Solar Power Projects – Bikaner-IV PS 400 kV S/c line on D/c tower (Suitable to carry minimum 950 MW at nominal voltage) C. <u>Transmission system for Connectivity under GNA</u>											

Sl. No.	Application ID	Applicant	Project Location	Application Date	Nature of Applicant	Criterion for applying	Injection at ISTS point (MW)	Start Date of Connectivity (As per Application)	Connectivity location	Conn BG requirement as per earlier intimation	Revised Conn BG requirement
- Establishment of 6x1500 MVA, 765/400 kV Bikaner-IV Pooling Station along with 2x240 MVar (765kV) & 2x125 MVar (420kV) Bus Reactors at a suitable location near Bikaner 400kV Sectionalizer bay (1 set) at Bikaner-IV PS - STATCOM (2x±300MVar) along with MSC (4x125 MVar) & MSR (2x125 MVar) along with 2 nos. 400kV bays at Bikaner-IV PS - LILO of both ckts of 400kV Bikaner II PS- Bikaner III PS (Quad) direct line at Bikaner-IV PS - Establishment of 765/400 kV, 6x1500 MVA S/s at suitable location near Siwani (Distt. Bhiwani) along with 2x240 MVar (765kV) Bus Reactor & 2x125 MVar (420kV) Bus Reactor - Bikaner-IV PS – Siwani 765 kV 2x D/c line along with 240 MVar switchable line reactor for each circuit at each end - Siwani – Jind (PG) 400 kV D/c line (Quad) - Siwani – Patran (Indi Grid) 400 kV D/c line (Quad) along with 80 MVar switchable line reactor for each circuit at Siwani S/s end - Siwani – Fatehabad (PG) 400 kV D/c line (Quad) - Siwani – Sonipat (PG) 400 kV D/c line (Quad) along with 63 MVar switchable line reactor for each circuit at Siwani S/s end - STATCOM (2x±300MVar) along with MSC (4x125 MVar) & MSR (2x125 MVar) along with 2 nos. 400kV bays at Siwani S/s 400kV Sectionalizer bay (1 set) at Siwani S/s D. Additional Transmission system for Inter-regional power transfer: - Establishment of 765kV Mandsaur Pooling Station along with 2x330 MVAR (765 kV) Bus Reactors - Beawar- Mandsaur PS 765 kV D/c line along with 240 MVar switchable line reactor for each circuit at each end a) Mandsaur PS – Indore(PG) 765 kV D/c Line along with 1x330 MVAR switchable line reactor on each ckt at Mandsaur end of Mandsaur PS – Indore(PG) 765 kV D/c Line OR b) Establishment of 765 kV Kurawar S/s with 2x330 MVar 765 kV bus reactor c) Mandsaur – Kurawar 765 kV D/c line along with 240 MVar switchable line reactors on each ckt at both ends of Mandsaur – Kurawar 765 kV D/c line d) LILO of Indore – Bhopal 765 kV S/c line at Kurawar Start Date of Connectivity under GNA: 01.09.2026(Interim). Final start date shall be confirmed upon award of the system. Note: As Bikaner-IV PS is still not closed for grant, it was not part of the reallocation exercise. M/s Avaada requested to shift its connectivity (50 MW) with another new application of Avaada (600 MW) while Bikaner-IV PS, therefore its grant as discussed in CMETS meeting shall continue.											

Sl. No.	Application ID	Applicant	Project Location	Application Date	Nature of Applicant	Criterion for applying	Injection at ISTS point (MW)	Start Date of Connectivity (As per Application)	Connectivity location	Conn BG requirement as per earlier intimation	Revised Conn BG requirement
2.	2200000179	NTPC Renewable Energy Limited	Bikaner distt., Rajasthan	02.08.2023	Generator (Solar)	REMCL LOA	500	28.06.2025	Bikaner-III PS	•Conn BG1: Rs. 50 lakh •Conn BG2: Rs. 6 Cr. •Conn BG3: Rs. 10 Cr.	•Conn BG1: Rs. 50 lakh •Conn BG2: Rs. 6 Cr. •Conn BG3: Rs. 10 Cr.

It was informed that, in the reallocation meeting held on 02.01.2024, NTPC opted to shift 500 MW connectivity from 400 kV Bikaner-IV PS to 220 kV Bikaner-III PS. Further, it was also informed that, the connectivity beyond 4000 MW at Bikaner-III PS (upto 4667 MW) shall be granted with Transmission system for Bikaner-III (7.7 GW) as well as Transmission system Bikaner-IV (3.6 GW). M/s NTPC agreed to shift connectivity of 500 MW to Bikaner-III PS with the above condition. Accordingly, in the present meeting it was proposed to grant connectivity of 500 MW to NTPC at 220 kV Bikaner-III PS through 2 no. of 220 kV line bays. NTPC agreed for the same. NTPC informed that 220 kV bay at Bikaner-III end shall be under ISTS. The same was noted. M/s NTPC vide mail dated 10/01/24 has submitted request for early bay completion date (before GNA effectiveness date) as 31/12/2025. It was informed that liability for bay charges will be on RE applicant in case of generation delay and power evacuation will be based on availability of real time margins only.

Accordingly, details of revised transmission system for connectivity under GNA for NTPC Renewable Energy Limited (500 MW) and revised Start date of connectivity is as below:

Details of Transmission system for Connectivity under GNA:

A. Associated Transmission System (ATS): NIL

B. Transmission System under applicant scope

- (i). NTPC Renewable Energy Limited Solar Power Project – Bikaner-III PS 220 kV D/c line (Suitable to carry minimum 300 MW per circuit at nominal voltage)

C. Transmission system for Connectivity under GNA

1. Augmentation of 1x1500 MVA (4th) 765/400kV ICT at Bikaner (PG)
2. Augmentation of 2x1500 MVA (5th & 6th) 765/400 kV ICTs at Bikaner-III Pooling Station
3. Augmentation of 2x500 MVA (3rd & 4th) 400/220 kV ICTs at Bikaner-III Pooling Station
4. LILO of both ckts of 400 kV Bikaner(PG)-Bikaner-II D/c line (Quad) at Bikaner-III PS
5. Bikaner-II PS – Bikaner-III PS 400 kV D/c line (Quad)
6. Bikaner-III - Neemrana-II 765 kV D/c line along with 330 MVar switchable line reactor for each circuit at each end
7. Augmentation of 2x1500 MVA (3rd & 4th), 765/400 kV ICTs at Neemrana-II S/s
8. Neemrana-II -Kotputli 400 kV D/c line (Quad)
9. LILO of both ckts of 400 kV Gurgaon (PG) - Sohna Road (GPTL) D/c line (Quad) at Neemrana-II S/s

Sl. No.	Application ID	Applicant	Project Location	Application Date	Nature of Applicant	Criterion for applying	Injection at ISTS point (MW)	Start Date of Connectivity (As per Application)	Connectivity location	Conn BG requirement as per earlier intimation	Revised Conn BG requirement
10. Bikaner-III - Neemrana-II 765 kV D/c line (2nd) along with 330 MVAR switchable line reactor for each circuit at each end 11. Neemrana-II- Bareilly (PG) 765 kV D/c line along with 330 MVAR switchable line reactor for each circuit at each end 12. Augmentation by 400/220 kV, 1x500 MVA ICT (3rd) at Kotputli (PG) 13. 400 kV Bareilly(765/400kV) – Bareilly(PG) D/c line (Quad) (2nd) 14. Augmentation with 1x1500 MVA, 765/400 kV ICT at Bareilly (765/400kV) S/s (3rd) 15. Establishment of 6x1500 MVA, 765/400 kV Bikaner-IV Pooling Station along with 2x240 MVAR (765kV) & 2x125 MVAR (420kV) Bus Reactors at a suitable location near Bikaner 16. 400kV Sectionalizer bay (1 set) at Bikaner-IV PS 17. STATCOM (2x±300MVAR) along with MSC (4x125 MVAR) & MSR (2x125 MVAR) along with 2 nos. 400kV bays at Bikaner-IV PS 18. LILO of both ckts of 400kV Bikaner II PS- Bikaner III PS (Quad) direct line at Bikaner-IV PS 19. Establishment of 765/400 kV, 6x1500 MVA S/s at suitable location near Siwani (Distt. Bhiwani) along with 2x240 MVAR (765kV) Bus Reactor & 2x125 MVAR (420kV) Bus Reactor 20. Bikaner-IV PS – Siwani 765 kV 2xD/c line along with 240 MVAR switchable line reactor for each circuit at each end 21. Siwani – Jind (PG) 400 kV D/c line (Quad) 22. Siwani – Patran (Indi Grid) 400 kV D/c line (Quad) along with 80 MVAR switchable line reactor for each circuit at Siwani S/s end 23. Siwani – Fatehabad (PG) 400 kV D/c line (Quad) 24. Siwani – Sonipat (PG) 400 kV D/c line (Quad) along with 63 MVAR switchable line reactor for each circuit at Siwani S/s end 25. STATCOM (2x±300MVAR) along with MSC (4x125 MVAR) & MSR (2x125 MVAR) along with 2 nos. 400kV bays at Siwani S/s 26. 400kV Sectionalizer bay (1 set) at at Siwani S/s D. Additional Transmission system for Inter-regional power transfer: 1. Establishment of 765kV Mandsaur Pooling Station along with 2x330 MVAR (765 kV) Bus Reactors 2. Beawar- Mandsaur PS 765 kV D/c line along with 240 MVAR switchable line reactor for each circuit at each end 3a) Mandsaur PS – Indore(PG) 765 kV D/c Line along with 1x330 MVAR switchable line reactor on each ckt at Mandsaur end of Mandsaur PS – Indore(PG) 765 kV D/c Line OR 3b) Establishment of 765 kV Kurawar S/s with 2x330 MVAR 765 kV bus reactor 3c) Mandsaur – Kurawar 765 kV D/c line along with 240 MVAR switchable line reactors on each ckt at both ends of Mandsaur – Kurawar 765 kV D/c line 3d) LILO of Indore – Bhopal 765 kV S/c line at Kurawar Start Date of Connectivity under GNA: 01.09.2026(Interim). Final start date shall be confirmed upon award of the system. Note: CERC vide order dated 19.01.2024 on petition no. 268/MP/2023 and 269/MP/2023 has directed CTU to carry out fresh reallocation exercise											

Sl. No.	Application ID	Applicant	Project Location	Application Date	Nature of Applicant	Criterion for applying	Injection at ISTS point (MW)	Start Date of Connectivity (As per Application)	Connectivity location	Conn BG requirement as per earlier intimation	Revised Conn BG requirement
---------	----------------	-----------	------------------	------------------	---------------------	------------------------	------------------------------	---	-----------------------	---	-----------------------------

for the meetings held on 20.6.2023 and 3.8.2023 or any subsequent reallocation meeting held for substations in Rajasthan. The same shall be carried out by CTU again and outcome shall be filed with CERC as per the order. ***In view of the above, the application may again need to be discussed in CMETS-NR meeting for scheme review and finalization.***

3.	2200000244	NTPC Renewable Energy Limited	Bikaner distt., Rajasthan	07.09.2023	Generator (Solar)	LoA or PPA	150	28.06.2025	Bikaner-III PS	•Conn BG1: Rs. 50 lakh •Conn BG2: Nil(Bay in sharing) •Conn BG3: Rs. 10 Cr.	•Conn BG1: Rs. 50 lakh •Conn BG2: Nil(Bay in sharing) •Conn BG3: Rs. 3 Cr.
----	------------	-------------------------------	---------------------------	------------	-------------------	------------	-----	------------	----------------	---	--

It was informed that NTPC Renewable Energy Limited had been granted connectivity of 650(500+150) MW at 400 kV Bikaner-IV PS. After reallocation of 500 MW to Bikaner-III PS, it was proposed that the connectivity granted for balance connectivity of 150 MW which is still at Bikaner-IV PS shall be revised and granted along with other applications of 700 MW, 400 MW & 250 MW by NTPC Renewable Energy Limited (which are being taken up for grant in the same meeting) through a 400 kV D/c line.

NTPC informed that 400kV bays at Bikaner-IV PS end shall be under ISTS. NTPC was asked to confirm lead application among these(150+700+250+400) for the allocation of bays and Conn BG2 submission. Regarding the lead entity, NTPC in the meeting had requested to consider application of 150 MW. As two no. of bays are being allocated to NTPC, it was decided to consider applications of 150 MW for Conn BG submission. However, subsequently, vide mail dated 18.01.2024 NTPC requested to consider application of 700 MW as the lead entity instead of 150 MW application. The same was agreed and accordingly application No. 2200000308(700 MW) shall be considered as lead entity for one no. of 400 kV bay and the application no. 220000373(400 MW) shall be considered as lead for 2nd Bay [NTPC may confirm if any change required for 2nd bay for lead entity within a week of issuance of minutes].

Accordingly, revised transmission system for NTPC Renewable Energy Limited (150 MW) is as below:

Details of Transmission system for Connectivity under GNA:

A. Associated Transmission System (ATS): NIL

B. Transmission System under applicant scope

- (i). Common pooling station for NTPC Renewable Energy Limited Solar Power Projects (App No. 2200000244(150 MW) , 2200000308(700 MW), 2200000373(400 MW) & 2200000358(250 MW) – Bikaner-IV PS 400 kV D/c line (Suitable to carry minimum 900 MW per circuit at nominal voltage)

C. Transmission system for Connectivity under GNA

Sl. No.	Application ID	Applicant	Project Location	Application Date	Nature of Applicant	Criterion for applying	Injection at ISTS point (MW)	Start Date of Connectivity (As per Application)	Connectivity location	Conn BG requirement as per earlier intimation	Revised Conn BG requirement
											- Establishment of 6x1500 MVA, 765/400 kV Bikaner-IV Pooling Station along with 2x240 MVar (765kV) & 2x125 MVar (420kV) Bus Reactors at a suitable location near Bikaner 400kV Sectionalizer bay (1 set) at Bikaner-IV PS - STATCOM (2x±300MVar) along with MSC (4x125 MVar) & MSR (2x125 MVar) along with 2 nos. 400kV bays at Bikaner-IV PS - LILO of both ckts of 400kV Bikaner II PS- Bikaner III PS (Quad) direct line at Bikaner-IV PS - Establishment of 765/400 kV, 6x1500 MVA S/s at suitable location near Siwani (Distt. Bhiwani) along with 2x240 MVar (765kV) Bus Reactor & 2x125 MVar (420kV) Bus Reactor - Bikaner-IV PS – Siwani 765 kV 2xD/c line along with 240 MVar switchable line reactor for each circuit at each end - Siwani – Jind (PG) 400 kV D/c line (Quad) - Siwani – Patran (Indi Grid) 400 kV D/c line (Quad) along with 80 MVar switchable line reactor for each circuit at Siwani S/s end - Siwani – Fatehabad (PG) 400 kV D/c line (Quad) - Siwani – Sonipat (PG) 400 kV D/c line (Quad) along with 63 MVar switchable line reactor for each circuit at Siwani S/s end - STATCOM (2x±300MVar) along with MSC (4x125 MVar) & MSR (2x125 MVar) along with 2 nos. 400kV bays at Siwani S/s 400kV Sectionalizer bay (1 set) at at Siwani S/s D. Additional Transmission system for Inter-regional power transfer: - Establishment of 765kV Mandsaur Pooling Station along with 2x330 MVAR (765 kV) Bus Reactors - Beawar- Mandsaur PS 765 kV D/c line along with 240 MVar switchable line reactor for each circuit at each end 3a) Mandsaur PS – Indore(PG) 765 kV D/c Line along with 1x330 MVAR switchable line reactor on each ckt at Mandsaur end of Mandsaur PS – Indore(PG) 765 kV D/c Line OR 3b) Establishment of 765 kV Kurawar S/s with 2x330 MVar 765 kV bus reactor 3c) Mandsaur – Kurawar 765 kV D/c line along with 240 MVar switchable line reactors on each ckt at both ends of Mandsaur – Kurawar 765 kV D/c line 3d) LILO of Indore – Bhopal 765 kV S/c line at Kurawar Start Date of Connectivity under GNA: 01.09.2026(Interim). Final date shall be confirmed upon award of the system Note: CERC vide order dated 19.01.2024 on petition no. 268/MP/2023 and 269/MP/2023 has directed CTU to carry out fresh reallocation exercise for the meetings held on 20.6.2023 and 3.8.2023 or any subsequent reallocation meeting held for substations in Rajasthan. The same shall be carried out by CTU again and outcome shall be filed with CERC as per the order. <i>In view of the above, the application may again need to be discussed in CMETS-NR meeting for scheme review and finalization.</i>

Sl. No.	Application ID	Applicant	Project Location	Application Date	Nature of Applicant	Criterion for applying	Injection at ISTS point (MW)	Start Date of Connectivity (As per Application)	Connectivity location	Conn BG requirement as per earlier intimation	Revised Conn BG requirement
4.	2200000184	Deshraj Solar Energy Private Limited	Bikaner distt., Rajasthan	11.08.2023	Generator (Solar)	Land Route	300	28.06.2025	Bikaner-III PS	•Conn BG1: Rs. 50 lakh •Conn BG2: Rs. 3 Cr. •Conn BG3: Rs. 6 Cr.	•Conn BG1: Rs. 50 lakh •Conn BG2: Rs. 3 Cr. •Conn BG3: Rs. 6 Cr

It was informed that in the reallocation meeting held on 02.01.2024, it was proposed to consider shifting of connectivity of 300 MW to Deshraj Solar Energy Private Limited from 220 kV Bikaner-IV PS to 220 kV Bikaner-III PS. Further, it was also informed that connectivity beyond 4000 MW at Bikaner-III PS (upto 4667 MW) shall be granted with Transmission system for Bikaner-III (7.7 GW) as well as Transmission system Bikaner-IV (3.6 GW). In the reallocation meeting, M/s Deshraj Solar agreed to shift connectivity of 300 MW to Bikaner-III PS with the above condition. Accordingly, in the present meeting, it was proposed to grant connectivity of 300 MW to M/s Deshraj Solar at 220 kV Bikaner-III PS through 1 no. of 220 kV line bay. M/s Deshraj vide mail dated 06/01/24 has submitted request for early bay completion date (before GNA effectiveness date) as 31/03/2026. It was informed that liability for bay charges will be on RE applicant in case of generation delay and power evacuation will be based on availability of real time margins only.

M/s Deshraj Solar informed that 220 kV bay at Bikaner-III end shall be under ISTS & tower configuration shall be S/c line on D/c tower. The same was noted & M/s Deshraj was asked to submit necessary undertaking for S/c line on D/c tower configurations. M/s Deshraj agreed for the same.

Accordingly, details of revised transmission system for connectivity under GNA for Deshraj Solar Energy Private Limited (300 MW) is as below:

Details of Transmission system for Connectivity under GNA:

A. Associated Transmission System (ATS): NIL

B. Transmission System under applicant scope

- (i). Deshraj Solar Energy Private Limited Solar Power Project – Bikaner-III PS 220 kV S/c line on D/c tower (Suitable to carry minimum 300 MW at nominal voltage)

C. Transmission system for Connectivity under GNA (tentative)

1. Augmentation of 1x1500 MVA (4th) 765/400kV ICT at Bikaner (PG)
2. Augmentation of 2x1500 MVA (5th & 6th) 765/400 kV ICTs at Bikaner-III Pooling Station
3. Augmentation of 2x500 MVA (4th & 5th) 400/220 kV ICTs at Bikaner-III Pooling Station
4. LILO of both ckts of 400 kV Bikaner(PG)-Bikaner-II D/c line (Quad) at Bikaner-III PS
5. Bikaner-II PS – Bikaner-III PS 400 kV D/c line (Quad)
6. Bikaner-III - Neemrana-II 765 kV D/c line along with 330 MVAR switchable line reactor for each circuit at each end

Sl. No.	Application ID	Applicant	Project Location	Application Date	Nature of Applicant	Criterion for applying	Injection at ISTS point (MW)	Start Date of Connectivity (As per Application)	Connectivity location	Conn BG requirement as per earlier intimation	Revised Conn BG requirement
7. Augmentation of 2x1500 MVA (3rd & 4th),765/400 kV ICTs at Neemrana-II S/s 8. Neemrana-II -Kotputli 400 kV D/c line (Quad) 9. LILO of both ckts of 400 kV Gurgaon (PG) - Sohna Road (GPTL) D/c line (Quad) at Neemrana-II S/s 10. Bikaner-III - Neemrana-II 765 kV D/c line (2nd) along with 330 MVAR switchable line reactor for each circuit at each end 11. Neemrana-II- Bareilly (PG) 765 kV D/c line along with 330 MVAR switchable line reactor for each circuit at each end 12. Augmentation by 400/220 kV, 1x500 MVA ICT (3rd) at Kotputli (PG) 13. 400 kV Bareilly(765/400kV) – Bareilly(PG) D/c line (Quad) (2nd) 14. Augmentation with 1x1500 MVA, 765/400 kV ICT at Bareilly (765/400kV) S/s (3rd) 15. Establishment of 6x1500 MVA, 765/400 kV Bikaner-IV Pooling Station along with 2x240 MVAR (765kV) & 2x125 MVAR (420kV) Bus Reactors at a suitable location near Bikaner 16. 400kV Sectionalizer bay (1 set) at Bikaner-IV PS 17. STATCOM (2x±300MVAR) along with MSC (4x125 MVAR) & MSR (2x125 MVAR) along with 2 nos. 400kV bays at Bikaner-IV PS 18. LILO of both ckts of 400kV Bikaner II PS- Bikaner III PS (Quad) direct line at Bikaner-IV PS 19. Establishment of 765/400 kV, 6x1500 MVA S/s at suitable location near Siwani (Distt. Bhiwani) along with 2x240 MVAR (765kV) Bus Reactor & 2x125 MVAR (420kV) Bus Reactor 20. Bikaner-IV PS – Siwani 765 kV 2xD/c line along with 240 MVAR switchable line reactor for each circuit at each end 21. Siwani – Jind (PG) 400 kV D/c line (Quad) 22. Siwani – Patran (Indi Grid) 400 kV D/c line (Quad) along with 80 MVAR switchable line reactor for each circuit at Siwani S/s end 23. Siwani – Fatehabad (PG) 400 kV D/c line (Quad) 24. Siwani – Sonipat (PG) 400 kV D/c line (Quad) along with 63 MVAR switchable line reactor for each circuit at Siwani S/s end 25. STATCOM (2x±300MVAR) along with MSC (4x125 MVAR) & MSR (2x125 MVAR) along with 2 nos. 400kV bays at Siwani S/s 26. 400kV Sectionalizer bay (1 set) at at Siwani S/s D. Additional Transmission system for Inter-regional power transfer: 1. Establishment of 765kV Mandsaur Pooling Station along with 2x330 MVAR (765 kV) Bus Reactors 2. Beawar- Mandsaur PS 765 kV D/c line along with 240 MVAR switchable line reactor for each circuit at each end 3a) Mandsaur PS – Indore(PG) 765 kV D/c Line along with 1x330 MVAR switchable line reactor on each ckt at Mandsaur end of Mandsaur PS – Indore(PG) 765 kV D/c Line OR 3b) Establishment of 765 kV Kurawar S/s with 2x330 MVAR 765 kV bus reactor 3c) Mandsaur – Kurawar 765 kV D/c line along with 240 MVAR switchable line reactors on each ckt at both ends of Mandsaur – Kurawar 765 kV D/c line 3d) LILO of Indore – Bhopal 765 kV S/c line at Kurawar											

Sl. No.	Application ID	Applicant	Project Location	Application Date	Nature of Applicant	Criterion for applying	Injection at ISTS point (MW)	Start Date of Connectivity (As per Application)	Connectivity location	Conn BG requirement as per earlier intimation	Revised Conn BG requirement
Start Date of Connectivity under GNA: 01.09.2026(Interim). Final start date shall be confirmed upon award of the system. Note: CERC vide order dated 19.01.2024 on petition no. 268/MP/2023 and 269/MP/2023 has directed CTU to carry out fresh reallocation exercise for the meetings held on 20.6.2023 and 3.8.2023 or any subsequent reallocation meeting held for substations in Rajasthan. The same shall be carried out by CTU again and outcome shall be filed with CERC as per the order. <i>In view of the above, the application may again need to be discussed in CMETS-NR meeting for scheme review and finalization.</i>											
5.	2200000077	Avaada Energy Private Limited	Barmer distt., Rajasthan	05.06.2023	Generator (Hybrid)	Land BG Route	50	30.09.2025	Barmer-I PS	•Conn BG1: 50 lakh •Conn BG2: Nil (Bay in sharing) •Conn BG3: Rs. 1 Cr.	•Conn BG1: 50 lakh •Conn BG2: Nil (Bay in sharing) •Conn BG3: Rs. 1 Cr.
It was informed that in the reallocation meeting held on 02.01.2024, based on the request of M/s Avaada, it was proposed that the connectivity of Avaada Energy Pvt. Ltd. (50 MW) granted at Fatehgarh-IV PS(Sec-II) in sharing with M/s Radiant (200 MW) shall be shifted to Barmer-I PS and granted connectivity along with another application (lead) of M/s Avaada(250 MW) which is being discussed in the present CMETS NR meeting. M/s Avaada had agreed for the change in its connectivity system with Barmer-I transmission system. Accordingly, it was proposed to grant connectivity to M/s Avaada at 220 kV Barmer-I PS through 1 no. of bay for combined capacity of 300(50+250) MW. M/s Avaada was asked to confirm 220 kV bay scope at Barmer-I end and tower configuration. M/s Avaada indicated 250 MW application as lead entity for above grant. M/s Avaada informed that 220 kV bay at Barmer-I PS end shall be under ISTS & tower configuration shall be S/c line on D/c tower. The same was noted & M/s Avaada was asked to submit necessary undertaking for S/c line on D/c tower configurations. M/s Avaada agreed for the same. Accordingly, details of revised transmission system for connectivity under GNA for Avaada Energy Private Limited (50 MW) is as below: <u>Details of Transmission system for Connectivity under GNA:</u> A. <u>Associated Transmission System (ATS):</u> NIL B. <u>Transmission System under applicant scope</u> (i). Common Pooling station for Avaada Energy Private Limited Solar Power Projects (App No. 2200000077(50 MW) & App. No. 22000000290(250 MW)) – Barmer-I PS 220 kV S/c line (Suitable to carry minimum 300 MW at nominal voltage) C. <u>Transmission system for Connectivity under GNA</u> 1. 2x1500 MVA (2nd & 3rd), 765/400 kV ICTs at Barmer-I Pooling Station											

Sl. No.	Application ID	Applicant	Project Location	Application Date	Nature of Applicant	Criterion for applying	Injection at ISTS point (MW)	Start Date of Connectivity (As per Application)	Connectivity location	Conn BG requirement as per earlier intimation	Revised Conn BG requirement
2. 1x500 MVA (1 st), 400/220 kV ICT at Barmer-I Pooling Station 3. Fatehgarh-III (Section-2) PS – Barmer-I PS 400 kV D/c line (Quad) 4. Establishment of 2x1500 MVA, 765/400 kV Substation at suitable location near Sirohi along with 2x240 MVAR (765 kV) & 2x125 MVAR (400 kV) Bus Reactor 5. Barmer-I PS– Sirohi PS 765 kV D/c line along with 240 MVAR switchable line reactor for each circuit at each end 6. Sirohi PS-Chittorgarh (PG) 400 kV D/c line along with 80 MVAR switchable line reactor for each circuit at Sirohi PS end (Quad) 7. Establishment of, 765 kV Substation at suitable location near Rishabdeo (Distt. Udaipur) along with 2x240 MVAR (765 kV) Bus Reactor 8. Sirohi PS- Rishabdeo 765 kV D/c line along with 330 MVAR switchable line reactor for each circuit at Sirohi end 9. Rishabdeo - Mandsaur PS 765 kV D/c line along with 240 MVAR switchable line reactor for each circuit at Rishabdeo end 10. LILO of one circuit of 765 kV Chittorgarh-Banaskanta D/c line at Rishabdeo S/s 11. Establishment of 765 kV Mandsaur PS along with 2x330 MVAR (765 kV) Bus Reactors 12. Mandsaur PS–Indore(PG) 765 kV D/c Line along with 1x330 MVAR switchable line reactor on each ckt at Mandsaur end of Mandsaur PS – Indore(PG) 765 kV D/c Line 13. Establishment of 765/400 kV (2x1500 MVA Kurawar S/s with 2x330 MVAR 765 kV bus reactor and 1x125 MVAR 420 kV bus reactor 14. Mandsaur – Kurawar 765 kV D/c line along with 240 MVAR switchable line reactors on each ckt at both ends of Mandsaur – Kurawar 765 kV D/c line 15. LILO of Indore – Bhopal 765 kV S/c line at Kurawar 16. Kurawar – Ashtha 400 kV D/c (Quad ACSR/AAAC/AL59 moose equivalent) line 17. LILO of one circuit of Indore – Itarsi 400 kV D/c line at Astha 18. Shujalpur – Kurawar 400 kV D/c (Quad ACSR/AAAC/AL59 moose equivalent) line Start Date of Connectivity under GNA: 01.03.2026(Interim). Final date shall be confirmed upon award of the system Note: As Barmer-1 was not part of the reallocation exercise and Avaada requested to shift its connectivity (50 MW) with another new application of Avaada (250 MW), therefore its grant as discussed in CMETS meeting shall continue.											
6.	2200000212	Radiant Star Solar Park Private Limited	Barmer distt., Rajasthan	28.08.2023	Renewable Power Park Developer (Solar)	Land Route	100	31.12.2025	Fatehgarh-IV PS(Sec-II)	•Conn BG1: Rs. 50 lakh •Conn BG2: Rs. 3 Cr. •Conn BG3: Rs. 2 Cr.	•Conn BG1: Rs. 50 lakh •Conn BG2: Nil(Bay in sharing) •Conn BG3: Rs. 2 Cr
In the reallocation meeting held on 02.01.2024, M/s Radiant Star (100 MW) granted connectivity at Barmer-I PS opted to shift its connectivity to Fatehgarh-IV PS(Sec-II). Accordingly, in the present meeting it was proposed to grant connectivity of 100 MW to M/s Radiant at 220 kV Fatehgarh-IV PS(Sec-II) through the same 220 kV bay in sharing with the other application of M/s Radiant Star(200 MW) [lead entity] for the combined capacity of 300 MW.											

Sl. No.	Application ID	Applicant	Project Location	Application Date	Nature of Applicant	Criterion for applying	Injection at ISTS point (MW)	Start Date of Connectivity (As per Application)	Connectivity location	Conn BG requirement as per earlier intimation	Revised Conn BG requirement
Accordingly, details of revised transmission system for connectivity under GNA for M/s Radiant Star (100 MW) is as below:											
<u>Details of Transmission system for Connectivity under GNA:</u>											
A. <u>Associated Transmission System (ATS): NIL</u>											
B. <u>Transmission System under applicant scope</u>											
(i). Common Pooling Station for Radiant Star Solar Park Private Limited Solar Power Park (App no.0212100037(200 MW) & 2200000212(100 MW))– Fatehgarh-IV PS(Sec-II) PS 220 kV S/c line on D/c tower(Suitable to carry minimum 300 MW at nominal voltage)											
C. <u>Transmission system for Connectivity under GNA</u>											
1. Augmentation of 1x500 MVA (7th), 400/220kV ICT at Fatehgarh-IV (Section-2) Pooling Station											
2. 2x1500 MVA (3 rd & 4 th), 765/400 kV ICT at Fatehgarh-IV (Section-2) Pooling Station											
3. Fatehgarh-IV PS(Sec-II) – Bhinmal (PG) 400 kV D/c line (Twin HTLS*) along with 50 MVAR switchable line reactor on each ckt at each end											
4. LILO of both ckts of 765 kV Fatehgarh-III- Beawar D/c line at Fatehgarh-IV (Section-2) PS along with 330 MVAR switchable line reactor at Fatehgarh-IV PS end of each ckt of 765 kV Fatehgarh-IV- Beawar D/c line (formed after LILO)											
5. Establishment of 765 kV Mandsaur PS along with 2x330 MVAR (765 kV) Bus Reactors											
6. Beawar- Mandsaur PS 765 kV D/c line along with 240 MVAR switchable line reactor for each circuit at each end											
7. Establishment of 2x1500 MVA, 765/400 kV Substation at suitable location near Sirohi along with 2x240 MVAR (765 kV) & 2x125 MVAR (400 kV) Bus Reactor											
8. Sirohi PS-Chittorgarh (PG) 400 kV D/c line along with 80 MVAR switchable line reactor for each circuit at Sirohi PS end (Quad)											
9. Fatehgarh-IV (Section-2) PS – Sirohi PS 765 kV D/c line along with 240 MVAR switchable line reactor for each circuit at each end											
10. Establishment of, 765 kV Substation at suitable location near Rishabdeo (Distt. Udaipur) along with 2x240 MVAR (765 kV) Bus Reactor											
11. Sirohi PS- Rishabdeo 765 kV D/c line along with 330 MVAR switchable line reactor for each circuit at Sirohi end											
12. Rishabdeo - Mandsaur PS 765 kV D/c line along with 240 MVAR switchable line reactor for each circuit at Rishabdeo end											
13. LILO of one circuit of 765 kV Chittorgarh-Banaskanta D/c line at Rishabdeo S/s											
14. Mandsaur PS–Indore(PG) 765 kV D/c Line along with 1x330 MVAR switchable line reactor on each ckt at Mandsaur end of Mandsaur PS – Indore(PG) 765 kV D/c Line											
15. Establishment of 765/400 kV (2x1500 MVA Kurawar S/s with 2x330 MVAR 765 kV bus reactor and 1x125 MVAR 420 kV bus reactor											
16. Mandsaur – Kurawar 765 kV D/c line along with 240 MVAR switchable line reactors on each ckt at both ends of Mandsaur – Kurawar 765 kV D/c line											
17. LILO of Indore – Bhopal 765 kV S/c line at Kurawar											

Sl. No.	Application ID	Applicant	Project Location	Application Date	Nature of Applicant	Criterion for applying	Injection at ISTS point (MW)	Start Date of Connectivity (As per Application)	Connectivity location	Conn BG requirement as per earlier intimation	Revised Conn BG requirement
18. Kurawar – Ashtha 400 kV D/c (Quad ACSR/AAAC/AL59 moose equivalent) line 19. LILO of one circuit of Indore – Itarsi 400 kV D/c line at Astha 20. Shujalpur – Kurawar 400 kV D/c (Quad ACSR/AAAC/AL59 moose equivalent) line <i>* with minimum capacity of 2100 MVA/ckt at nominal voltage</i> Start Date of Connectivity under GNA: 01.03.2026(Interim). Final date shall be confirmed upon award of the system Note: CERC vide order dated 19.01.2024 on petition no. 268/MP/2023 and 269/MP/2023 has directed CTU to carry out fresh reallocation exercise for the meetings held on 20.6.2023 and 3.8.2023 or any subsequent reallocation meeting held for substations in Rajasthan. The same shall be carried out by CTU again and outcome shall be filed with CERC as per the order. <i>In view of the above, the application may again need to be discussed in CMETS-NR meeting for scheme review and finalization.</i>											
7.	2200000221	First Energy Private Limited	Bikaner distt., Rajasthan	11.09.2023	Captive Generating Plant (Solar)	Land BG Route	100	31.05.2025	Bikaner-III	•Conn BG1: Rs. 50 lakh •Conn BG2: NIL(Bay under applicant scope) •Conn BG3: Rs. 2 Cr.	•Conn BG1: Rs. 50 lakh •Conn BG2: NIL(Bay in sharing) •Conn BG3: Rs. 2 Cr.
It was informed that, in the reallocation meeting held on 02.01.2024, it was proposed to shift connectivity of 100 MW granted to M/s First Energy to Bikaner-III PS in sharing with M/s Juniper considering the existing sharing arrangement of M/s Sunsare(200 MW) & M/s Saimaa(150 MW). However, in case M/s Saimaa agrees for sharing with M/s Juniper(150 MW), then the connectivity to M/s First Energy shall be granted in sharing with M/s Proteus(217 MW). In view of no further communication from Sunsare or Saimaa regarding the revision in existing sharing arrangement, it was proposed to grant connectivity of M/s First Energy in sharing with M/s Juniper Cosmic for which sharing agreement to be submitted as decided in the reallocation meeting. Accordingly, details of revised transmission system for connectivity under GNA for M/s First Energy (100 MW) is as below: A. <u>Associated Transmission System (ATS):</u> NIL B. <u>Transmission System under applicant scope</u> (i). Through sharing of dedicated transmission system of M/s Juniper Green Cosmic Pvt. Ltd.– Bikaner-III PS 220 kV S/c line on D/c tower (Suitable to carry minimum 300 MW at nominal voltage) (ii). Infrastructure required for sharing shall be under the scope of M/s First Energy Pvt. Ltd.											

Sl. No.	Application ID	Applicant	Project Location	Application Date	Nature of Applicant	Criterion for applying	Injection at ISTS point (MW)	Start Date of Connectivity (As per Application)	Connectivity location	Conn BG requirement as per earlier intimation	Revised Conn BG requirement
C. <u>Transmission system for Connectivity under GNA</u> 1. Augmentation of 1x1500 MVA (4th) 765/400kV ICT at Bikaner (PG) 2. Augmentation of 2x1500 MVA (5th & 6th) 765/400 kV ICTs at Bikaner-III Pooling Station 3. Augmentation of 1x500 MVA (5th) 400/220 kV ICT at Bikaner-III Pooling Station 4. LILO of both ckts of 400 kV Bikaner(PG)-Bikaner-II D/c line (Quad) at Bikaner-III PS 5. Bikaner-II PS – Bikaner-III PS 400 kV D/c line (Quad) 6. Bikaner-III - Neemrana-II 765 kV D/c line along with 330 MVar switchable line reactor for each circuit at each end 7. Augmentation of 2x1500 MVA (3rd & 4th), 765/400 kV ICTs at Neemrana-II S/s 8. Neemrana-II -Kotputli 400 kV D/c line (Quad) 9. LILO of both ckts of 400 kV Gurgaon (PG) - Sohna Road (GPTL) D/c line (Quad) at Neemrana-II S/s 10. Bikaner-III - Neemrana-II 765 kV D/c line (2nd) along with 330 MVar switchable line reactor for each circuit at each end 11. Neemrana-II- Bareilly (PG) 765 kV D/c line along with 330 MVar switchable line reactor for each circuit at each end 12. Augmentation by 400/220 kV, 1x500 MVA ICT (3rd) at Kotputli (PG) 13. 400 kV Bareilly(765/400kV) – Bareilly(PG) D/c line (Quad) (2nd) 14. Augmentation with 1x1500 MVA, 765/400 kV ICT at Bareilly (765/400kV) S/s (3rd) 15. Establishment of 6x1500 MVA, 765/400 kV Bikaner-IV Pooling Station along with 2x240 MVar (765kV) & 2x125 MVar (420kV) Bus Reactors at a suitable location near Bikaner 16. 400kV Sectionalizer bay (1 set) at Bikaner-IV PS 17. STATCOM (2x±300MVar) along with MSC (4x125 MVar) & MSR (2x125 MVar) along with 2 nos. 400kV bays at Bikaner-IV PS 18. LILO of both ckts of 400kV Bikaner II PS- Bikaner III PS (Quad) direct line at Bikaner-IV PS 19. Establishment of 765/400 kV, 6x1500 MVA S/s at suitable location near Siwani (Distt. Bhiwani) along with 2x240 MVar (765kV) Bus Reactor & 2x125 MVar (420kV) Bus Reactor 20. Bikaner-IV PS – Siwani 765 kV 2xD/c line along with 240 MVar switchable line reactor for each circuit at each end 21. Siwani – Jind (PG) 400 kV D/c line (Quad) 22. Siwani – Patran (Indi Grid) 400 kV D/c line (Quad) along with 80 MVar switchable line reactor for each circuit at Siwani S/s end 23. Siwani – Fatehabad (PG) 400 kV D/c line (Quad) 24. Siwani – Sonipat (PG) 400 kV D/c line (Quad) along with 63 MVar switchable line reactor for each circuit at Siwani S/s end 25. STATCOM (2x±300MVar) along with MSC (4x125 MVar) & MSR (2x125 MVar) along with 2 nos. 400kV bays at Siwani S/s 26. 400kV Sectionalizer bay (1 set) at at Siwani S/s A. <u>Additional Transmission system for Inter-regional power transfer:</u> 1. Establishment of 765kV Mandsaur Pooling Station along with 2x330 MVar (765 kV) Bus Reactors											

Sl. No.	Application ID	Applicant	Project Location	Application Date	Nature of Applicant	Criterion for applying	Injection at ISTS point (MW)	Start Date of Connectivity (As per Application)	Connectivity location	Conn BG requirement as per earlier intimation	Revised Conn BG requirement
2. Beawar- Mandsaur PS 765 kV D/c line along with 240 MVAR switchable line reactor for each circuit at each end 3a) Mandsaur PS – Indore(PG) 765 kV D/c Line along with 1x330 MVAR switchable line reactor on each ckt at Mandsaur end of Mandsaur PS – Indore(PG) 765 kV D/c Line OR 3b) Establishment of 765 kV Kurawar S/s with 2x330 MVAR 765 kV bus reactor 3c) Mandsaur – Kurawar 765 kV D/c line along with 240 MVAR switchable line reactors on each ckt at both ends of Mandsaur – Kurawar 765 kV D/c line 3d) LILO of Indore – Bhopal 765 kV S/c line at Kurawar Start Date of Connectivity under GNA: 01.09.2026(Interim). Final start date shall be confirmed upon award of the system. Note: CERC vide order dated 19.01.2024 on petition no. 268/MP/2023 and 269/MP/2023 has directed CTU to carry out fresh reallocation exercise for the meetings held on 20.6.2023 and 3.8.2023 or any subsequent reallocation meeting held for substations in Rajasthan. The same shall be carried out by CTU again and outcome shall be filed with CERC as per the order. In view of the above, the application may again need to be discussed in CMETS-NR meeting for scheme review and finalization.											

C4. Applications for Connectivity to ISTS under CERC (Connectivity and General Network Access to the inter-State Transmission System) Regulations, 2022 received in Oct'23:

Sl. No.	Application ID	Applicant	Project Location	Submission Date	Nature of Applicant	Criterion for applying	Injection at ISTS point (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)
1.	2200000254	Avaada RJBikaner Private Limited	Bikaner distt., Rajasthan	03.10.2023	Generator (Solar)	Land Route	600	31.03.2025	Bikaner-III PS
2.	2200000275	Renew Solar (Shakti Six) Private Limited	Jodhpur distt., Rajasthan	05.10.2023	Generator (Solar)	Land Route	400	30.06.2026	Bhadla-III PS
3.	2200000281	Anboto Solar Private Limited	Barmer distt., Rajasthan	06.10.2023	Generator (Solar)	Land BG Route	250	30.11.2025	Barmer PS

Minutes for 27th Consultation meeting for Evolving Transmission Schemes in NR

Sl. No.	Application ID	Applicant	Project Location	Submission Date	Nature of Applicant	Criterion for applying	Injection at ISTS point (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)
4.	2200000266	Malur Renewables Private Limited	Bikaner distt., Rajasthan	06.10.2023	Renewable Power Park Developer (Solar)	Land Route	650	30.05.2025	Bhadla-III PS
5.	2200000300 (Enhancement)	Solarcraft Power India 4 Private Limited	Jaisalmer distt., Rajasthan	08.10.2023	Generator (Solar)	Land Route	50	16.06.2025	Bhadla-III PS
6.	2200000305	Juniper Green Energy Private Limited	Barmer distt., Rajasthan	11.10.2023	Generator (Solar)	Land BG Route	300	31.03.2027	Barmer-I PS
7.	2200000289	Avaada Energy Private Limited	Bikaner distt., Rajasthan	12.10.2023	Generator (Solar)	Land BG Route	300	30.11.2025	Bhadla-III PS
8.	2200000309	Renew Solar (Shakti Six) Private Limited	Jodhpur & Bikaner distt., Rajasthan	13.10.2023	Generator (Solar)	Land Route	585	30.06.2026	Bhadla-III PS
9.	2200000284	AK Renewable Infra Private Limited	Bikaner distt., Rajasthan	13.10.2023	Generator (Solar)	Land Route	300	30.06.2025	Bikaner-I PS
10.	2200000312	Auxo Sunlight Private Limited	Barmer distt., Rajasthan	14.10.2023	Generator (Solar)	Land BG Route	300	30.09.2026	Barmer-I PS
11.	2200000290 (Enhancement)	Avaada Energy Private Limited	Barmer distt., Rajasthan	17.10.2023	Generator (Hybrid)	Land BG Route	250 (Solar-125 Wind-125)	30.09.2025	Fatehgarh-IV PS
12.	2200000316	Renew Sun Power Private Limited	Barmer & Jaisalmer distt., Rajasthan	17.10.2023	Generator (Solar)	Land BG Route	300	30.09.2026	Barmer-I PS
13.	2200000327	Hazel Hybren Private Limited	Jaisalmer distt.,	20.10.2023	Generator (Solar)	Land BG Route	300	31.01.2026	Ramgarh PS

Minutes for 27th Consultation meeting for Evolving Transmission Schemes in NR

Sl. No.	Application ID	Applicant	Project Location	Submission Date	Nature of Applicant	Criterion for applying	Injection at ISTS point (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)
			Rajasthan						
14.	2200000306	Shudh Solar Power Private Limited	Bikaner distt., Rajasthan	21.10.2023	Generator (Solar)	Land Route	250	31.05.2025	Bikaner-III PS
15.	2200000308	NTPC Renewable Energy Limited	Bikaner distt., Rajasthan	21.10.2023	Generator (Solar)	Land Route	700	31.03.2026	Bikaner-III PS
16.	2200000334	Hazel Hybren Private Limited	Bikaner distt., Rajasthan	23.10.2023	Generator (Solar)	Land BG Route	300	31.01.2026	Bikaner-III PS
17.	2200000319	AM Green Energy Private Limited	Bikaner distt., Rajasthan	23.10.2023	Generator (Solar)	Land BG Route	300	31.05.2025	Bikaner-II PS
18.	2200000333	Furies Solren Private Limited	Jaisalmer distt., Rajasthan	27.10.2023	Generator (Solar)	Land BG Route	300	31.01.2026	Ramgarh PS
19.	2200000355	Gamma Renewables India Project One Private Limited	Barmer distt., Rajasthan	31.10.2023	Renewable Power Park Developer (Solar)	Land BG Route	300	01.06.2026	Barmer-I PS

Sl. No.	Application ID	Applicant	Project Location	Application Date	Nature of Applicant	Criterion for applying	Injection at ISTS point (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)	Conn BGs to be submitted
1.	2200000254	Avaada RJBikaner Private Limited	Bikaner distt., Rajasthan	03.10.2023	Generator (Solar)	Land Route	600	31.03.2025	Bikaner-III PS	• Conn BG1: 50 lakh • Conn BG2: Rs. 6 Cr. • Conn BG3: 2 lakh/MW

It was informed that M/s Avaada RJ Bikaner Private Limited has applied for connectivity for 600 MW at Bikaner-III PS. After reallocation meeting, connectivity of 4667 MW is being granted at Bikaner-III PS. Therefore, there is no margin available for grant of connectivity at Bikaner-III PS including enhancements. M/s Avaada was earlier granted connectivity of 50 MW at Bikaner-III PS in sharing with M/s Proteus Energy. As proposed in the reallocation meeting held on

Sl. No.	Application ID	Applicant	Project Location	Application Date	Nature of Applicant	Criterion for applying	Injection at ISTS point (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)	Conn BGs to be submitted
02.01.2024, based on M/s Avaada request total connectivity of 650 MW(including the connectivity of 50) shall be granted at Bikaner-IV PS. However, it may be noted that as informed in the reallocation meeting held on 02.01.2024, for the 50 MW connectivity, the transmission system shall be revised with Bikaner-IV transmission system and will not be as per application priority due to shifting of connectivity to Bikaner-IV PS. M/s Avaada agreed for the same in the reallocation meeting. Accordingly, it is proposed to grant connectivity of 650 MW at Bikaner-IV PS at 400 kV level through 1 no. of 400 kV line bay. Applicant may confirm the 400kV bay scope at Bikaner-IV PS end and the tower configuration. M/s Avaada initially intended to obtain connectivity at 220 kV for 650 MW. However, subsequently in the same meeting, M/s Avaada requested to grant additional connectivity of 300 MW at Bikaner-IV PS for application No. 2200000289(300 MW) which was applied in Land BG route for connectivity at Bhadla-III PS. Considering the same, total connectivity at Bikaner-IV PS shall be 950 MW for M/s Avaada. Therefore, it was decided to grant connectivity at 400 kV through 1 no. of 400 kV line bay. M/s Avaada agreed for the same. Further, it was also informed that M/s Avaada RJ Bikaner Private Limited has requested Start Date of Connectivity under GNA from 31/03/2025. However, considering the same and expected commissioning date of transmission system (i.e. 31.08.2025), the start date of Connectivity under GNA shall be 01.09.2026 (interim). M/s Avaada noted the same. M/s Avaada requested to consider App. No. 2200000254(600 MW) as lead. Proposed Transmission system for connectivity of Avaada RJ Bikaner Private Limited for 600 MW under GNA is as below: <u>Details of Transmission system for Connectivity under GNA:</u> A. <u>Associated Transmission System (ATS):</u> NIL B. <u>Transmission System under applicant scope</u> (i). Common Pooling station for Avaada RJ Bikaner Private Limited Solar Power Projects (App No. 220000013(50 MW), App. No. 2200000254(600 MW)) & Avaada Energy Private Limited (App No. 2200000289(300 MW)) Solar Power Projects – Bikaner-IV PS 400 kV S/c line on D/c tower (Suitable to carry minimum 950 MW at nominal voltage) C. <u>Transmission system for Connectivity under GNA</u> 1. Establishment of 6x1500 MVA, 765/400 kV Bikaner-IV Pooling Station along with 2x240 MVar (765kV) & 2x125 MVar (420kV) Bus Reactors at a suitable location near Bikaner 2. 400kV Sectionalizer bay (1 set) at Bikaner-IV PS 3. STATCOM (2x+300MVar) along with MSC (4x125 MVar) & MSR (2x125 MVar) along with 2 nos. 400kV bays at Bikaner-IV PS 4. LILO of both ckts of 400kV Bikaner II PS- Bikaner III PS (Quad) direct line at Bikaner-IV PS 5. Establishment of 765/400 kV, 6x1500 MVA S/s at suitable location near Siwani (Distt. Bhiwani) along with 2x240 MVar (765kV) Bus Reactor & 2x125 MVar (420kV) Bus Reactor										

Sl. No.	Application ID	Applicant	Project Location	Application Date	Nature of Applicant	Criterion for applying	Injection at ISTS point (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)	Conn BGs to be submitted
6. Bikaner-IV PS – Siwani 765 kV 2xD/c line along with 240 MVAR switchable line reactor for each circuit at each end 7. Siwani – Jind (PG) 400 kV D/c line (Quad) 8. Siwani – Patran (Indi Grid) 400 kV D/c line (Quad) along with 80 MVAR switchable line reactor for each circuit at Siwani S/s end 9. Siwani – Fatehabad (PG) 400 kV D/c line (Quad) 10. Siwani – Sonipat (PG) 400 kV D/c line (Quad) along with 63 MVAR switchable line reactor for each circuit at Siwani S/s end 11. STATCOM (2x300MVAR) along with MSC (4x125 MVAR) & MSR (2x125 MVAR) along with 2 nos. 400kV bays at Siwani S/s 12. 400kV Sectionalizer bay (1 set) at Siwani S/s D. Additional Transmission system for Inter-regional power transfer: 1. Establishment of 765kV Mandsaur Pooling Station along with 2x330 MVAR (765 kV) Bus Reactors 2. Beawar- Mandsaur PS 765 kV D/c line along with 240 MVAR switchable line reactor for each circuit at each end 3a) Mandsaur PS – Indore(PG) 765 kV D/c Line along with 1x330 MVAR switchable line reactor on each ckt at Mandsaur end of Mandsaur PS – Indore(PG) 765 kV D/c Line OR 3b) Establishment of 765 kV Kurawar S/s with 2x330 MVAR 765 kV bus reactor 3c) Mandsaur – Kurawar 765 kV D/c line along with 240 MVAR switchable line reactors on each ckt at both ends of Mandsaur – Kurawar 765 kV D/c line 3d) LILO of Indore – Bhopal 765 kV S/c line at Kurawar Start Date of Connectivity under GNA: 01.09.2026(Interim). Final start date shall be confirmed upon award of the system.										
2.	2200000275	Renew Solar (Shakti Six) Private Limited	Jodhpur distt., Rajasthan	05.10.2023	Generator (Solar)	Land Route	400	30.06.2026	Bhadla-III PS	• Conn BG1:50 lakh • Conn BG2: Rs. 6 Cr. • Conn BG3: 2 lakh/MW
It was informed that M/s Renew Solar (Shakti Six) Private Limited has applied for connectivity for 400 MW at Bhadla-III PS. M/s Renew Solar (Shakti Six) Pvt. Ltd. is already granted connectivity of 1000 MW (550+450) at Bhadla-III PS. Accordingly, it is proposed to grant connectivity of 400 MW at 400kV Bhadla-III PS through 400 kV D/c line for combined capacity of 1400 MW with an additional 400 kV line bay(1 No.) with the present application. Applicant was asked to confirm 400kV bay scope at Bhadla-III PS. M/s Renew informed that 400 kV bay at Bhadla-III PS end shall be under ISTS. The same was noted. It was also informed that M/s Renew Solar (Shakti Six) Private Limited has requested Start Date of Connectivity under GNA from 30/06/2026. However, considering the same and expected commissioning date of transmission system(i.e. 31.08.2028), the start date of Connectivity under GNA shall be 01.09.2028 (interim). M/s Renew noted the same.										

Sl. No.	Application ID	Applicant	Project Location	Application Date	Nature of Applicant	Criterion for applying	Injection at ISTS point (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)	Conn BGs to be submitted
Transmission system for connectivity of Renew Solar (Shakti Six) Private Limited under GNA is as below:										
<u>Details of Transmission system for Connectivity under GNA:</u>										
A. <u>Associated Transmission System (ATS):</u> NIL										
B. <u>Transmission System under applicant scope</u>										
(i). Common Pooling Station for Renew Solar (Shakti Six) Private Limited (App. No. 120003848(550 MW), 0312100004(450 MW) & 2200000275(400 MW)) – Bhadla-III PS 400 kV D/c line (Suitable to carry minimum 900 MW per circuit at nominal voltage)										
C. <u>Transmission system for Connectivity under GNA:</u>										
1. Augmentation of 2x1500 MVA (3rd & 4th), 765/400kV ICT at Bhadla-III PS										
2. 765 kV Bhadla-III – Bikaner-III D/c line along with 240 MVA switchable line reactor for each circuit at Bhadla-III end										
3. Establishment of 6000 MW, ± 800 kV Bhadla (HVDC) [LCC] terminal station (4x1500 MW) at a suitable location near Bhadla-III substation										
4. Establishment of 6000 MW, ±800 kV Fatehpur (HVDC) [LCC] terminal station (4x1500 MW) at suitable location near Fatehpur (UP)										
5. Bhadla-III – Bhadla (HVDC) 400 kV 2xD/c quad moose line along with the line bays at both substations										
6. Bhadla-III – Sikar-II 765 kV D/c line along with 330 MVA Switchable line reactor for each circuit at each end of Bhadla-III – Sikar-II 765 kV D/c line										
7. Fatehgarh-III – Bhadla-III 400kV D/c line (Quad) along with 63 MVA Switchable line reactor for each circuit at both ends of Fatehgarh-III- Bhadla-III 400kV D/c line										
8. Augmentation of 5x1500 MVA (1 st to 5 th), 765/400 kV ICTs at Fatehpur (HVDC) along with 2x330 MVA (765kV) bus reactor										
9. LILO of both ckts of 765 kV Varanasi – Kanpur (GIS) D/c line at Fatehpur										
10. Augmentation by 1x1500 MVA ICT at 765/400kV Kanpur (GIS) substation										
Start Date of Connectivity under GNA: 01.09.2028(Interim). Final date shall be confirmed upon award of the system										
3.	2200000281	Anboto Solar Private Limited	Barmer distt., Rajasthan	06.10.2023	Generator (Solar)	Land BG Route	250	30.11.2025	Barmer-I PS	• Conn BG1:50 lakh • Conn BG2: Rs. 3 Cr. • Conn BG3: 2 lakh/MW
It was informed that M/s Anboto Solar Private Limited has applied for connectivity for 250 MW at Barmer-I PS. Accordingly, it was proposed to grant connectivity at Barmer-I PS at 220 kV level through 1 no. of 220 kV line bay.										
M/s Anboto was asked to confirm the 220kV bay scope at Barmer-I PS end and the tower configuration. M/s Anboto informed that 220 kV line bay at Barmer-I end shall be under ISTS and tower configuration shall be S/c line on D/c tower. The same was noted & M/s Anboto was asked to submit necessary										

Sl. No.	Application ID	Applicant	Project Location	Application Date	Nature of Applicant	Criterion for applying	Injection at ISTS point (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)	Conn BGs to be submitted
undertaking for S/c line on D/c tower configurations. M/s Anboto agreed for the same. It was also informed that presently, total connectivity granted at Fatehgarh-IV (4080MW) & Barmer-I PS is 1350 MW (after reallocation of Radiant(100 MW) & Avaada(50 MW)). Transmission system for evacuation of 5.5 GW from Fatehgarh-IV PS(Sec-II) (4 GW) & Barmer-I PS(1.5 GW) is already under bidding. Considering the present application of M/s Anboto Solar, total connectivity at Barmer-I PS is exceeding 1.5 GW as only 70 MW margin is left out of 5.5 GW tr. system. Therefore, additional system beyond the 5.5 GW scheme is required. Additional transmission system for 2.5 GW from Barmer-I PS & 1 GW from Fatehgarh-IV PS was discussed and agreed in the same meeting. Accordingly, tentative scheme for the connectivity of M/s Anboto is being proposed. Further, it was informed that M/s Anboto Solar Private Limited has requested Start Date of Connectivity under GNA from 30/11/2025. Considering the same and expected commissioning date of transmission system(i.e. 31.08.2026), the start date of Connectivity under GNA shall be 01.09.2026(interim). M/s Anboto noted the same. Transmission system for connectivity of M/s Anboto Solar Private Limited under GNA is as below: <u>Details of Transmission system for Connectivity under GNA:</u> A. <u>Associated Transmission System (ATS): NIL</u> B. <u>Transmission System under applicant scope</u> (i). Anboto Solar Private Limited Solar Power Project– Barmer-I PS 220 kV S/c line on D/c tower (Suitable to carry minimum 300 MW at nominal voltage) C. <u>Transmission system for Connectivity under GNA:</u> 1. Augmentation with 765/400 kV, 3x1500 MVA Transformer (3rd, 4th & 5th) at Barmer-I PS 2. Augmentation of 2x500 MVA (3rd & 4th), 400/220 kV ICTs at Barmer-I PS 3. Fatehgarh-III (Section-2) PS – Barmer-I PS 400 kV D/c line (Quad) 4. Establishment of 2x1500 MVA, 765/400 kV Substation at suitable location near Sirohi along with 2x240 MVAR (765 kV) & 2x125 MVAR (400 kV) Bus Reactor 5. Barmer-I PS– Sirohi PS 765 kV D/c line along with 240 MVAR switchable line reactor for each circuit at each end 6. Sirohi PS-Chittorgarh (PG) 400 kV D/c line along with 80 MVAR switchable line reactor for each circuit at Sirohi PS end (Quad) 7. Establishment of, 765 kV Substation at suitable location near Rishabdeo (Distt. Udaipur) along with 2x240 MVAR (765 kV) Bus Reactor 8. Sirohi PS- Rishabdeo 765 kV D/c line along with 330 MVAR switchable line reactor for each circuit at Sirohi end 9. Rishabdeo - Mandsaur PS 765 kV D/c line along with 240 MVAR switchable line reactor for each circuit at Rishabdeo end 10. LILO of one circuit of 765 kV Chittorgarh-Banaskanta D/c line at Rishabdeo S/s										

Sl. No.	Application ID	Applicant	Project Location	Application Date	Nature of Applicant	Criterion for applying	Injection at ISTS point (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)	Conn BGs to be submitted
11. Establishment of 765 kV Mandsaur PS along with 2x330 MVAR (765 kV) Bus Reactors 12. Mandsaur PS–Indore(PG) 765 kV D/c Line along with 1x330 MVAR switchable line reactor on each ckt at Mandsaur end of Mandsaur PS – Indore(PG) 765 kV D/c Line 13. Establishment of 765/400 kV (2x1500 MVA Kurawar S/s with 2x330 MVAR 765 kV bus reactor and 1x125 MVAR 420 kV bus reactor 14. Mandsaur – Kurawar 765 kV D/c line along with 240 MVAR switchable line reactors on each ckt at both ends of Mandsaur – Kurawar 765 kV D/c line 15. Kurawar – Ashtha 400 kV D/c (Quad ACSR/AAAC/AL59 moose equivalent) line 16. LILO of Indore – Bhopal 765 kV S/c line at Kurawar 17. LILO of one circuit of Indore – Itarsi 400 kV D/c line at Astha 18. Shujalpur – Kurawar 400 kV D/c (Quad ACSR/AAAC/AL59 moose equivalent) line 19. Establishment of 765/400kV, 2x1500 MVA S/s at suitable location near Merta (Merta-II Substation) along with 2x125 MVAR & 2x240 MVAR bus reactor at Merta-II S/s 20. STATCOM (2x±300MVAR) along with MSC (4x125 MVAR) & MSR (2x125 MVAR) along with 2 nos. 400kV bays at Barmer-I PS 21. 400kV Sectionalizer bay (1 set) at Barmer-I S/s 22. Barmer-I PS – Merta-II 765 kV D/c line along with 240 MVAR switchable line reactor at Barmer-I PS end and 330MVAR switchable line reactor at Merta-II PS end for each circuit of Barmer-I PS – Merta-II 765 kV D/c line 23. Merta-II – Beawar 400 kV D/c line (Quad) 24. Merta-II – Dausa 765 kV D/c line along with 240 MVAR switchable line reactor for each circuit at each end of Merta-II – Dausa 765 kV D/c line 25. Establishment of 765/400kV, 2x1500 MVA S/s at suitable location near Ghiror (Distt. Mainpuri) along with 2x125 MVAR & 2x240 MVAR bus reactor at Ghiror S/s (UP) 26. Dausa - Ghiror 765 kV D/c line along with 240 MVAR switchable line reactor for each circuit at each end of Dausa - Ghiror 765 kV D/c line 27. LILO of both ckt of 765 kV Aligarh (PG) -Orai (PG) D/c line at Ghiror S/s along with 240 MVAR switchable line reactor for each circuit at Ghiror S/s end of 765 kV Ghiror -Orai D/c line 28. LILO of one ckt of 765kV Agra (PG) – Fatehpur(PG) 765kV D/c line at Ghiror along with 240 MVAR switchable line reactor at Ghiror S/s end of 765 kV Ghiror -Fatehpur line 29. 400kV Ghiror-Firozabad (UPPTCL) D/c line (Quad) Start Date of Connectivity under GNA: 01.09.2026(Interim). Final date shall be confirmed upon award of the system										
4.	2200000266	Malur Renewables	Bikaner distt.,	06.10.2023	Renewable Power Park	Land Route	650	30.05.2025	Bhadla-III PS	• Conn BG1:50 lakh

Sl. No.	Application ID	Applicant	Project Location	Application Date	Nature of Applicant	Criterion for applying	Injection at ISTS point (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)	Conn BGs to be submitted
		Private Limited	Rajasthan		Developer (Solar)					• Conn BG2: Nil(Bay under applicant scope) • Conn BG3: 2 lakh/MW
It was informed that M/s Malur Renewables Private Limited has applied for connectivity for 650 MW at 400 kV at Bhadla-III PS. M/s Malur informed that the connectivity start date of year 2028 is not matching with their project timeline. Therefore, M/s Malur requested to shift their application for grant of connectivity at Barmer-I PS. CTUIL informed that M/s Malur Renewables has applied for connectivity through Land Route. Barmer-I PS is about 250 km away from the project location of M/s Malur. Therefore, technically it is not feasible to grant connectivity at such a far off substation as it shall not be able to meet the requirements of CEA technical standards. M/s Malur renewables informed that they will change the land closer to Barmer-I PS after grant as the land change is allowed by CERC. Regarding, this CTUIL informed that as per 11/SM/2023 CERC order : <i>“RE Developers that sometimes the location of the land for which Connectivity has been granted needs some changes based on changes in location coordinates of the substation (which was under bidding/yet to be awarded), proximity to the ISTS substation which reduces the length of dedicated transmission line, higher capacity utilisation factor at different land location as calculated at a later point in time and unseen practical difficulty in acquiring the complete land parcel near to the earlier land location. Considering the issues highlighted by the RE Developers, we direct that change in the location of the land parcel without any change in (a) the point of Connectivity with ISTS and (b) the start date of Connectivity shall be allowed”</i> As per above, RE developer to whom connectivity has been granted, land change has been allowed keeping the point of connectivity as same. However, in this instant application the substation as well as land both are being changed and more so applicant is not yet a connectivity grantee. At the application stage, application is made based on land documents for Bhadla pockets and Barmer is very far off. Though this may be allowed for applications with LOA or Land BG applications where land is not firmed up. Therefore, it is not feasible for grant of connectivity at Barmer-I PS with land based application at Bhadla. After discussions, M/s Malur agreed for connectivity at Bhadla-III PS at 400 kV. Accordingly, it was proposed to grant connectivity of 650 MW to M/s Malur Renewables at 400 kV at Bhadla-III PS through 1 no. of 400 kV line bay. M/s Malur renewables was asked to confirm bay scope and tower configuration. M/s Malur informed that bay shall be under their scope and tower configuration shall be S/c line on D/c. The same was noted and M/s Malur was asked to submit necessary undertaking for S/c line on D/c tower. M/s Malur renewables agreed for the same. Further, it was also informed that, M/s Malur Renewables Private Limited has requested Start Date of Connectivity under GNA from 30/05/2025. However, considering the same and expected commissioning date of transmission system (i.e 31.08.2028), the start date of Connectivity under GNA shall be 01.09.2028(interim).M/s Malur renewables noted the same.										

Sl. No.	Application ID	Applicant	Project Location	Application Date	Nature of Applicant	Criterion for applying	Injection at ISTS point (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)	Conn BGs to be submitted
---------	----------------	-----------	------------------	------------------	---------------------	------------------------	------------------------------	---	--	--------------------------

Transmission system for connectivity of M/s Malur Renewables Private Limited under GNA is as below:

Details of Transmission system for Connectivity under GNA:

A. Associated Transmission System (ATS): NIL

B. Transmission System under applicant scope

- (i). M/s Malur Renewables Private Limited Renewable Power Park– Bhadla-III PS 400 kV S/c line on D/c tower (Suitable to carry minimum 900 MW at nominal voltage)

C. Transmission system for Connectivity under GNA

1. Augmentation of 2x1500 MVA (3rd & 4th), 765/400kV ICT at Bhadla-III PS
2. 765 kV Bhadla-III – Bikaner-III D/c line along with 240 MVar switchable line reactor for each circuit at Bhadla-III end
3. Establishment of 6000 MW, $\pm$ 800 kV Bhadla (HVDC) [LCC] terminal station (4x1500 MW) at a suitable location near Bhadla-III substation
4. Establishment of 6000 MW, $\pm$ 800 kV Fatehpur (HVDC) [LCC] terminal station (4x1500 MW) at suitable location near Fatehpur (UP)
5. Bhadla-III – Bhadla (HVDC) 400 kV D/c (1st & 2nd) quad moose line along with the line bays at both substations
6. Bhadla-III – Sikar-II 765 kV D/c line along with 330 MVar Switchable line reactor for each circuit at each end of Bhadla-III – Sikar-II 765 kV D/c line
7. Fatehgarh-III – Bhadla-III 400kV D/c line (Quad) along with 63 MVar Switchable line reactor for each circuit at both ends of Fatehgarh-III- Bhadla-III 400kV D/c line
8. Augmentation of 5x1500 MVA (1st to 5th), 765/400 kV ICTs at Fatehpur (HVDC) along with 2x330 MVar (765kV) bus reactor
9. LILO of both ckts of 765 kV Varanasi – Kanpur (GIS) D/c line at Fatehpur
10. Augmentation by 1x1500 MVA ICT at 765/400kV Kanpur (GIS) substation

Start Date of Connectivity under GNA: 01.09.2028(Interim). Final date shall be confirmed upon award of the system

5.	2200000300 (Enhancement)	Solarcraft Power India 4 Private Limited	Jaisalmer distt., Rajasthan	08.10.2023	Generator (Solar)	Land Route	50	16.06.2025	Bhadla-III PS	• Conn BG1:50 lakh • Conn BG2: NIL(Bay in sharing) • Conn BG3: 2 lakh/MW
----	-----------------------------	--	-----------------------------	------------	-------------------	------------	----	------------	---------------	--

It was informed that M/s Solarcraft Power India 4 Private Limited has applied for enhancement of connectivity for 50 MW at Bhadla-III PS. M/s Solarcraft Power India 4 is already granted connectivity of 260 MW(150+110) at Bhadla-III PS. Accordingly, it was proposed to grant enhancement of connectivity of 50 MW at Bhadla-III PS at 220 kV level through same 220 kV line bay earlier granted with 150 MW application. M/s Solarcraft agreed for the same.

Further it was also informed that, M/s Solarcraft Power India 4 Private Limited has requested Start Date of Connectivity under GNA from 16/06/2025. However,

Sl. No.	Application ID	Applicant	Project Location	Application Date	Nature of Applicant	Criterion for applying	Injection at ISTS point (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)	Conn BGs to be submitted
considering the same and expected commissioning date of transmission system, the start date of Connectivity under GNA shall be 31.08.2028 (interim). M/s Solarcraft agreed for the same. Accordingly, detailed Transmission system for connectivity of Solarcraft Power India 4 Private Limited for 50 MW under GNA is as below: <u>Details of Transmission system for Connectivity under GNA:</u> A. <u>Associated Transmission System (ATS):</u> NIL B. <u>Transmission System under applicant scope</u> (i). Common Pooling station of M/s Solarcraft Power India 4 Pvt. Ltd. for Solar Power Projects (Appl. Nos.2200000152 (150 MW), 2200000192 (110 MW) & 220000300(50 MW) – Bhadla-III PS 220 kV S/c line on D/c tower (Suitable to carry minimum 310 MW at nominal voltage) C. <u>Transmission system for Connectivity under GNA</u> 1. Augmentation of 1x500 MVA (9th), 400/220 kV ICT at Bhadla-III PS 2. Augmentation of 2x1500 MVA (3rd & 4th), 765/400kV ICTs at Bhadla-III PS 3. 765 kV Bhadla-III – Bikaner-III D/c line along with 240 MVAR switchable line reactor for each circuit at Bhadla-III end 4. Establishment of 6000 MW, $\pm$ 800 kV Bhadla (HVDC) [LCC] terminal station (4x1500 MW) at a suitable location near Bhadla-III substation 5. Establishment of 6000 MW, $\pm$800 kV Fatehpur (HVDC) [LCC] terminal station (4x1500 MW) at suitable location near Fatehpur (UP) 6. Bhadla-III – Bhadla (HVDC) 400 kV D/c (1st & 2nd) quad moose line along with the line bays at both substations 7. Bhadla-III – Sikar-II 765 kV D/c line along with 330 MVAR Switchable line reactor for each circuit at each end of Bhadla-III – Sikar-II 765 kV D/c line 8. Fatehgarh-III – Bhadla-III 400kV D/c line (Quad) along with 63 MVAR Switchable line reactor for each circuit at both ends of Fatehgarh-III- Bhadla-III 400kV D/c line 9. Augmentation of 5x1500 MVA (1st to 5th), 765/400 kV ICTs at Fatehpur (HVDC) along with 2x330 MVAR (765kV) bus reactor 10. LILO of both ckts of 765 kV Varanasi – Kanpur (GIS) D/c line at Fatehpur 11. Augmentation by 1x1500 MVA ICT at 765/400kV Kanpur (GIS) substation Start Date of Connectivity under GNA: 01.09.2028(Interim). Final date shall be confirmed upon award of the system										
6.	2200000305	Juniper Green Energy Private Limited	Barmer distt., Rajasthan	11.10.2023	Generator (Solar)	Land BG Route	300	31.03.2027	Barmer-I PS	• Conn BG1:Rs.50 lakh • Conn BG2: Rs. 3 Cr. • Conn BG3: Rs. 2 lakh/MW

Sl. No.	Application ID	Applicant	Project Location	Application Date	Nature of Applicant	Criterion for applying	Injection at ISTS point (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)	Conn BGs to be submitted
It was informed that M/s Juniper Green Energy Private Limited has applied for connectivity of 300 MW at Barmer-I PS. As informed earlier, additional transmission system for 2.5 GW from Barmer-I PS & 1 GW from Fatehgarh-IV PS which is discussed and agreed in the present meeting shall be required for the grant of above application. Accordingly, it was proposed to grant connectivity at 220kV Barmer-I PS. M/s Juniper Green was asked to confirm the 220kV bay scope at Barmer-I PS end and the tower configuration. M/s Juniper Green informed that 220 kV line bay at Barmer-I end shall be under ISTS and tower configuration shall be S/c line on D/c tower. Same was noted & M/s Juniper Green was asked to submit necessary undertaking for S/c line on D/c tower configuration. M/s Juniper Green agreed for the same. Further, it was also informed that M/s Juniper Green Energy Private Limited has requested start Date of Connectivity under GNA from 31/03/2027. Considering the same and expected commissioning date of transmission system(i.e. 01.09.2026), the start date of Connectivity under GNA shall be 31.03.2027(interim). M/s Juniper noted the same. Accordingly, transmission system for connectivity of M/s Juniper Green Energy Private Limited under GNA is as below: <u>Details of Transmission system for Connectivity under GNA:</u> A. <u>Associated Transmission System (ATS):</u> NIL B. <u>Transmission System under applicant scope</u> (i). Juniper Green Energy Private Limited Solar Power Project– Barmer-I PS 220 kV S/c line on D/c tower (Suitable to carry minimum 300 MW at nominal voltage) C. <u>Transmission system for Connectivity under GNA (tentative)</u> 1. Augmentation of 1x500 MVA (4th), 400/220 kV ICT at Barmer-I PS 2. Augmentation with 765/400 kV, 3x1500 MVA Transformer (3rd, 4th & 5th) at Barmer-I PS 3. Fatehgarh-III (Section-2) PS – Barmer-I PS 400 kV D/c line (Quad) 4. Establishment of 2x1500 MVA, 765/400 kV Substation at suitable location near Sirohi along with 2x240 MVar (765 kV) & 2x125 MVar (400 kV) Bus Reactor 5. Barmer-I PS– Sirohi PS 765 kV D/c line along with 240 MVAR switchable line reactor for each circuit at each end 6. Sirohi PS-Chittorgarh (PG) 400 kV D/c line along with 80 MVar switchable line reactor for each circuit at Sirohi PS end (Quad) 7. Establishment of, 765 kV Substation at suitable location near Rishabdeo (Distt. Udaipur) along with 2x240 MVar (765 kV) Bus Reactor 8. Sirohi PS- Rishabdeo 765 kV D/c line along with 330 MVar switchable line reactor for each circuit at Sirohi end										

Sl. No.	Application ID	Applicant	Project Location	Application Date	Nature of Applicant	Criterion for applying	Injection at ISTS point (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)	Conn BGs to be submitted
9. Rishabdeo - Mandsaur PS 765 kV D/c line along with 240 MVAr switchable line reactor for each circuit at Rishabdeo end 10. LILO of one circuit of 765 kV Chittorgarh-Banaskanta D/c line at Rishabdeo S/s 11. Establishment of 765 kV Mandsaur PS along with 2x330 MVAR (765 kV) Bus Reactors 12. Mandsaur PS-Indore(PG) 765 kV D/c Line along with 1x330 MVAR switchable line reactor on each ckt at Mandsaur end of Mandsaur PS – Indore(PG) 765 kV D/c Line 13. Establishment of 765/400 kV (2x1500 MVA Kurawar S/s with 2x330 MVAr 765 kV bus reactor and 1x125 MVAr 420 kV bus reactor 14. Mandsaur – Kurawar 765 kV D/c line along with 240 MVAr switchable line reactors on each ckt at both ends of Mandsaur – Kurawar 765 kV D/c line 15. Kurawar – Ashtha 400 kV D/c (Quad ACSR/AAAC/AL59 moose equivalent) line 16. LILO of Indore – Bhopal 765 kV S/c line at Kurawar 17. LILO of one circuit of Indore – Itarsi 400 kV D/c line at Astha 18. Shujalpur – Kurawar 400 kV D/c (Quad ACSR/AAAC/AL59 moose equivalent) line 19. Establishment of 765/400kV, 2x1500 MVA S/s at suitable location near Merta (Merta-II Substation) along with 2x125 MVAr & 2x240 MVAr bus reactor at Merta-II S/s 20. STATCOM (2x±300MVAr) along with MSC (4x125 MVAr) & MSR (2x125 MVAr) along with 2 nos. 400kV bays at Barmer-I PS 21. 400kV Sectionalizer bay (1 set) at Barmer-I S/s 22. Barmer-I PS – Merta-II 765 kV D/c line along with 240 MVAr switchable line reactor at Barmer-I PS end and 330MVAr switchable line reactor at Merta-II PS end for each circuit of Barmer-I PS – Merta-II 765 kV D/c line 23. Merta-II – Beawar 400 kV D/c line (Quad) 24. Merta-II – Dausa 765 kV D/c line along with 240 MVAr switchable line reactor for each circuit at each end of Merta-II – Dausa 765 kV D/c line 25. Establishment of 765/400kV, 2x1500 MVA S/s at suitable location near Ghiror (Distt. Mainpuri) along with 2x125 MVAr & 2x240 MVAr bus reactor at Ghiror S/s (UP) 26. Dausa - Ghiror 765 kV D/c line along with 240 MVAr switchable line reactor for each circuit at each end of Dausa - Ghiror 765 kV D/c line 27. LILO of both ckt of 765 kV Aligarh (PG) -Orai (PG) D/c line at Ghiror S/s along with 240 MVAr switchable line reactor for each circuit at Ghiror S/s end of 765 kV Ghiror -Orai D/c line 28. LILO of one ckt of 765kV Agra (PG) – Fatehpur(PG) 765kV D/c line at Ghiror along with 240 MVAr switchable line reactor at Ghiror S/s end of 765 kV Ghiror -Fatehpur line (29. 400kV Ghiror-Firozabad (UPPTCL) D/c line (Quad)										
Start Date of Connectivity under GNA: 31.03.2027(Interim). Final date shall be confirmed upon award of the system										

Sl. No.	Application ID	Applicant	Project Location	Application Date	Nature of Applicant	Criterion for applying	Injection at ISTS point (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)	Conn BGs to be submitted
7.	2200000289	Avaada Energy Private Limited	Bikaner distt., Rajasthan	12.10.2023	Generator (Solar)	Land BG Route	300	30.11.2025	Bhadla-III PS	• Conn BG1:Rs.50 lakh • Conn BG2: Nil(Bay in sharing) • Conn BG3: Rs. 2 lakh/MW

It was informed that M/s Avaada Energy Private Limited has applied for connectivity for 300 MW at Bhadla-III PS. Accordingly, it was proposed to grant connectivity of 300 MW at Bhadla-III PS at 220kV level through 1 no. of 220 kV line bay. M/s Avaada informed that Start date of Connectivity of 2028 timeframe is not suitable for their project timeline. As their application is through Land BG route, M/s Avaada requested to grant connectivity of 300 MW at Bikaner-IV PS in sharing with its other applications. CTUIL informed that as the application is through land BG route, it was agreed to grant connectivity at Bikaner-IV PS. Further, CTUIL also informed that as the total connectivity of 950 MW to M/s Avaada(50+600[lead]+300) at Bikaner-III PS, considering the techno-economical conditions, connectivity shall be granted at 400 kV through 1 no. of bay. M/s Avaada agreed for the same.

Further, it was also informed that M/s Avaada Energy Private Limited has requested Start Date of Connectivity under GNA from 31/03/2025. However, considering the same and expected commissioning date of transmission system (i.e. 31.08.2026), the start date of Connectivity under GNA shall be 01.09.2026 (interim). Proposed Transmission system for connectivity of Avaada Energy Private Limited for 300 MW under GNA is as below:

Details of Transmission system for Connectivity under GNA:

A. Associated Transmission System (ATS): NIL

B. Transmission System under applicant scope

- (i). Common Pooling station for Avaada RJ Bikaner Private Limited (App No. 220000013(50 MW), App. No. 2200000254(600 MW)) & Avaada Energy Private Limited (App No. 2200000289(300 MW)) Solar Power Projects – Bikaner-IV PS 400 kV S/c line on D/c tower (Suitable to carry minimum 950 MW at nominal voltage)

A. Transmission system for Connectivity under GNA

1. Establishment of 6x1500 MVA, 765/400 kV Bikaner-IV Pooling Station along with 2x240 MVar (765kV) & 2x125 MVar (420kV) Bus Reactors at a suitable location near Bikaner
2. 400kV Sectionalizer bay (1 set) at Bikaner-IV PS
3. STATCOM (2x±300MVar) along with MSC (4x125 MVar) & MSR (2x125 MVar) along with 2 nos. 400kV bays at Bikaner-IV PS
4. LILO of both ckts of 400kV Bikaner II PS- Bikaner III PS (Quad) direct line at Bikaner-IV PS
5. Establishment of 765/400 kV, 6x1500 MVA S/s at suitable location near Siwani (Distt. Bhiwani) along with 2x240 MVar (765kV) Bus Reactor & 2x125 MVar (420kV) Bus Reactor

Sl. No.	Application ID	Applicant	Project Location	Application Date	Nature of Applicant	Criterion for applying	Injection at ISTS point (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)	Conn BGs to be submitted
6. Bikaner-IV PS – Siwani 765 kV 2xD/c line along with 240 MVar switchable line reactor for each circuit at each end 7. Siwani – Jind (PG) 400 kV D/c line (Quad) 8. Siwani – Patran (Indi Grid) 400 kV D/c line (Quad) along with 80 MVar switchable line reactor for each circuit at Siwani S/s end 9. Siwani – Fatehabad (PG) 400 kV D/c line (Quad) 10. Siwani – Sonipat (PG) 400 kV D/c line (Quad) along with 63 MVar switchable line reactor for each circuit at Siwani S/s end 11. STATCOM (2x±300MVar) along with MSC (4x125 MVar) & MSR (2x125 MVar) along with 2 nos. 400kV bays at Siwani S/s 12. 400kV Sectionalizer bay (1 set) at at Siwani S/s B. Additional Transmission system for Inter-regional power transfer: 1. Establishment of 765kV Mandsaur Pooling Station along with 2x330 MVAR (765 kV) Bus Reactors 2. Beawar- Mandsaur PS 765 kV D/c line along with 240 MVar switchable line reactor for each circuit at each end 3a) Mandsaur PS – Indore(PG) 765 kV D/c Line along with 1x330 MVAR switchable line reactor on each ckt at Mandsaur end of Mandsaur PS – Indore(PG) 765 kV D/c Line OR 3b) Establishment of 765 kV Kurawar S/s with 2x330 MVar 765 kV bus reactor 3c) Mandsaur – Kurawar 765 kV D/c line along with 240 MVar switchable line reactors on each ckt at both ends of Mandsaur – Kurawar 765 kV D/c line 3d) LILO of Indore – Bhopal 765 kV S/c line at Kurawar Start Date of Connectivity under GNA: 01.09.2026(Interim). Final start date shall be confirmed upon award of the system.										
8.	2200000309	Renew Solar (Shakti Six) Private Limited	Jodhpur & Bikaner distt., Rajasthan	13.10.2023	Generator (Solar)	Land Route	585	30.06.2026	Bhadla-III PS	• Conn BG1:Rs.50 lakh • Conn BG2: Nil (Bay in sharing) • Conn BG3: Rs. 2 lakh/MW
It was informed that M/s Renew Solar (Shakti Six) Private Limited has applied for connectivity for 585 MW at Bhadla-III PS. It was also informed that the total connectivity agreed/granted at Bhadla-III PS (including 600 MW reserved capacity) is 6100 MW. As Bhadla-III PS was planned for evacuation of 6.5 GW as part of Transmission system for Rajasthan REZ Ph-III scheme (20 GW), only 400 MW connectivity margin is left out. In view of the above, due to technical limitations, connectivity of Renew Solar (Shakti Six) of 585 MW can be granted connectivity at Bhadla-IV PS. M/s Renew informed in the meeting that they wish to reduce the connectivity quantum & installed capacity to 400 MW so that they can be accommodated at Bhadla-III PS at 400 kV bays(2 no.) already granted at Bhadla-III PS										

Sl. No.	Application ID	Applicant	Project Location	Application Date	Nature of Applicant	Criterion for applying	Injection at ISTS point (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)	Conn BGs to be submitted
M/s Renew was asked to submit the formal request in this regard. CTU stated that the proposal may be agreed subject to M/s Renew submitting a formal request for reduction of connectivity quantum & installed capacity for the above application in this regard. M/s Renew agreed for the same and submitted the request on 11/01/24 for capacity reduction to 400 MW. Accordingly, it was proposed to grant connectivity of 400 MW (reduced from 585 MW) to M/s Renew through the earlier granted 400 kV D/c line for the combined capacity of 1800 MW. M/s Renew Solar (Shakti Six) Private Limited has requested Start Date of Connectivity under GNA from 30/06/2026. However, considering the same and expected commissioning date of transmission system (i.e. 31.08.2028), the start date of Connectivity under GNA shall be 01.09.2028 (interim). Proposed Transmission system for connectivity of M/s Renew Solar(Shakti Six) Pvt. Ltd. for 400 MW under GNA is as below: <u>Details of Transmission system for Connectivity under GNA:</u> A. <u>Associated Transmission System (ATS): NIL</u> B. <u>Transmission System under applicant scope</u> (i). Common Pooling Station for Renew Solar (Shakti Six) Private Limited (App. No. 120003848(550 MW), 0312100004(450 MW), 2200000275(400 MW) & 2200000309(400 MW)) Solar power projects – Bhadla-III PS 400 kV D/c line (Suitable to carry minimum 900 MW per circuit at nominal voltage) C. <u>Transmission system for Connectivity under GNA(Tentative)</u> 1. Augmentation of 2x1500 MVA (3rd & 4th), 765/400kV ICT at Bhadla-III PS 2. 765 kV Bhadla-III – Bikaner-III D/c line along with 240 MVar switchable line reactor for each circuit at Bhadla-III end 3. Establishment of 6000 MW, ± 800 kV Bhadla (HVDC) [LCC] terminal station (4x1500 MW) at a suitable location near Bhadla-III substation 4. Establishment of 6000 MW, ±800 kV Fatehpur (HVDC) [LCC] terminal station (4x1500 MW) at suitable location near Fatehpur (UP) 5. Bhadla-III – Bhadla (HVDC) 400 kV D/c (1st & 2nd) quad moose line along with the line bays at both substations 6. Bhadla-III – Sikar-II 765 kV D/c line along with 330 MVar Switchable line reactor for each circuit at each end of Bhadla-III – Sikar-II 765 kV D/c line 7. Fatehgarh-III – Bhadla-III 400kV D/c line (Quad) along with 63 MVar Switchable line reactor for each circuit at both ends of Fatehgarh-III- Bhadla-III 400kV D/c line 8. Augmentation of 5x1500 MVA (1st to 5th), 765/400 kV ICTs at Fatehpur (HVDC) along with 2x330 MVar (765kV) bus reactor 9. LILO of both ckts of 765 kV Varanasi – Kanpur (GIS) D/c line at Fatehpur 10. Augmentation by 1x1500 MVA ICT at 765/400kV Kanpur (GIS) substation Start Date of Connectivity under GNA: 01.09.2028(Interim). Final date shall be confirmed upon award of the system										

Sl. No.	Application ID	Applicant	Project Location	Application Date	Nature of Applicant	Criterion for applying	Injection at ISTS point (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)	Conn BGs to be submitted
9.	2200000284	AK Renewable Infra Private Limited	Bikaner distt., Rajasthan	13.10.2023	Generator (Solar)	Land Route	300	30.06.2025	Bikaner-I PS	• Conn BG1:Rs.50 lakh • Conn BG2: Nil(Bay in sharing) • Conn BG3: Rs. 2 lakh/MW

It was informed that M/s AK Renewable Infra Private Limited has applied for connectivity for 300 MW at Bikaner-I PS. It was also informed that at Bikaner-I PS, margin for grant of connectivity of 350 MW is available which can be granted only in sharing with M/s Renew Surya Ravi (550 MW) or M/s Ayana Renewable (600 MW) or with M/s Azure(600 MW) at 400 kV level. M/s AK Renewable informed that location of their project is closer to M/s Renew Surya Ravi and therefore requested CTUIL to grant connectivity at Bikaner PS in sharing with M/s Renew Surya Ravi(550 MW) and submit the sharing agreement M/s Renew also agreed for the same. Accordingly, it was proposed to grant connectivity of 300 MW to M/s AK Renewable Infra in sharing with M/s Renew Surya Ravi(550 MW) at 400kV Bikaner-I PS (Bikaner (PG). M/s AK Renewable infra agreed for the same.

Further, it was also informed that M/s AK Renewable Infra Private Limited has requested Start Date of Connectivity under GNA from 30/06/2025. However, considering the same and expected commissioning date of transmission system (i.e. 28.02.2026), the start date of Connectivity under GNA shall be 01.03.2026 (interim). M/s AK Renewable infra noted the same.

Details of Transmission system for connectivity of AK Renewable Infra Private Limited for 300 MW under GNA is as below:

Details of Transmission system for Connectivity under GNA:

A. Associated Transmission System (ATS): NIL

B. Transmission System under applicant scope

- Through sharing of dedicated transmission system of M/s Renew Surya Ravi Pvt. Ltd.– Bikaner (PG) 400 kV S/c line (Suitable to carry minimum 900 MW at nominal voltage)
- Infrastructure required for sharing shall be under the scope of M/s AK Renewable Infra Pvt. Ltd.

C. Transmission system for Connectivity under GNA

- Augmentation of 1x1500 MVA (4th) 765/400kV ICT at Bikaner (PG)
- Augmentation of 2x1500 MVA (5th & 6th) 765/400 kV ICTs at Bikaner-III Pooling Station
- LILO of both ckts of 400 kV Bikaner(PG)-Bikaner-II D/c line (Quad) at Bikaner-III PS
- Bikaner-II PS – Bikaner-III PS 400 kV D/c line (Quad)
- Bikaner-III - Neemrana-II 765 kV D/c line along with 330 MVar switchable line reactor for each circuit at each end
- Augmentation of 2x1500 MVA (3rd & 4th),765/400 kV ICTs at Neemrana-II S/s

Sl. No.	Application ID	Applicant	Project Location	Application Date	Nature of Applicant	Criterion for applying	Injection at ISTS point (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)	Conn BGs to be submitted
7. Neemrana-II -Kotputli 400 kV D/c line (Quad) 8. LILO of both ckts of 400 kV Gurgaon (PG) - Sohna Road (GPTL) D/c line (Quad) at Neemrana-II S/s 9. Bikaner-III - Neemrana-II 765 kV D/c line (2nd) along with 330 MVAR switchable line reactor for each circuit at each end 10. Neemrana-II- Bareilly (PG) 765 kV D/c line along with 330 MVAR switchable line reactor for each circuit at each end 11. Augmentation by 400/220 kV, 1x500 MVA ICT (3rd) at Kotputli (PG) 12. 400 kV Bareilly(765/400kV) – Bareilly(PG) D/c line (Quad) (2nd) 13. Augmentation with 1x1500 MVA, 765/400 kV ICT at Bareilly (765/400kV) S/s (3rd) D. Additional Transmission system for Inter-regional power transfer: 1. Establishment of 765kV Mandsaur Pooling Station along with 2x330 MVAR (765 kV) Bus Reactors 2. Beawar- Mandsaur PS 765 kV D/c line along with 240 MVAR switchable line reactor for each circuit at each end 3a) Mandsaur PS – Indore(PG) 765 kV D/c Line along with 1x330 MVAR switchable line reactor on each ckt at Mandsaur end of Mandsaur PS – Indore(PG) 765 kV D/c Line OR 3b) Establishment of 765 kV Kurawar S/s with 2x330 MVAR 765 kV bus reactor 3c) Mandsaur – Kurawar 765 kV D/c line along with 240 MVAR switchable line reactors on each ckt at both ends of Mandsaur – Kurawar 765 kV D/c line 3d) LILO of Indore – Bhopal 765 kV S/c line at Kurawar Start Date of Connectivity under GNA: 01.03.2026(Interim). Final date shall be confirmed upon award of the system										
10.	2200000312	Auxo Sunlight Private Limited	Barmer distt., Rajasthan	14.10.2023	Generator (Solar)	Land BG Route	300	30.09.2026	Barmer-I PS	• Conn BG1:Rs.50 lakh • Conn BG2: Rs. 3 Cr. • Conn BG3: Rs. 2 lakh/MW
It was informed that M/s Auxo Sunlight Private Limited has applied for connectivity for 300 MW at Barmer-I PS. Accordingly, it was proposed to grant connectivity at Barmer-I PS at 220 kV level through 1 no. of 220 kV line bay. M/s Auxo Sunlight Private Limited was asked to confirm the 220kV bay scope at Barmer-I PS end and the tower configuration. M/s Auxo Sunlight Private Limited informed that 220 kV line bay at Barmer-I end shall be under ISTS and tower configuration shall be S/c line on D/c tower. The same was noted & M/s Auxo Sunlight Private Limited was asked to submit necessary undertaking for S/c line on D/c tower configurations. M/s Auxo Sunlight Private Limited agreed for the same. Further, it was also informed that M/s Auxo Sunlight Private Limited has requested start Date of Connectivity under GNA from 30/09/2026. Considering the										

Sl. No.	Application ID	Applicant	Project Location	Application Date	Nature of Applicant	Criterion for applying	Injection at ISTS point (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)	Conn BGs to be submitted
same and expected commissioning date of transmission system(31.08.2026), the start date of Connectivity under GNA shall be 30.09.2026(interim). M/s Auxo sunlight noted the same. Accordingly, Transmission system for connectivity of M/s Auxo Sunlight Private Limited under GNA is as below: <u>Details of Transmission system for Connectivity under GNA:</u> A. <u>Associated Transmission System (ATS): NIL</u> B. <u>Transmission System under applicant scope</u> (i). Auxo Sunlight Private Limited Solar Power Project – Barmer-I PS 220 kV S/c line on D/c tower (Suitable to carry minimum 300 MW at nominal voltage) C. <u>Transmission system for Connectivity under GNA(tentative)</u> 1. Augmentation of 6x500 MVA (4th to 9th), 400/220 kV ICTs at Barmer-I PS 2. Augmentation with 765/400 kV, 3x1500 MVA Transformer (3rd, 4th & 5th) at Barmer-I PS 3. Fatehgarh-III (Section-2) PS – Barmer-I PS 400 kV D/c line (Quad) 4. Establishment of 2x1500 MVA, 765/400 kV Substation at suitable location near Sirohi along with 2x240 MVAR (765 kV) & 2x125 MVAR (400 kV) Bus Reactor 5. Barmer-I PS– Sirohi PS 765 kV D/c line along with 240 MVAR switchable line reactor for each circuit at each end 6. Sirohi PS-Chittorgarh (PG) 400 kV D/c line along with 80 MVAR switchable line reactor for each circuit at Sirohi PS end (Quad) 7. Establishment of, 765 kV Substation at suitable location near Rishabdeo (Distt. Udaipur) along with 2x240 MVAR (765 kV) Bus Reactor 8. Sirohi PS- Rishabdeo 765 kV D/c line along with 330 MVAR switchable line reactor for each circuit at Sirohi end 9. Rishabdeo - Mandsaur PS 765 kV D/c line along with 240 MVAR switchable line reactor for each circuit at Rishabdeo end 10. LILO of one circuit of 765 kV Chittorgarh-Banaskanta D/c line at Rishabdeo S/s 11. Establishment of 765 kV Mandsaur PS along with 2x330 MVAR (765 kV) Bus Reactors 12. Mandsaur PS–Indore(PG) 765 kV D/c Line along with 1x330 MVAR switchable line reactor on each ckt at Mandsaur end of Mandsaur PS – Indore(PG) 765 kV D/c Line 13. Establishment of 765/400 kV (2x1500 MVA Kurawar S/s with 2x330 MVAR 765 kV bus reactor and 1x125 MVAR 420 kV bus reactor 14. Mandsaur – Kurawar 765 kV D/c line along with 240 MVAR switchable line reactors on each ckt at both ends of Mandsaur – Kurawar 765 kV D/c line 15. Kurawar – Ashtha 400 kV D/c (Quad ACSR/AAAC/AL59 moose equivalent) line 16. LILO of Indore – Bhopal 765 kV S/c line at Kurawar										

Sl. No.	Application ID	Applicant	Project Location	Application Date	Nature of Applicant	Criterion for applying	Injection at ISTS point (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)	Conn BGs to be submitted
17. LILO of one circuit of Indore – Itarsi 400 kV D/c line at Astha 18. Shujalpur – Kurawar 400 kV D/c (Quad ACSR/AAAC/AL59 moose equivalent) line 19. Establishment of 765/400kV, 2x1500 MVA S/s at suitable location near Merta (Merta-II Substation) along with 2x125 MVar & 2x240 MVar bus reactor at Merta-II S/s 20. STATCOM (2x±300MVar) along with MSC (4x125 MVar) & MSR (2x125 MVar) along with 2 nos. 400kV bays at Barmer-I PS 21. 400kV Sectionalizer bay (1 set) at Barmer-I S/s 22. Barmer-I PS – Merta-II 765 kV D/c line along with 240 MVar switchable line reactor at Barmer-I PS end and 330MVar switchable line reactor at Merta-II PS end for each circuit of Barmer-I PS – Merta-II 765 kV D/c line 23. Merta-II – Beawar 400 kV D/c line (Quad) 24. Merta-II – Dausa 765 kV D/c line along with 240 MVar switchable line reactor for each circuit at each end of Merta-II – Dausa 765 kV D/c line line 25. Establishment of 765/400kV, 2x1500 MVA S/s at suitable location near Ghiror (Distt. Mainpuri) along with 2x125 MVar & 2x240 MVar bus reactor at Ghiror S/s (UP) 26. Dausa - Ghiror 765 kV D/c line along with 240 MVar switchable line reactor for each circuit at each end of Dausa - Ghiror 765 kV D/c line 27. LILO of both ckt of 765 kV Aligarh (PG) -Orai (PG) D/c line at Ghiror S/s along with 240 MVar switchable line reactor for each circuit at Ghiror S/s end of 765 kV Ghiror -Orai D/c line 28. LILO of one ckt of 765kV Agra (PG) – Fatehpur(PG) 765kV D/c line at Ghiror along with 240 MVar switchable line reactor at Ghiror S/s end of 765 kV Ghiror -Fatehpur line (29. 400kV Ghiror-Firozabad (UPPTCL) D/c line (Quad)										
Start Date of Connectivity under GNA: 30.09.2026(Interim). Final date shall be confirmed upon award of the system										
11.	2200000290 (Enhancement)	Avaada Energy Private Limited	Barmer distt., Rajasthan	17.10.2023	Generator (Hybrid)	Land BG Route	250 (Solar-125 Wind-125)	30.09.2025	Fatehgarh-IV PS	• Conn BG1:Rs.50 lakh • Conn BG2: Rs. 3 Cr. • Conn BG3: Rs. 2 lakh/MW
It was informed that M/s Avaada Energy Private Limited has applied for connectivity for 250 MW at Fatehgarh-IV PS. As informed earlier, in the reallocation meeting held on 02.01.2024, based on the request of M/s Avaada, it was decided that the connectivity of Avaada Energy Pvt. Ltd. (50 MW) granted at Fatehgarh-IV PS(Sec-II) in sharing with M/s Radiant(200 MW) shall be shifted to Barmer-I PS and granted connectivity along with present application of M/s Avaada(250 MW). Considering margins non availability at Fatehgarh-IV PS (220kV), it was proposed to grant connectivity at Barmer-I PS at 220 kV through 1 no. of 220 kV bay for combined capacity of 300 MW (50 +250 MW), M/s Avaada informed that 220kV bay shall be in ISTS with DTL having S/c line on D/c tower and 250 MW application will be the lead. M/s Avaada agreed to submit undertaking for above DTL configuration (S/c on D/c).										

Sl. No.	Application ID	Applicant	Project Location	Application Date	Nature of Applicant	Criterion for applying	Injection at ISTS point (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)	Conn BGs to be submitted
---------	----------------	-----------	------------------	------------------	---------------------	------------------------	------------------------------	---	--	--------------------------

Further, it was also informed that M/s Avaada Energy Private Limited has requested Start Date of Connectivity under GNA from 30/09/2025. Considering the same and expected commissioning date of transmission system (i.e. 31.08.2026), the start date of Connectivity under GNA shall be 01.09.2026(interim). M/s Avaada noted the same.

Accordingly, transmission system for connectivity of M/s Avaada Energy Private Limited under GNA is as below:

Details of Transmission system for Connectivity under GNA:

A. Associated Transmission System (ATS): NIL

B. Transmission System under applicant scope

- (i). Avaada Energy Private Limited Hybrid Power Project – Barmer-I PS 220 kV S/c line on D/c tower(Suitable to carry minimum 300 MW at nominal voltage)

C. Transmission system for Connectivity under GNA (tentative)

1. Augmentation of 5x500 MVA (5th to 9th), 400/220 kV ICTs at Barmer-I PS
2. Augmentation with 765/400 kV, 3x1500 MVA Transformer (3rd, 4th & 5th) at Barmer-I PS
3. Fatehgarh-III (Section-2) PS – Barmer-I PS 400 kV D/c line (Quad)
4. Establishment of 2x1500 MVA, 765/400 kV Substation at suitable location near Sirohi along with 2x240 MVAR (765 kV) & 2x125 MVAR (400 kV) Bus Reactor
5. Barmer-I PS– Sirohi PS 765 kV D/c line along with 240 MVAR switchable line reactor for each circuit at each end
6. Sirohi PS-Chittorgarh (PG) 400 kV D/c line along with 80 MVAR switchable line reactor for each circuit at Sirohi PS end (Quad)
7. Establishment of, 765 kV Substation at suitable location near Rishabdeo (Distt. Udaipur) along with 2x240 MVAR (765 kV) Bus Reactor
8. Sirohi PS- Rishabdeo 765 kV D/c line along with 330 MVAR switchable line reactor for each circuit at Sirohi end
9. Rishabdeo - Mandsaur PS 765 kV D/c line along with 240 MVAR switchable line reactor for each circuit at Rishabdeo end
10. LILO of one circuit of 765 kV Chittorgarh-Banaskanta D/c line at Rishabdeo S/s
11. Establishment of 765 kV Mandsaur PS along with 2x330 MVAR (765 kV) Bus Reactors
12. Mandsaur PS–Indore(PG) 765 kV D/c Line along with 1x330 MVAR switchable line reactor on each ckt at Mandsaur end of Mandsaur PS – Indore(PG) 765 kV D/c Line
13. Establishment of 765/400 kV (2x1500 MVA Kurawar S/s with 2x330 MVAR 765 kV bus reactor and 1x125 MVAR 420 kV bus reactor
14. Mandsaur – Kurawar 765 kV D/c line along with 240 MVAR switchable line reactors on each ckt at both ends of Mandsaur – Kurawar 765 kV D/c line
15. Kurawar – Ashtha 400 kV D/c (Quad ACSR/AAAC/AL59 moose equivalent) line

Sl. No.	Application ID	Applicant	Project Location	Application Date	Nature of Applicant	Criterion for applying	Injection at ISTS point (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)	Conn BGs to be submitted
16. LILO of Indore – Bhopal 765 kV S/c line at Kurawar 17. LILO of one circuit of Indore – Itarsi 400 kV D/c line at Astha 18. Shujalpur – Kurawar 400 kV D/c (Quad ACSR/AAAC/AL59 moose equivalent) line 19. Establishment of 765/400kV, 2x1500 MVA S/s at suitable location near Merta (Merta-II Substation) along with 2x125 MVar & 2x240 MVar bus reactor at Merta-II S/s 20. STATCOM (2x±300MVar) along with MSC (4x125 MVar) & MSR (2x125 MVar) along with 2 nos. 400kV bays at Barmer-I PS 21. 400kV Sectionalizer bay (1 set) at Barmer-I S/s 22. Barmer-I PS – Merta-II 765 kV D/c line along with 240 MVar switchable line reactor at Barmer-I PS end and 330MVar switchable line reactor at Merta-II PS end for each circuit of Barmer-I PS – Merta-II 765 kV D/c line 23. Merta-II – Beawar 400 kV D/c line (Quad) 24. Merta-II – Dausa 765 kV D/c line along with 240 MVar switchable line reactor for each circuit at each end of Merta-II – Dausa 765 kV D/c line 25. Establishment of 765/400kV, 2x1500 MVA S/s at suitable location near Ghiror (Distt. Mainpuri) along with 2x125 MVar & 2x240 MVar bus reactor at Ghiror S/s (UP) 26. Dausa - Ghiror 765 kV D/c line along with 240 MVar switchable line reactor for each circuit at each end of Dausa - Ghiror 765 kV D/c line 27. LILO of both ckt of 765 kV Aligarh (PG) -Orai (PG) D/c line at Ghiror S/s along with 240 MVar switchable line reactor for each circuit at Ghiror S/s end of 765 kV Ghiror -Orai D/c line 28. LILO of one ckt of 765kV Agra (PG) – Fatehpur(PG) 765kV D/c line at Ghiror along with 240 MVar switchable line reactor at Ghiror S/s end of 765 kV Ghiror -Fatehpur line (29. 400kV Ghiror-Firozabad (UPPTCL) D/c line (Quad)										
Start Date of Connectivity under GNA: 01.09.2026(Interim). Final date shall be confirmed upon award of the system										
12.	2200000316	Renew Sun Power Private Limited	Barmer & Jaisalmer distt., Rajasthan	17.10.2023	Generator (Solar)	Land BG Route	300	30.09.2026	Barmer-I PS	• Conn BG1:50 lakh • Conn BG2: Rs. 3 Cr. • Conn BG3: 2 lakh/MW
It was informed that M/s Renew Sun Power Private Limited has applied for connectivity for 300 MW at Barmer-I PS. Accordingly, it was proposed to grant connectivity at Barmer-I PS at 220 kV level through 1 no. of 220 kV line bay. M/s Renew Sun Power Private Limited was asked to confirm the 220kV bay scope at Barmer-I PS end and the tower configuration. M/s Renew Sun Power Private Limited informed that 220 kV line bay at Barmer-I end shall be under ISTS and tower configuration shall be S/c line on D/c tower. The same was noted & M/s Renew Sun Power Private Limited was asked to submit the necessary undertaking for S/c on D/c tower configuration. M/s Renew Sun Power Private Limited agreed for the same. Further, it was also informed that M/s Renew Sun Power Private Limited has requested start Date of Connectivity under GNA from 30/09/2026. Considering the same and expected commissioning date of transmission system (31.08.2026), the start date of Connectivity under GNA shall be 30.09.2026(interim). M/s										

Sl. No.	Application ID	Applicant	Project Location	Application Date	Nature of Applicant	Criterion for applying	Injection at ISTS point (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)	Conn BGs to be submitted
Renew Sun Power noted the same.										
Accordingly, Transmission system for connectivity of M/s Renew Sun Power under GNA is as below:										
<u>Details of Transmission system for Connectivity under GNA:</u>										
<u>A. Associated Transmission System (ATS): NIL</u>										
<u>B. Transmission System under applicant scope</u>										
(i). Renew Sun Power Private Limited Solar Power Project– Barmer-I PS 220 kV S/c line on D/c tower(Suitable to carry minimum 300 MW at nominal voltage)										
<u>C. Transmission system for Connectivity under GNA(tentative)</u>										
1. Augmentation of 5x500 MVA (5th to 9th), 400/220 kV ICTs at Barmer-I PS										
2. Augmentation with 765/400 kV, 3x1500 MVA Transformer (3rd, 4th & 5th) at Barmer-I PS										
3. Fatehgarh-III (Section-2) PS – Barmer-I PS 400 kV D/c line (Quad)										
4. Establishment of 2x1500 MVA, 765/400 kV Substation at suitable location near Sirohi along with 2x240 MVar (765 kV) & 2x125 MVar (400 kV) Bus Reactor										
5. Barmer-I PS– Sirohi PS 765 kV D/c line along with 240 MVAR switchable line reactor for each circuit at each end										
6. Sirohi PS-Chittorgarh (PG) 400 kV D/c line along with 80 MVar switchable line reactor for each circuit at Sirohi PS end (Quad)										
7. Establishment of, 765 kV Substation at suitable location near Rishabdeo (Distt. Udaipur) along with 2x240 MVar (765 kV) Bus Reactor										
8. Sirohi PS- Rishabdeo 765 kV D/c line along with 330 MVar switchable line reactor for each circuit at Sirohi end										
9. Rishabdeo - Mandsaur PS 765 kV D/c line along with 240 MVar switchable line reactor for each circuit at Rishabdeo end										
10. LILO of one circuit of 765 kV Chittorgarh-Banaskanta D/c line at Rishabdeo S/s										
11. Establishment of 765 kV Mandsaur PS along with 2x330 MVAR (765 kV) Bus Reactors										
12. Mandsaur PS–Indore(PG) 765 kV D/c Line along with 1x330 MVAR switchable line reactor on each ckt at Mandsaur end of Mandsaur PS – Indore(PG) 765 kV D/c Line										
13. Establishment of 765/400 kV (2x1500 MVA Kurawar S/s with 2x330 MVar 765 kV bus reactor and 1x125 MVar 420 kV bus reactor										
14. Mandsaur – Kurawar 765 kV D/c line along with 240 MVar switchable line reactors on each ckt at both ends of Mandsaur – Kurawar 765 kV D/c line										
15. Kurawar – Ashtha 400 kV D/c (Quad ACSR/AAAC/AL59 moose equivalent) line										
16. LILO of Indore – Bhopal 765 kV S/c line at Kurawar										
17. LILO of one circuit of Indore – Itarsi 400 kV D/c line at Astha										

Sl. No.	Application ID	Applicant	Project Location	Application Date	Nature of Applicant	Criterion for applying	Injection at ISTS point (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)	Conn BGs to be submitted
18. Shujalpur – Kurawar 400 kV D/c (Quad ACSR/AAAC/AL59 moose equivalent) line 19. Establishment of 765/400kV, 2x1500 MVA S/s at suitable location near Merta (Merta-II Substation) along with 2x125 MVar & 2x240 MVar bus reactor at Merta-II S/s 20. STATCOM (2x+300MVar) along with MSC (4x125 MVar) & MSR (2x125 MVar) along with 2 nos. 400kV bays at Barmer-I PS 21. 400kV Sectionalizer bay (1 set) at Barmer-I S/s 22. Barmer-I PS – Merta-II 765 kV D/c line along with 240 MVar switchable line reactor at Barmer-I PS end and 330MVar switchable line reactor at Merta-II PS end for each circuit of Barmer-I PS – Merta-II 765 kV D/c line 23. Merta-II – Beawar 400 kV D/c line (Quad) 24. Merta-II – Dausa 765 kV D/c line along with 240 MVar switchable line reactor for each circuit at each end of Merta-II – Dausa 765 kV D/c line line 25. Establishment of 765/400kV, 2x1500 MVA S/s at suitable location near Ghiror (Distt. Mainpuri) along with 2x125 MVar & 2x240 MVar bus reactor at Ghiror S/s (UP) 26. Dausa - Ghiror 765 kV D/c line along with 240 MVar switchable line reactor for each circuit at each end of Dausa - Ghiror 765 kV D/c line 27. LILO of both ckt of 765 kV Aligarh (PG) -Orai (PG) D/c line at Ghiror S/s along with 240 MVar switchable line reactor for each circuit at Ghiror S/s end of 765 kV Ghiror -Orai D/c line 28. LILO of one ckt of 765kV Agra (PG) – Fatehpur(PG) 765kV D/c line at Ghiror along with 240 MVar switchable line reactor at Ghiror S/s end of 765 kV Ghiror -Fatehpur line (29. 400kV Ghiror-Firozabad (UPPTCL) D/c line (Quad)										
Start Date of Connectivity under GNA: 30.09.2026(Interim). Final date shall be confirmed upon award of the system										
13.	2200000327	Hazel Hybren Private Limited	Jaisalmer distt., Rajasthan	20.10.2023	Generator (Solar)	Land BG Route	300	31.01.2026	Ramgarh PS	• Conn BG1:NA • Conn BG2: NA • Conn BG3: NA
It was informed that M/s Hazel Hybren Private Limited has applied for connectivity for 300 MW at Ramgarh PS. Accordingly, it is proposed to grant connectivity at Ramgarh PS at 220 kV level through 1 no. of 220 kV line bay. M/s Hazel Hybren informed that the 2028 timeline of Start date of Connectivity with Bhadla(HVDC) system is not viable for their project. In view of the above, M/s Hazel Hybren informed that they withdraw their application for grant of connectivity. The same was communicated vide mail dated 10.01.2024. Accordingly, the application shall be closed and treatment shall be as per GNA Regulations.										
14.	2200000306	Shudh Solar Power Private Limited	Bikaner distt., Rajasthan	21.10.2023	Generator (Solar)	Land Route	250	31.05.2025	Bikaner-III PS	• Conn BG1:50 lakh • Conn BG2: Rs. 3 Cr. • Conn BG3: 2 lakh/MW

Sl. No.	Application ID	Applicant	Project Location	Application Date	Nature of Applicant	Criterion for applying	Injection at ISTS point (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)	Conn BGs to be submitted
It was informed that M/s Shudh Solar Power Private Limited has applied for connectivity for 250 MW at Bikaner-III PS. However, as there is no margin available for grant of connectivity at Bikaner-III PS, therefore it was proposed to grant connectivity of 250 MW at Bikaner-IV PS at 220 kV level through 1 no. of 220 kV line bay. M/s Shudh Solar Power Private Limited was asked to confirm the 220kV bay scope at Bikaner-IV PS end and the tower configuration. M/s Shudh Solar Power Private Limited informed that 220 kV line bay at Bikaner-IV end shall be under ISTS and tower configuration shall be S/c line on D/c tower. The same was noted & M/s Shudh Solar Power Private Limited was asked to submit the necessary undertaking for S/c line on D/c tower configuration. M/s Shudh Solar Power Private Limited agreed for the same. Further, it was also informed that M/s Shudh Solar Power Private Limited has requested start Date of Connectivity under GNA from 31/05/2025. Considering the same and expected commissioning date of transmission system(31.08.2026), the start date of Connectivity under GNA shall be 01.09.2026(interim). M/s Shudh Solar noted the same. Accordingly, Transmission system for connectivity of M/s Shudh Solar Power under GNA is as below: <u>Details of Transmission system for Connectivity under GNA:</u> A. <u>Associated Transmission System (ATS): NIL</u> B. <u>Transmission System under applicant scope</u> (i). Shudh Solar Power Private Limited Solar Power Project– Bikaner-IV PS 220 kV S/c line on D/c tower(Suitable to carry minimum 300 MW at nominal voltage) C. <u>Transmission system for Connectivity under GNA (tentative)</u> 1. Establishment of 6x1500 MVA, 765/400 kV & 6x500 MVA, 400/220 kV Bikaner-IV Pooling Station along with 2x240 MVA (765kV) & 2x125 MVA (420kV) Bus Reactors at a suitable location near Bikaner 2. 400kV Sectionalizer bay (1 set) at Bikaner-IV PS 3. STATCOM (2x+300MVA) along with MSC (4x125 MVA) & MSR (2x125 MVA) along with 2 nos. 400kV bays at Bikaner-IV PS 4. LILO of both ckts of 400kV Bikaner II PS- Bikaner III PS (Quad) direct line at Bikaner-IV PS 5. Establishment of 765/400 kV, 6x1500 MVA S/s at suitable location near Siwani (Distt. Bhiwani) along with 2x240 MVA (765kV) Bus Reactor & 2x125 MVA (420kV) Bus Reactor 6. Bikaner-IV PS – Siwani 765 kV 2xD/c line along with 240 MVA switchable line reactor for each circuit at each end 7. Siwani – Jind (PG) 400 kV D/c line (Quad) 8. Siwani – Patran (Indi Grid) 400 kV D/c line (Quad) along with 80 MVA switchable line reactor for each circuit at Siwani S/s end 9. Siwani – Fatehabad (PG) 400 kV D/c line (Quad)										

Sl. No.	Application ID	Applicant	Project Location	Application Date	Nature of Applicant	Criterion for applying	Injection at ISTS point (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)	Conn BGs to be submitted
10. Siwani – Sonipat (PG) 400 kV D/c line (Quad) along with 63 MVAR switchable line reactor for each circuit at Siwani S/s end 11. STATCOM (2x±300MVAR) along with MSC (4x125 MVAR) & MSR (2x125 MVAR) along with 2 nos. 400kV bays at Siwani S/s 12. 400kV Sectionalizer bay (1 set) at at Siwani S/s D. <u>Additional Transmission system for Inter-regional power transfer:</u> 1. Establishment of 765kV Mandsaur Pooling Station along with 2x330 MVAR (765 kV) Bus Reactors 2. Beawar- Mandsaur PS 765 kV D/c line along with 240 MVAR switchable line reactor for each circuit at each end 3a) Mandsaur PS – Indore(PG) 765 kV D/c Line along with 1x330 MVAR switchable line reactor on each ckt at Mandsaur end of Mandsaur PS – Indore(PG) 765 kV D/c Line OR 3b) Establishment of 765 kV Kurawar S/s with 2x330 MVAR 765 kV bus reactor 3c) Mandsaur – Kurawar 765 kV D/c line along with 240 MVAR switchable line reactors on each ckt at both ends of Mandsaur – Kurawar 765 kV D/c line 3d) LILO of Indore – Bhopal 765 kV S/c line at Kurawar Start Date of Connectivity under GNA: 01.09.2026(Interim). Final date shall be confirmed upon award of the system										
15.	2200000308	NTPC Renewable Energy Limited	Bikaner distt., Rajasthan	21.10.2023	Generator (Solar)	Land Route	700	31.03.2026	Bikaner-III PS	• Conn BG1:Rs. 50 lakh • Conn BG2: Rs. 6 Cr. • Conn BG3: Rs. 2 lakh/MW
It was informed that M/s NTPC Renewable Energy Limited(NTPCREL) has applied for connectivity for 700 MW at Bikaner-III PS. As discussed earlier, it was proposed to grant connectivity of 700 MW along with other applications of 150 MW, 400 MW & 250 MW by NTPCREL(total 1500 MW) at Bikaner-IV PS through 400 kV D/c line. Further, it was also informed that M/s NTPC Renewable Energy Limited has requested Start Date of Connectivity under GNA from 31/03/2026. However, considering the same and expected commissioning date of transmission system (i.e. 31.08.2026), the start date of Connectivity under GNA shall be 01.09.2026 (interim). M/s NTPC Renewables noted the same. Accordingly, transmission system for connectivity of NTPC Renewable Energy Limited for 700 MW (lead application for 700+150 MW) under GNA is as below: <u>Details of Transmission system for Connectivity under GNA:</u> A. <u>Associated Transmission System (ATS):</u> NIL										

Sl. No.	Application ID	Applicant	Project Location	Application Date	Nature of Applicant	Criterion for applying	Injection at ISTS point (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)	Conn BGs to be submitted
B. <u>Transmission System under applicant scope</u> (i). Common pooling station for NTPC Renewable Energy Limited Solar Power Projects (App No. 2200000244(150 MW) , 2200000308(700 MW), 2200000373(400 MW) & 2200000358(250 MW) – Bikaner-IV PS 400 kV D/c line (Suitable to carry minimum 900 MW per circuit at nominal voltage) C. <u>Transmission system for Connectivity under GNA (for 700 MW) (tentative)</u> 1. Establishment of 6x1500 MVA, 765/400 kV Bikaner-IV Pooling Station along with 2x240 MVar (765kV) & 2x125 MVar (420kV) Bus Reactors at a suitable location near Bikaner 2. 400kV Sectionalizer bay (1 set) at Bikaner-IV PS 3. STATCOM (2x+300MVar) along with MSC (4x125 MVar) & MSR (2x125 MVar) along with 2 nos. 400kV bays at Bikaner-IV PS 4. LILO of both ckts of 400kV Bikaner II PS- Bikaner III PS (Quad) direct line at Bikaner-IV PS 5. Establishment of 765/400 kV, 6x1500 MVA S/s at suitable location near Siwani (Distt. Bhiwani) along with 2x240 MVar (765kV) Bus Reactor & 2x125 MVar (420kV) Bus Reactor 6. Bikaner-IV PS – Siwani 765 kV 2xD/c line along with 240 MVar switchable line reactor for each circuit at each end 7. Siwani – Jind (PG) 400 kV D/c line (Quad) 8. Siwani – Patran (Indi Grid) 400 kV D/c line (Quad) along with 80 MVar switchable line reactor for each circuit at Siwani S/s end 9. Siwani – Fatehabad (PG) 400 kV D/c line (Quad) 10. Siwani – Sonipat (PG) 400 kV D/c line (Quad) along with 63 MVar switchable line reactor for each circuit at Siwani S/s end 11. STATCOM (2x+300MVar) along with MSC (4x125 MVar) & MSR (2x125 MVar) along with 2 nos. 400kV bays at Siwani S/s 12. 400kV Sectionalizer bay (1 set) at at Siwani S/s D. <u>Additional Transmission system for Inter-regional power transfer:</u> 1. Establishment of 765kV Mandsaur Pooling Station along with 2x330 MVAR (765 kV) Bus Reactors 2. Beawar- Mandsaur PS 765 kV D/c line along with 240 MVar switchable line reactor for each circuit at each end 3a) Mandsaur PS – Indore(PG) 765 kV D/c Line along with 1x330 MVAR switchable line reactor on each ckt at Mandsaur end of Mandsaur PS – Indore(PG) 765 kV D/c Line OR 3b) Establishment of 765 kV Kurawar S/s with 2x330 MVar 765 kV bus reactor 3c) Mandsaur – Kurawar 765 kV D/c line along with 240 MVar switchable line reactors on each ckt at both ends of Mandsaur – Kurawar 765 kV D/c line										

Sl. No.	Application ID	Applicant	Project Location	Application Date	Nature of Applicant	Criterion for applying	Injection at ISTS point (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)	Conn BGs to be submitted
3d) LILO of Indore – Bhopal 765 kV S/c line at Kurawar										
Start Date of Connectivity under GNA: 01.09.2026(Interim). Final date shall be confirmed upon award of the system										
16.	2200000334	Hazel Hybren Private Limited	Bikaner distt., Rajasthan	23.10.2023	Generator (Solar)	Land BG Route	300	31.01.2026	Bikaner-III PS	• Conn BG1:50 lakh • Conn BG2: Rs. 3 Cr. • Conn BG3: 2 lakh/MW
It was informed that M/s Hazel Hybren Private Limited has applied for connectivity for 300 MW at Bikaner-III PS. As there is no margin available for grant of connectivity at Bikaner-III PS, it was proposed to grant connectivity of 300 MW at Bikaner-IV PS at 220 kV level through 1 no. of 220 kV line bay. M/s Hazel Hybren agreed for the same. M/s Hazel Hybren Private Limited was asked to confirm the 220kV bay scope at Bikaner-IV PS end and the tower configuration. M/s Hazel Hybren Private Limited informed that 220 kV line bay at Bikaner-IV end shall be under ISTS and tower configuration shall be S/c line on D/c tower. The same was noted & M/s Hazel Hybren Private Limited was asked to submit the necessary undertaking for S/c line on D/c tower configurations. M/s Hazel Hybren agreed for the same. Further, it was also informed that M/s Hazel Hybren Private Limited has requested start Date of Connectivity under GNA from 31/01/2026. Considering the same and expected commissioning date of transmission system (31.08.2026), the start date of Connectivity under GNA shall be 01.09.2026(interim). M/s Hazel Hybren noted the same. Accordingly, Transmission system for connectivity of M/s Hazel Hybren under GNA is as below: <u>Details of Transmission system for Connectivity under GNA:</u> A. <u>Associated Transmission System (ATS):</u> NIL B. <u>Transmission System under applicant scope</u> (i). Hazel Hybren Private Limited Solar Power Project– Bikaner-IV PS 220 kV S/c line on D/c tower (Suitable to carry minimum 300 MW at nominal voltage) A. <u>Transmission system for Connectivity under GNA</u> 1. Establishment of 6x1500 MVA, 765/400 kV & 6x500 MVA, 400/220 kV Bikaner-IV Pooling Station along with 2x240 MVA (765kV) & 2x125 MVA (420kV) Bus Reactors at a suitable location near Bikaner 2. 400kV Sectionalizer bay (1 set) at Bikaner-IV PS										

Sl. No.	Application ID	Applicant	Project Location	Application Date	Nature of Applicant	Criterion for applying	Injection at ISTS point (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)	Conn BGs to be submitted
3. STATCOM (2x±300MVar) along with MSC (4x125 MVar) & MSR (2x125 MVar) along with 2 nos. 400kV bays at Bikaner-IV PS 4. LILO of both ckts of 400kV Bikaner II PS- Bikaner III PS (Quad) direct line at Bikaner-IV PS 5. Establishment of 765/400 kV, 6x1500 MVA S/s at suitable location near Siwani (Distt. Bhiwani) along with 2x240 MVar (765kV) Bus Reactor & 2x125 MVar (420kV) Bus Reactor 6. Bikaner-IV PS – Siwani 765 kV 2xD/c line along with 240 MVar switchable line reactor for each circuit at each end 7. Siwani – Jind (PG) 400 kV D/c line (Quad) 8. Siwani – Patran (Indi Grid) 400 kV D/c line (Quad) along with 80 MVar switchable line reactor for each circuit at Siwani S/s end 9. Siwani – Fatehabad (PG) 400 kV D/c line (Quad) 10. Siwani – Sonipat (PG) 400 kV D/c line (Quad) along with 63 MVar switchable line reactor for each circuit at Siwani S/s end 11. STATCOM (2x±300MVar) along with MSC (4x125 MVar) & MSR (2x125 MVar) along with 2 nos. 400kV bays at Siwani S/s 12. 400kV Sectionalizer bay (1 set) at at Siwani S/s C. Additional Transmission system for Inter-regional power transfer: 1. Establishment of 765kV Mandsaur Pooling Station along with 2x330 MVAR (765 kV) Bus Reactors 2. Beawar- Mandsaur PS 765 kV D/c line along with 240 MVar switchable line reactor for each circuit at each end 3a) Mandsaur PS – Indore(PG) 765 kV D/c Line along with 1x330 MVAR switchable line reactor on each ckt at Mandsaur end of Mandsaur PS – Indore(PG) 765 kV D/c Line OR 3b) Establishment of 765 kV Kurawar S/s with 2x330 MVar 765 kV bus reactor 3c) Mandsaur – Kurawar 765 kV D/c line along with 240 MVar switchable line reactors on each ckt at both ends of Mandsaur – Kurawar 765 kV D/c line 3d) LILO of Indore – Bhopal 765 kV S/c line at Kurawar Start Date of Connectivity under GNA: 01.09.2026(Interim). Final date shall be confirmed upon award of the system										
17.	2200000319	AM Green Energy Private Limited	Bikaner distt., Rajasthan	23.10.2023	Generator (Solar)	Land BG Route	300	31.05.2025	Bikaner-II PS	• Conn BG1:50 lakh • Conn BG2: Rs. 3 Cr. • Conn BG3: 2 lakh/MW
It was informed that M/s AM Green Energy Private Limited has applied for connectivity for 300 MW at Bikaner-II PS. However, as informed in the 23rd CMETS NR meeting, 675 MW of margin available at Bikaner-II PS cannot be granted to any entity till further order by CERC. Further, there is no margin available for grant of connectivity at Bikaner-III PS as well. Therefore, it was proposed to grant connectivity at Bikaner-IV PS at 220 kV level through 1 no. of 220 kV line bay. M/s AM Green Energy agreed for the same. M/s AM Green Energy Private Limited was asked to confirm the 220kV bay scope at Bikaner-IV PS end										

Sl. No.	Application ID	Applicant	Project Location	Application Date	Nature of Applicant	Criterion for applying	Injection at ISTS point (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)	Conn BGs to be submitted
and the tower configuration. M/s AM Green Energy Private Limited informed that 220 kV line bay at Bikaner-IV end shall be under ISTS and tower configuration shall be S/c line on D/c tower. The same was noted & M/s AM Green Energy Private Limited was asked to submit the necessary undertaking for S/c on D/c tower configuration. M/s AM Green Energy Private Limited agreed for the same. Further, it was also informed that M/s AM Green Energy Private Limited has requested start Date of Connectivity under GNA from 31/05/2025. Considering the same and expected commissioning date of transmission system (31.08.2026), the start date of Connectivity under GNA shall be 01.09.2026(interim). M/s AM Green Energy noted the same. Accordingly, Transmission system for connectivity of M/s AM Green Energy under GNA is as below: <u>Details of Transmission system for Connectivity under GNA:</u> A. <u>Associated Transmission System (ATS): NIL</u> B. <u>Transmission System under applicant scope</u> (i). AM Green Energy Private Limited Solar Power Project – Bikaner-IV PS 220 kV S/c line on D/c tower (Suitable to carry minimum 300 MW at nominal voltage) C. <u>Transmission system for Connectivity under GNA</u> 1. Establishment of 6x1500 MVA, 765/400 kV & 6x500 MVA, 400/220 kV Bikaner-IV Pooling Station along with 2x240 MVA (765kV) & 2x125 MVA (420kV) Bus Reactors at a suitable location near Bikaner 2. 400kV Sectionalizer bay (1 set) at Bikaner-IV PS 3. STATCOM (2x+300MVA) along with MSC (4x125 MVA) & MSR (2x125 MVA) along with 2 nos. 400kV bays at Bikaner-IV PS 4. LILO of both ckts of 400kV Bikaner II PS- Bikaner III PS (Quad) direct line at Bikaner-IV PS 5. Establishment of 765/400 kV, 6x1500 MVA S/s at suitable location near Siwani (Distt. Bhiwani) along with 2x240 MVA (765kV) Bus Reactor & 2x125 MVA (420kV) Bus Reactor 6. Bikaner-IV PS – Siwani 765 kV 2xD/c line along with 240 MVA switchable line reactor for each circuit at each end 7. Siwani – Jind (PG) 400 kV D/c line (Quad) 8. Siwani – Patran (Indi Grid) 400 kV D/c line (Quad) along with 80 MVA switchable line reactor for each circuit at Siwani S/s end 9. Siwani – Fatehabad (PG) 400 kV D/c line (Quad) 10. Siwani – Sonipat (PG) 400 kV D/c line (Quad) along with 63 MVA switchable line reactor for each circuit at Siwani S/s end 11. STATCOM (2x+300MVA) along with MSC (4x125 MVA) & MSR (2x125 MVA) along with 2 nos. 400kV bays at Siwani S/s										

Sl. No.	Application ID	Applicant	Project Location	Application Date	Nature of Applicant	Criterion for applying	Injection at ISTS point (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)	Conn BGs to be submitted
12. 400kV Sectionalizer bay (1 set) at at Siwani S/s										
D. <u>Additional Transmission system for Inter-regional power transfer:</u>										
1. Establishment of 765kV Mandsaur Pooling Station along with 2x330 MVAR (765 kV) Bus Reactors 2. Beawar- Mandsaur PS 765 kV D/c line along with 240 MVAR switchable line reactor for each circuit at each end 3a) Mandsaur PS – Indore(PG) 765 kV D/c Line along with 1x330 MVAR switchable line reactor on each ckt at Mandsaur end of Mandsaur PS – Indore(PG) 765 kV D/c Line OR 3b) Establishment of 765 kV Kurawar S/s with 2x330 MVAR 765 kV bus reactor 3c) Mandsaur – Kurawar 765 kV D/c line along with 240 MVAR switchable line reactors on each ckt at both ends of Mandsaur – Kurawar 765 kV D/c line 3d) LILO of Indore – Bhopal 765 kV S/c line at Kurawar										
Start Date of Connectivity under GNA: 01.09.2026(Interim). Final date shall be confirmed upon award of the system										
18.	2200000333	Furies Solren Private Limited	Jaisalmer distt., Rajasthan	27.10.2023	Generator (Solar)	Land BG Route	300	31.01.2026	Ramgarh PS	• Conn BG1:50 lakh • Conn BG2: Rs. 3 Cr. • Conn BG3: 2 lakh/MW
It was informed that M/s Furies Solren Private Limited has applied for connectivity for 300 MW at Ramgarh PS in Land BG route. Accordingly, it was proposed to grant connectivity at Ramgarh PS at 220 kV level through 1 no. of 220 kV line bay. M/s Furies Solren informed that the 2028 timeline of Start date of Connectivity with Bhadla(HVDC) system is not viable for their project. Therefore, in the meeting and confirmation vide email dated 10.01.2024, M/s Furies Solren requested to grant connectivity at Bikaner-IV PS. As the application was through Land BG route, the same was agreed by CTUIL. Accordingly, it was proposed to grant connectivity at Bikaner-IV PS through 1no. of 220 kV line bay. M/s Furies Solren Private Limited was asked to confirm the 220kV bay scope at Bikaner-IV PS end and the tower configuration. M/s Furies Solren Private Limited informed that 220 kV line bay at Bikaner-IV end shall be under ISTS and tower configuration shall be S/c line on D/c tower. The same was noted & M/s Furies Solren Private Limited was asked to submit the necessary undertaking for S/c line on D/c tower configuration. M/s Furies Solren Private Limited agreed for the same. Further, it was also informed that M/s Furies Solren Private Limited has requested start Date of Connectivity under GNA from 31/01/2026. However, considering the same and expected commissioning date of transmission system(31.08.2026), the start date of Connectivity under GNA shall be 01.09.2026(interim). M/s Furies Solren noted the same.										

Sl. No.	Application ID	Applicant	Project Location	Application Date	Nature of Applicant	Criterion for applying	Injection at ISTS point (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)	Conn BGs to be submitted
---------	----------------	-----------	------------------	------------------	---------------------	------------------------	------------------------------	---	--	--------------------------

Accordingly, Transmission system for connectivity of M/s Furies Solren under GNA is as below:

Details of Transmission system for Connectivity under GNA:

A. Associated Transmission System (ATS): NIL

B. Transmission System under applicant scope

- (i). Furies Solren Private Limited Solar Power Project– Bikaner-IV PS 220 kV S/c line on D/c tower (Suitable to carry minimum 300 MW at nominal voltage)

A. Transmission system for Connectivity under GNA

1. Establishment of 6x1500 MVA, 765/400 kV & 6x500 MVA, 400/220 kV Bikaner-IV Pooling Station along with 2x240 MVAR (765kV) & 2x125 MVAR (420kV) Bus Reactors at a suitable location near Bikaner
2. 400kV Sectionalizer bay (1 set) at Bikaner-IV PS
3. STATCOM (2x+300MVAR) along with MSC (4x125 MVAR) & MSR (2x125 MVAR) along with 2 nos. 400kV bays at Bikaner-IV PS
4. LILO of both ckts of 400kV Bikaner II PS- Bikaner III PS (Quad) direct line at Bikaner-IV PS
5. Establishment of 765/400 kV, 6x1500 MVA S/s at suitable location near Siwani (Distt. Bhiwani) along with 2x240 MVAR (765kV) Bus Reactor & 2x125 MVAR (420kV) Bus Reactor
6. Bikaner-IV PS – Siwani 765 kV 2xD/c line along with 240 MVAR switchable line reactor for each circuit at each end
7. Siwani – Jind (PG) 400 kV D/c line (Quad)
8. Siwani – Patran (Indi Grid) 400 kV D/c line (Quad) along with 80 MVAR switchable line reactor for each circuit at Siwani S/s end
9. Siwani – Fatehabad (PG) 400 kV D/c line (Quad)
10. Siwani – Sonipat (PG) 400 kV D/c line (Quad) along with 63 MVAR switchable line reactor for each circuit at Siwani S/s end
11. STATCOM (2x+300MVAR) along with MSC (4x125 MVAR) & MSR (2x125 MVAR) along with 2 nos. 400kV bays at Siwani S/s
12. 400kV Sectionalizer bay (1 set) at at Siwani S/s

B. Additional Transmission system for Inter-regional power transfer:

1. Establishment of 765kV Mandsaur Pooling Station along with 2x330 MVAR (765 kV) Bus Reactors
2. Beawar- Mandsaur PS 765 kV D/c line along with 240 MVAR switchable line reactor for each circuit at each end
- 3a) Mandsaur PS – Indore(PG) 765 kV D/c Line along with 1x330 MVAR switchable line reactor on each ckt at Mandsaur end of Mandsaur PS – Indore(PG) 765 kV D/c Line

Sl. No.	Application ID	Applicant	Project Location	Application Date	Nature of Applicant	Criterion for applying	Injection at ISTS point (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)	Conn BGs to be submitted
OR 3b) Establishment of 765 kV Kurawar S/s with 2x330 MVA 765 kV bus reactor 3c) Mandsaur – Kurawar 765 kV D/c line along with 240 MVA switchable line reactors on each ckt at both ends of Mandsaur – Kurawar 765 kV D/c line 3d) LILO of Indore – Bhopal 765 kV S/c line at Kurawar Start Date of Connectivity under GNA: 01.09.2026(Interim). Final date shall be confirmed upon award of the system										
19.	2200000355	Gamma Renewables India Project One Private Limited	Barmer distt., Rajasthan	31.10.2023	Renewable Power Park Developer (Solar)	Land BG Route	300	01.06.2026	Barmer-I PS	• Conn BG1:50 lakh • Conn BG2: Rs. 3 Cr. • Conn BG3: 2 lakh/MW
It was informed that M/s Gamma Renewables India Project One Private Limited has applied for connectivity for 300 MW at Barmer-I PS. Accordingly, it was proposed to grant connectivity at Barmer-I PS at 220 kV level through 1 no. of 220 kV line bay. M/s Gamma Renewables India agreed for the same. M/s Gamma Renewables India was asked to confirm the 220kV bay scope at Barmer-I PS end and the tower configuration. M/s Gamma Renewables India informed that 220 kV line bay at Barmer-I end shall be under ISTS and tower configuration shall be S/c line on D/c tower. The same was noted & M/s Gamma Renewables India was asked to submit the necessary undertaking for S/c line on D/c tower configurations. M/s Gamma Renewables India agreed for the same. Further, it was also informed that M/s Gamma Renewables India has requested start Date of Connectivity under GNA from 01/06/2026. However, considering the same and expected commissioning date of transmission system (31.08.2026), the start date of Connectivity under GNA shall be 01.09.2026(interim). M/s Gamma Renewables India noted the same. Accordingly, Transmission system for connectivity of M/s Gamma Renewables India under GNA is as below: <u>Details of Transmission system for Connectivity under GNA:</u> A. <u>Associated Transmission System (ATS):</u> NIL B. <u>Transmission System under applicant scope</u> (i). Gamma Renewables India Project Private Limited Renewable Power Park– Barmer-I PS 220 kV S/c line on D/c tower (Suitable to carry minimum 300 MW at nominal voltage) C. <u>Transmission system for Connectivity under GNA (tentative)</u>										

Sl. No.	Application ID	Applicant	Project Location	Application Date	Nature of Applicant	Criterion for applying	Injection at ISTS point (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)	Conn BGs to be submitted
										1. Augmentation of 5x500 MVA (5th to 9th), 400/220 kV ICTs at Barmer-I PS 2. Augmentation with 765/400 kV, 3x1500 MVA Transformer (3rd, 4th & 5th) at Barmer-I PS 3. Fatehgarh-III (Section-2) PS – Barmer-I PS 400 kV D/c line (Quad) 4. Establishment of 2x1500 MVA, 765/400 kV Substation at suitable location near Sirohi along with 2x240 MVAR (765 kV) & 2x125 MVAR (400 kV) Bus Reactor 5. Barmer-I PS– Sirohi PS 765 kV D/c line along with 240 MVAR switchable line reactor for each circuit at each end 6. Sirohi PS-Chittorgarh (PG) 400 kV D/c line along with 80 MVAR switchable line reactor for each circuit at Sirohi PS end (Quad) 7. Establishment of, 765 kV Substation at suitable location near Rishabdeo (Distt. Udaipur) along with 2x240 MVAR (765 kV) Bus Reactor 8. Sirohi PS- Rishabdeo 765 kV D/c line along with 330 MVAR switchable line reactor for each circuit at Sirohi end 9. Rishabdeo - Mandsaur PS 765 kV D/c line along with 240 MVAR switchable line reactor for each circuit at Rishabdeo end 10. LILO of one circuit of 765 kV Chittorgarh-Banaskanta D/c line at Rishabdeo S/s 11. Establishment of 765 kV Mandsaur PS along with 2x330 MVAR (765 kV) Bus Reactors 12. Mandsaur PS–Indore(PG) 765 kV D/c Line along with 1x330 MVAR switchable line reactor on each ckt at Mandsaur end of Mandsaur PS – Indore(PG) 765 kV D/c Line 13. Establishment of 765/400 kV (2x1500 MVA Kurawar S/s with 2x330 MVAR 765 kV bus reactor and 1x125 MVAR 420 kV bus reactor 14. Mandsaur – Kurawar 765 kV D/c line along with 240 MVAR switchable line reactors on each ckt at both ends of Mandsaur – Kurawar 765 kV D/c line 15. Kurawar – Ashtha 400 kV D/c (Quad ACSR/AAAC/AL59 moose equivalent) line 16. LILO of Indore – Bhopal 765 kV S/c line at Kurawar 17. LILO of one circuit of Indore – Itarsi 400 kV D/c line at Astha 18. Shujalpur – Kurawar 400 kV D/c (Quad ACSR/AAAC/AL59 moose equivalent) line 19. Establishment of 765/400kV, 2x1500 MVA S/s at suitable location near Merta (Merta-II Substation) along with 2x125 MVAR & 2x240 MVAR bus reactor at Merta-II S/s 20. STATCOM (2x±300MVAR) along with MSC (4x125 MVAR) & MSR (2x125 MVAR) along with 2 nos. 400kV bays at Barmer-I PS 21. 400kV Sectionalizer bay (1 set) at Barmer-I S/s 22. Barmer-I PS – Merta-II 765 kV D/c line along with 240 MVAR switchable line reactor at Barmer-I PS end and 330MVAR switchable line reactor at Merta-II PS end for each circuit of Barmer-I PS – Merta-II 765 kV D/c line 23. Merta-II – Beawar 400 kV D/c line (Quad) 24. Merta-II – Dausa 765 kV D/c line along with 240 MVAR switchable line reactor for each circuit at each end of Merta-II – Dausa 765 kV D/c line 25. Establishment of 765/400kV, 2x1500 MVA S/s at suitable location near Ghiror (Distt. Mainpuri) along with 2x125 MVAR & 2x240 MVAR bus reactor at Ghiror S/s (UP) 26. Dausa - Ghiror 765 kV D/c line along with 240 MVAR switchable line reactor for each circuit at each end of Dausa - Ghiror 765 kV D/c line

Sl. No.	Application ID	Applicant	Project Location	Application Date	Nature of Applicant	Criterion for applying	Injection at ISTS point (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)	Conn BGs to be submitted
27. LILO of both ckt of 765 kV Aligarh (PG) -Orai (PG) D/c line at Ghiror S/s along with 240 MVAr switchable line reactor for each circuit at Ghiror S/s end of 765 kV Ghiror -Orai D/c line 28. LILO of one ckt of 765kV Agra (PG) – Fatehpur(PG) 765kV D/c line at Ghiror along with 240 MVAr switchable line reactor at Ghiror S/s end of 765 kV Ghiror -Fatehpur line (29. 400kV Ghiror-Firozabad (UPPTCL) D/c line (Quad) Start Date of Connectivity under GNA: 01.09.2026(Interim). Final date shall be confirmed upon award of the system.										

C5. RE Applications for Connectivity to ISTS under CERC (Connectivity and General Network Access to the inter-State Transmission System) Regulations, 2022 received in Nov'23:

Sl. No.	Application ID	Applicant	Project Location	Submission Date	Nature of Applicant	Criterion for applying	Injection at ISTS point (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)
1.	2200000343	Iraax International (Opc) Private Limited	Bikaner distt., Rajasthan	09.11.2023	Renewable Power Park developer (Solar)	Land Route	300	31.03.2026	Bikaner-III PS
2.	2200000348	NTPC Renewable Energy Limited	Jaisalmer distt., Rajasthan	09.11.2023	Generator (Solar)	Land BG Route	900	31.03.2026	Fatehgarh-IV PS
3.	2200000344	Clean Max Gamma Private Limited	Bikaner distt., Rajasthan	09.11.2023	Renewable Power Park developer (Solar)	Land Route	300	28.06.2025	Bikaner-II PS
4.	2200000345	Bundelkhand Saur Urja Limited	Jalaun distt., Uttar Pradesh	11.11.2023	Renewable Power Park developer (Solar)	Land Route	1200	02.10.2025	PGCIL substation at Ait, Tehsil Orai dist. Jalaun ,UP
5.	2200000373 (Enhancement)	NTPC Renewable Energy Limited	Bikaner distt., Rajasthan	17.11.2023	Generator (Solar)	Land Route	400	31.03.2026	Bikaner-III PS
6.	2200000360	Prerak Solar Urja Private Limited	Bikaner distt., Rajasthan	17.11.2023	Renewable Power Park developer (Solar)	Land Route	400	31.05.2025	Bhadla-III PS
7.	2200000358 (Enhancement)	NTPC Renewable Energy Limited	Bikaner distt., Rajasthan	22.11.2023	Generator (Solar)	Land Route	250	30.06.2026	Bikaner-III/IV PS
8.	2200000387	Acme Cleantech Solutions Private Limited	Jaisalmer distt., Rajasthan	29.11.2023	Generator (Solar)	Land BG route	600	31.03.2026	Fatehgarh-II PS
9.	2200000396	Acme Cleantech Solutions Private Limited	Jaisalmer distt., Rajasthan	30.11.2023	Generator (Hybrid)	LOA (SJVN)	250	31.03.2026	Fatehgarh-II PS

Sl. No.	Application ID	Applicant	Project Location	Submission Date	Nature of Applicant	Criterion for applying	Injection at ISTS point (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)	Conn BGs to be submitted
1.	2200000343	Iraax International (Opc) Private Limited	Bikaner distt., Rajasthan	09.11.2023	Renewable Power Park developer (Solar)	Land Route	300	31.03.2026	Bikaner-III PS	• Conn BG1:50 lakh • Conn BG2: Nil(Bay under applicant scope) • Conn BG3: 2 lakh/MW
It was informed that M/s Iraax International (Opc) Private Limited has applied for connectivity for 300 MW at Bikaner-III PS. However, there is no margin available for grant of connectivity at Bikaner-III PS. Therefore, it was proposed to grant connectivity at Bikaner-IV PS at 220 kV level through 1 no. of 220 kV line bay. M/s Iraax International (Opc) agreed for the same. M/s Iraax International (Opc) was asked to confirm the 220kV bay scope at Bikaner-IV PS end and the tower configuration. M/s Iraax International (Opc) informed that 220 kV line bay at Bikaner-IV end shall be under their scope and tower configuration shall be S/c line on D/c tower. The same was noted & M/s Iraax International (Opc) was asked to submit the necessary undertaking for S/c line on D/c tower configuration. M/s Iraax International (Opc) agreed for the same. Further, it was also informed that M/s Iraax International (Opc) has requested start Date of Connectivity under GNA from 31/03/2026. Considering the same and expected commissioning date of transmission system(31.08.2026), the start date of Connectivity under GNA shall be 01.09.2026(interim). M/s Iraax International (Opc) noted the same. Accordingly, Transmission system for connectivity of M/s Iraax International (Opc) under GNA is as below: <u>Details of Transmission system for Connectivity under GNA:</u> A. <u>Associated Transmission System (ATS):</u> NIL B. <u>Transmission System under applicant scope</u> (i). Iraax International (Opc) Private Limited Renewable Power Park– Bikaner-IV PS 220 kV S/c line on D/c tower(Suitable to carry minimum 300 MW at nominal voltage) A. <u>Transmission system for Connectivity under GNA</u> 1. Establishment of 6x1500 MVA, 765/400 kV & 6x500 MVA, 400/220 kV Bikaner-IV Pooling Station along with 2x240 MVar (765kV) & 2x125 MVar (420kV) Bus Reactors at a suitable location near Bikaner 2. 400kV Sectionalizer bay (1 set) at Bikaner-IV PS										

Sl. No.	Application ID	Applicant	Project Location	Submission Date	Nature of Applicant	Criterion for applying	Injection at ISTS point (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)	Conn BGs to be submitted
3. STATCOM (2x+300MVA) along with MSC (4x125 MVA) & MSR (2x125 MVA) along with 2 nos. 400kV bays at Bikaner-IV PS 4. LILO of both ckts of 400kV Bikaner II PS- Bikaner III PS (Quad) direct line at Bikaner-IV PS 5. Establishment of 765/400 kV, 6x1500 MVA S/s at suitable location near Siwani (Distt. Bhiwani) along with 2x240 MVA (765kV) Bus Reactor & 2x125 MVA (420kV) Bus Reactor 6. Bikaner-IV PS – Siwani 765 kV 2xD/c line along with 240 MVA switchable line reactor for each circuit at each end 7. Siwani – Jind (PG) 400 kV D/c line (Quad) 8. Siwani – Patran (Indi Grid) 400 kV D/c line (Quad) along with 80 MVA switchable line reactor for each circuit at Siwani S/s end 9. Siwani – Fatehabad (PG) 400 kV D/c line (Quad) 10. Siwani – Sonipat (PG) 400 kV D/c line (Quad) along with 63 MVA switchable line reactor for each circuit at Siwani S/s end 11. STATCOM (2x+300MVA) along with MSC (4x125 MVA) & MSR (2x125 MVA) along with 2 nos. 400kV bays at Siwani S/s 12. 400kV Sectionalizer bay (1 set) at at Siwani S/s C. Additional Transmission system for Inter-regional power transfer: 1. Establishment of 765kV Mandsaur Pooling Station along with 2x330 MVA (765 kV) Bus Reactors 2. Beawar- Mandsaur PS 765 kV D/c line along with 240 MVA switchable line reactor for each circuit at each end 3a) Mandsaur PS – Indore(PG) 765 kV D/c Line along with 1x330 MVA switchable line reactor on each ckt at Mandsaur end of Mandsaur PS – Indore(PG) 765 kV D/c Line OR 3b) Establishment of 765 kV Kurawar S/s with 2x330 MVA 765 kV bus reactor 3c) Mandsaur – Kurawar 765 kV D/c line along with 240 MVA switchable line reactors on each ckt at both ends of Mandsaur – Kurawar 765 kV D/c line 3d) LILO of Indore – Bhopal 765 kV S/c line at Kurawar Start Date of Connectivity under GNA: 01.09.2026(Interim). Final date shall be confirmed upon award of the system										
2.	2200000348	NTPC Renewable Energy Limited	Jaisalmer distt., Rajasthan	09.11.2023	Generator (Solar)	Land BG Route	900	31.03.2026	Fatehgarh-IV PS	• Conn BG1:50 lakh • Conn BG2: Rs. 6 Cr. • Conn BG3: 2 lakh/MW
It was informed that M/s NTPC Renewable Energy Limited has applied for connectivity for 900 MW at Fatehgarh-IV PS (400kV).Considering availability of margins at 400kV level, it was proposed to grant connectivity to M/s NTPC Limited at Fatehgarh-IV PS(Sec-II) thought 1 no. of 400 kV line bay.										

Sl. No.	Application ID	Applicant	Project Location	Submission Date	Nature of Applicant	Criterion for applying	Injection at ISTS point (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)	Conn BGs to be submitted
Further, it was also informed that, with the above grant of connectivity to M/s NTPC(900 MW), total connectivity granted at Fatehgarh-IV PS(Sec-II) shall be 4980 MW (3480 MW at 220 kV & 1500 MW) at 400 kV. As Fatehgarh-IV PS was planned for maximum capacity of 5 GW Fatehgarh-IV PS(Sec-II) shall be closed for further grant of connectivity beyond 4980 MW including enhancement. M/s NTPC Renewable was asked to confirm the 400kV bay scope at Fatehgarh-IV PS(Sec-II) end and the tower configuration. M/ NTPC Renewable informed that 400 kV line bay at Fatehgarh-IV PS(Sec-II) end shall be under ISTS and tower configuration shall be S/c line on D/c tower. The same was noted & M/s NTPC Renewable was asked to submit the necessary undertaking for S/c line on D/c tower configuration. M/s NTPC Renewable agreed for the same. Further, it was also informed that M/s NTPC Renewable has requested start Date of Connectivity under GNA from 31/03/2026. Considering the same and expected commissioning date of transmission system(31.08.2026), the start date of Connectivity under GNA shall be 01.09.2026(interim). M/s NTPC Renewable noted the same. Accordingly, Transmission system for connectivity of M/s NTPC Renewable under GNA is as below: <u>Details of Transmission system for Connectivity under GNA:</u> A. <u>Associated Transmission System (ATS): NIL</u> B. <u>Transmission System under applicant scope</u> (i). NTPC Renewable Energy Limited Solar Power Project– Fatehgarh-IV PS(Sec-II) 400 kV S/c line on D/c tower (Suitable to carry minimum 900 MW at nominal voltage) C. <u>Transmission system for Connectivity under GNA</u> 1. Augmentation with 765/400kV, 2x1500MVA Transformer (3rd & 4th) at Fatehgarh-IV PS (Sec-II) 2. Augmentation with 765/400 kV, 3x1500 MVA Transformer (3rd, 4th & 5th) at Barmer-I PS 3. Fatehgarh-IV PS(Sec-II) – Bhinmal (PG) 400 kV D/c line (Twin HTLS*) along with 50 MVAR switchable line reactor on each ckt at each end 4. LILO of both ckts of 765 kV Fatehgarh-III- Beawar D/c line at Fatehgarh-IV (Section-2) PS along with 330 MVAR switchable line reactor at Fatehgarh-IV PS end of each ckt of 765 kV Fatehgarh-IV- Beawar D/c line (formed after LILO) 5. Fatehgarh-III (Section-2) PS – Barmer-I PS 400 kV D/c line (Quad) 6. Establishment of 2x1500 MVA, 765/400 kV Substation at suitable location near Sirohi along with 2x240 MVAR (765 kV) & 2x125 MVAR (400 kV) Bus Reactor 7. Barmer-I PS– Sirohi PS 765 kV D/c line along with 240 MVAR switchable line reactor for each circuit at each end										

Sl. No.	Application ID	Applicant	Project Location	Submission Date	Nature of Applicant	Criterion for applying	Injection at ISTS point (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)	Conn BGs to be submitted
8. Sirohi PS-Chittorgarh (PG) 400 kV D/c line along with 80 MVA switchable line reactor for each circuit at Sirohi PS end (Quad) 9. Fatehgarh-IV (Sec-II) PS – Sirohi PS 765 kV D/c line along with 240 MVA switchable line reactor for each circuit at each end 10. Establishment of, 765 kV Substation at suitable location near Rishabdeo (Distt. Udaipur) along with 2x240 MVA (765 kV) Bus Reactor 11. Sirohi PS- Rishabdeo 765 kV D/c line along with 330 MVA switchable line reactor for each circuit at Sirohi end 12. Rishabdeo - Mandsaur PS 765 kV D/c line along with 240 MVA switchable line reactor for each circuit at Rishabdeo end 13. LILO of one circuit of 765 kV Chittorgarh-Banaskanta D/c line at Rishabdeo S/s 14. Establishment of 765 kV Mandsaur PS along with 2x330 MVA (765 kV) Bus Reactors 15. Beawar- Mandsaur PS 765 kV D/c line along with 240 MVA switchable line reactor for each circuit at each end 16. Mandsaur PS–Indore(PG) 765 kV D/c Line along with 1x330 MVA switchable line reactor on each ckt at Mandsaur end of Mandsaur PS – Indore(PG) 765 kV D/c Line 17. Establishment of 765/400 kV (2x1500 MVA Kurawar S/s with 2x330 MVA 765 kV bus reactor and 1x125 MVA 420 kV bus reactor 18. Mandsaur – Kurawar 765 kV D/c line along with 240 MVA switchable line reactors on each ckt at both ends of Mandsaur – Kurawar 765 kV D/c line 19. Kurawar – Ashtha 400 kV D/c (Quad ACSR/AAAC/AL59 moose equivalent) line 20. LILO of Indore – Bhopal 765 kV S/c line at Kurawar 21. LILO of one circuit of Indore – Itarsi 400 kV D/c line at Ashtha 22. Shujalpur – Kurawar 400 kV D/c (Quad ACSR/AAAC/AL59 moose equivalent) line 23. Establishment of 765/400kV, 2x1500 MVA S/s at suitable location near Merta (Merta-II Substation) along with 2x125 MVA & 2x240 MVA bus reactor at Merta-II S/s 24. Fatehgarh-IV (Sec-II) PS – Barmer-I PS 400 kV D/c line (Quad) 25. STATCOM (2x±300MVA) along with MSC (4x125 MVA) & MSR (2x125 MVA) along with 2 nos. 400kV bays at Barmer-I PS 26. 400kV Sectionalizer bay (1 set) at Barmer-I S/s 27. Barmer-I PS – Merta-II 765 kV D/c line along with 240 MVA switchable line reactor at Barmer-I PS end and 330MVA switchable line reactor at Merta-II PS end for each circuit of Barmer-I PS – Merta-II 765 kV D/c line 28. Merta-II – Beawar 400 kV D/c line (Quad) 29. Merta-II – Dausa 765 kV D/c line along with 240 MVA switchable line reactor for each circuit at each end of Merta-II – Dausa 765 kV D/c line 30. Establishment of 765/400kV, 2x1500 MVA S/s at suitable location near Ghiror (Distt. Mainpuri) along with 2x125 MVA & 2x240 MVA bus reactor at Ghiror S/s (UP) 31. Dausa - Ghiror 765 kV D/c line along with 240 MVA switchable line reactor for each circuit at each end of Dausa - Ghiror 765 kV D/c line 32. LILO of both ckt of 765 kV Aligarh (PG) -Orai (PG) D/c line at Ghiror S/s along with 240 MVA switchable line reactor for each circuit at Ghiror S/s end of 765 kV Ghiror -Orai D/c line 33. LILO of one ckt of 765kV Agra (PG) – Fatehpur(PG) 765kV D/c line at Ghiror along with 240 MVA switchable line reactor at Ghiror S/s end of 765 kV Ghiror -Fatehpur line (										

Sl. No.	Application ID	Applicant	Project Location	Submission Date	Nature of Applicant	Criterion for applying	Injection at ISTS point (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)	Conn BGs to be submitted
34. 400kV Ghiror-Firozabad (UPPTCL) D/c line (Quad) <i>* with minimum capacity of 2100 MVA/ckt at nominal voltage</i>										
Start Date of Connectivity under GNA: 01.09.2026(Interim). Final date shall be confirmed upon award of the system										
3.	2200000344	Clean Max Gamma Private Limited	Bikaner distt., Rajasthan	09.11.2023	Renewable Power Park developer (Solar)	Land Route	300	28.06.2025	Bikaner-II PS	• Conn BG1:50 lakh • Conn BG2: Rs. 3 Cr. • Conn BG3: 2 lakh/MW
It was informed that M/s Clean Max Gamma Private Limited has applied for connectivity for 300 MW at Bikaner-II PS. However, as informed in the 23rd CMETS NR meeting, 675 MW of margin available at Bikaner-II PS cannot be granted to any entity till further order by CERC. Further, any vacancy created at Bikaner-II PS shall also be offered to applicants as per application priority under the reallocation process. Further, as informed earlier, there is no margin available for grant of connectivity at Bikaner-III PS as well. Therefore, it was proposed to grant connectivity at Bikaner-IV PS at 220 kV level through 1 no. of 220 kV line bay. M/s Clean Max enquired whether they can be granted connectivity at Bikaner-I PS. Regarding this, CTUIL informed that Bikaner-I PS is an old substation and due to limitations of ICT, the margin available at 400kV Bikaner-I PS is 350 MW. Out of which, 300 MW is already granted to M/s AK Renewable in this meeting. Therefore, only 50 MW of margin is available for grant. Accordingly, connectivity of 300 MW cannot be granted at Bikaner-I PS, however Bikaner-IV PS is available. M/s Clean Max agreed for the connectivity at Bikaner-IV PS. M/s Clean Max Gamma was asked to confirm the 220kV bay scope at Bikaner-IV PS end and the tower configuration. M/s Clean Max Gamma informed that 220 kV line bay at Bikaner-IV PS end shall be under ISTS and tower configuration shall be S/c line on D/c tower. The same was noted & M/s Clean Max Gamma was asked to submit the necessary undertaking for S/c line on D/c tower configurations. M/s Clean Max Gamma agreed for the same. M/s Clean Max Gamma Private Limited has requested Start Date of Connectivity under GNA from 28/06/2025. However, considering the same and expected commissioning date of transmission system (31.08.2026), the start date of Connectivity under GNA shall be 01.09.2026 (interim). Proposed Transmission system for connectivity of Clean Max Gamma Private Limited for 300 MW under GNA is as below: <u>Details of Transmission system for Connectivity under GNA:</u> A. <u>Associated Transmission System (ATS):</u> NIL B. <u>Transmission System under applicant scope</u> (i). Clean Max Gamma Private Limited Renewable Power Park– Bikaner-IV PS 220 kV S/c line on D/c tower (Suitable to carry minimum 300 MW at nominal voltage)										

Sl. No.	Application ID	Applicant	Project Location	Submission Date	Nature of Applicant	Criterion for applying	Injection at ISTS point (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)	Conn BGs to be submitted
C. <u>Transmission system for Connectivity under GNA:</u> 1. Establishment of 6x1500 MVA, 765/400 kV & 6x500 MVA, 400/220 kV Bikaner-IV Pooling Station along with 2x240 MVar (765kV) & 2x125 MVar (420kV) Bus Reactors at a suitable location near Bikaner 2. 400kV Sectionalizer bay (1 set) at Bikaner-IV PS 3. STATCOM (2x±300MVar) along with MSC (4x125 MVar) & MSR (2x125 MVar) along with 2 nos. 400kV bays at Bikaner-IV PS 4. LILO of both ckts of 400kV Bikaner II PS- Bikaner III PS (Quad) direct line at Bikaner-IV PS 5. Establishment of 765/400 kV, 6x1500 MVA S/s at suitable location near Siwani (Distt. Bhiwani) along with 2x240 MVar (765kV) Bus Reactor & 2x125 MVar (420kV) Bus Reactor 6. Bikaner-IV PS – Siwani 765 kV 2xD/c line along with 240 MVar switchable line reactor for each circuit at each end 7. Siwani – Jind (PG) 400 kV D/c line (Quad) 8. Siwani – Patran (Indi Grid) 400 kV D/c line (Quad) along with 80 MVar switchable line reactor for each circuit at Siwani S/s end 9. Siwani – Fatehabad (PG) 400 kV D/c line (Quad) 10. Siwani – Sonipat (PG) 400 kV D/c line (Quad) along with 63 MVar switchable line reactor for each circuit at Siwani S/s end 11. STATCOM (2x±300MVar) along with MSC (4x125 MVar) & MSR (2x125 MVar) along with 2 nos. 400kV bays at Siwani S/s 12. 400kV Sectionalizer bay (1 set) at at Siwani S/s D. <u>Additional Transmission system for Inter-regional power transfer:</u> 1. Establishment of 765kV Mandsaur Pooling Station along with 2x330 MVar (765 kV) Bus Reactors 2. Beawar- Mandsaur PS 765 kV D/c line along with 240 MVar switchable line reactor for each circuit at each end 3a) Mandsaur PS – Indore(PG) 765 kV D/c Line along with 1x330 MVar switchable line reactor on each ckt at Mandsaur end of Mandsaur PS – Indore(PG) 765 kV D/c Line OR 3b) Establishment of 765 kV Kurawar S/s with 2x330 MVar 765 kV bus reactor 3c) Mandsaur – Kurawar 765 kV D/c line along with 240 MVar switchable line reactors on each ckt at both ends of Mandsaur – Kurawar 765 kV D/c line 3d) LILO of Indore – Bhopal 765 kV S/c line at Kurawar Start Date of Connectivity under GNA: 01.09.2026(Interim). Final date shall be confirmed upon award of the system										
4.	2200000345	Bundelkhand Saur Urja Limited	Jalaun distt.,	11.11.2023	Renewable Power Park	Land Route	1200	02.10.2025	Orai (PG) S/s (GIS)	• Conn BG1:50 lakh

Sl. No.	Application ID	Applicant	Project Location	Submission Date	Nature of Applicant	Criterion for applying	Injection at ISTS point (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)	Conn BGs to be submitted
			Uttar Pradesh		developer (Solar)					• Conn BG2: Nil(Bay under applicant scope) • Conn BG3: 2 lakh/MW
It was informed that M/s Bundelkhand Saur Urja Limited has applied for connectivity for 1200 MW at Orai (PG) S/s (GIS) [400kV]. Accordingly, it was proposed to grant connectivity at Orai (PG) S/s (GIS) at 400 kV level through 400 kV S/c (high capacity) line to Orai (PG) S/s (GIS). It was also informed that in view of the margin availability, the connectivity is being granted with existing system beyond 400kV Orai (PG) S/s. M/s Bundelkhand Saur Urja was asked to confirm the 400kV bay scope at Orai(PG) S/s(GIS) end and the tower configuration. M/s Bundelkhand Saur Urja informed that 400 kV line bay (GIS) at Orai PS end shall be under their scope and line configuration shall be S/c line on D/c tower. The same was noted & M/s Bundelkhand Saur Urja was asked to submit the necessary undertaking for S/c line on D/c tower configuration. M/s Bundelkhand Saur Urja agreed for the same. Further, M/s Bundelkhand Saur Urja Limited had requested start date of Connectivity under GNA from 02/10/2025. In the meeting M/s Bundelkhand Saur Urja informed that they shall confirm the revised Start Date vide mail. Subsequently, vide mail dated 12.01.2024 confirmed their revised start date of connectivity as 31.12.2026 Accordingly, the start date of Connectivity under GNA shall be considered as 31.12.2026. Transmission system for connectivity of M/s Bundelkhand Saur Urja Limited under GNA is as below: <u>Details of Transmission system for Connectivity under GNA:</u> A. <u>Transmission System under applicant scope</u> (i). Bundelkhand Saur Urja Limited Renewable Power Park– Orai(PG) 400 kV S/c (high capacity) line on D/c tower (Suitable to carry minimum 1200 MW per circuit at nominal voltage) B. <u>Transmission system for Connectivity under GNA:</u> Existing system beyond 400 kV Orai (PG) S/s (GIS). Start Date of Connectivity under GNA: 31.12.2026										
5.	2200000373 (Enhancement)	NTPC Renewable Energy Limited	Bikaner distt., Rajasthan	17.11.2023	Generator (Solar)	Land Route	400	31.03.2026	Bikaner-III PS	• Conn BG1:50 lakh • Conn BG2: Rs. 6 Cr. • Conn BG3: 2 lakh/MW

Sl. No.	Application ID	Applicant	Project Location	Submission Date	Nature of Applicant	Criterion for applying	Injection at ISTS point (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)	Conn BGs to be submitted
It was informed that M/s NTPC Renewable Energy Limited (NTPCREL) has applied for enhancement of connectivity for 400 MW. As decided earlier with the previous NTPC applications of 700 MW & 150 MW, it is proposed to grant connectivity to NTPCREL at Bikaner-IV PS through 400 kV D/c line. M/s NTPC Renewable Energy Limited has requested Start Date of Connectivity under GNA from 31/03/2026. However, considering the same and expected commissioning date of transmission system(i.e. 31.08.2026), the start date of Connectivity under GNA shall be 01.09.2026 (interim). Proposed Transmission system for connectivity of NTPC Renewable Energy Limited for 400 MW under GNA is as below: <u>Details of Transmission system for Connectivity under GNA:</u> A. <u>Associated Transmission System (ATS):</u> NIL B. <u>Transmission System under applicant scope</u> (i). Common pooling station for NTPC Renewable Energy Limited Solar Power Projects (App No. 2200000244(150 MW), 2200000308(700 MW), 2200000373(400 MW) & 2200000358(250 MW) – Bikaner-IV PS 400 kV D/c line (Suitable to carry minimum 900 MW per circuit at nominal voltage) C. <u>Transmission system for Connectivity under GNA</u> - Establishment of 6x1500 MVA, 765/400 kV Bikaner-IV Pooling Station along with 2x240 MVar (765kV) & 2x125 MVar (420kV) Bus Reactors at a suitable location near Bikaner 400kV Sectionalizer bay (1 set) at Bikaner-IV PS - STATCOM (2x±300MVar) along with MSC (4x125 MVar) & MSR (2x125 MVar) along with 2 nos. 400kV bays at Bikaner-IV PS - LILO of both ckts of 400kV Bikaner II PS- Bikaner III PS (Quad) direct line at Bikaner-IV PS - Establishment of 765/400 kV, 6x1500 MVA S/s at suitable location near Siwani (Distt. Bhiwani) along with 2x240 MVar (765kV) Bus Reactor & 2x125 MVar (420kV) Bus Reactor - Bikaner-IV PS – Siwani 765 kV 2xD/c line along with 240 MVar switchable line reactor for each circuit at each end - Siwani – Jind (PG) 400 kV D/c line (Quad) - Siwani – Patran (Indi Grid) 400 kV D/c line (Quad) along with 80 MVar switchable line reactor for each circuit at Siwani S/s end - Siwani – Fatehabad (PG) 400 kV D/c line (Quad) - Siwani – Sonipat (PG) 400 kV D/c line (Quad) along with 63 MVar switchable line reactor for each circuit at Siwani S/s end - STATCOM (2x±300MVar) along with MSC (4x125 MVar) & MSR (2x125 MVar) along with 2 nos. 400kV bays at Siwani S/s 400kV Sectionalizer bay (1 set) at at Siwani S/s D. <u>Additional Transmission system for Inter-regional power transfer:</u>										

Sl. No.	Application ID	Applicant	Project Location	Submission Date	Nature of Applicant	Criterion for applying	Injection at ISTS point (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)	Conn BGs to be submitted
1. Establishment of 765kV Mandsaur Pooling Station along with 2x330 MVAR (765 kV) Bus Reactors 2. Beawar- Mandsaur PS 765 kV D/c line along with 240 MVAR switchable line reactor for each circuit at each end 3a) Mandsaur PS – Indore(PG) 765 kV D/c Line along with 1x330 MVAR switchable line reactor on each ckt at Mandsaur end of Mandsaur PS – Indore(PG) 765 kV D/c Line OR 3b) Establishment of 765 kV Kurawar S/s with 2x330 MVAR 765 kV bus reactor 3c) Mandsaur – Kurawar 765 kV D/c line along with 240 MVAR switchable line reactors on each ckt at both ends of Mandsaur – Kurawar 765 kV D/c line 3d) LILO of Indore – Bhopal 765 kV S/c line at Kurawar Start Date of Connectivity under GNA: 01.09.2026(Interim). Final date shall be confirmed upon award of the system										
6.	2200000360	Prerak Solar Urja Private Limited	Bikaner distt., Rajasthan	17.11.2023	Renewable Power Park developer (Solar)	Land Route	400	31.05.2025	Bhadla-III PS	• Conn BG1:50 lakh • Conn BG2: Rs. 3 Cr. • Conn BG3: 2 lakh/MW
It was informed that M/s Prerak Solar Urja Private Limited has applied for connectivity for 400 MW at Bhadla-III PS. However, as Bhadla-III PS is already closed after grant of 6500 MW(including 2 bays -600 MW on hold due to CERC order), it was proposed to grant connectivity at Bhadla-IV PS at 220 kV level through 1 no. of 220 kV line bay. M/s Prerak Solar Urja was asked to confirm the 220kV bay scope at Bhadla-IV PS end and the tower configuration. M/s Prerak Solar Urja informed that 220 kV line bay at Bhadla-IV PS end shall be under ISTS and tower configuration shall be S/c line on D/c tower. The same was noted & M/s Prerak Solar Urja was asked to submit the necessary undertaking for S/c line on D/c tower configurations. M/s Prerak Solar Urja agreed for the same. It was also informed that the transmission system for evacuation of power from Bhadla-IV PS is presently under discussion. The scheme is currently tentative, which shall be finalized in a separate joint study meeting among CEA, CTUIL, Grid India & Stakeholders followed by the subsequent CMETS meeting. The exact transmission system for shall be informed upon finalization and approval of the scheme. Further, it was also informed that M/s Prerak Solar Urja Private Limited has requested Start Date of Connectivity under GNA from 31/05/2025. However, considering the same and expected commissioning date of transmission system (i.e. 31.03.2029), the start date of Connectivity under GNA shall be 01.04.2029 (interim). M/s Prerak Solar noted the same. Accordingly, tentative transmission system for connectivity of M/s Prerak Solar Urja Limited under GNA is as below:										

Sl. No.	Application ID	Applicant	Project Location	Submission Date	Nature of Applicant	Criterion for applying	Injection at ISTS point (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)	Conn BGs to be submitted
Details of Transmission system for Connectivity under GNA: A. <u>Associated Transmission System (ATS):</u> NIL B. <u>Transmission System under applicant scope</u> (i). Prerak Solar Urja Private Limited Renewable Power Park– Bhadla-IV PS 220 kV S/c (high capacity) line on D/c tower (Suitable to carry minimum 400 MW at nominal voltage) C. <u>Transmission system for Connectivity under GNA (tentative)</u> 1. Establishment of 765/400kV, 2x1500 MVA & 400/220kV 1x500MVA (1 st) pooling station at suitable location near Bhadla (Bhadla-IV PS) along with 2x125 MVar & 2x240 MVar bus reactor 2. Establishment of 6000 MW, ± 800 kV Merta (HVDC) [LCC] terminal station (4x1500 MW) in Merta-II substation 3. Establishment of 765/400kV, 3x1500 MVA S/s & 400/220kV 3x500MVA pooling station at suitable location near Bhadla (Bhadla-IV PS) along with 2x125 MVar & 2x240 MVar bus reactor 4. Bhadla-IV PS – Merta-II 765KV D/c line along with 240 MVar switchable line reactor for each circuit at each end of Bhadla-IV PS – Merta-II 765KV D/c line 5. Augmentation of 765/400 kV, 3x1500 MVA ICTs (1 st to 3 rd) at Merta-II S/s 6. 6000 MW, ±800 kV Merta (HVDC) [LCC] terminal station (4x1500 MW) 7. Establishment of 6000 MW, ±800 kV Baripada (HVDC) [LCC] terminal station (4x1500 MW) at suitable location near Baripada (Distt. Mayurbhanj) 8. ±800 kV HVDC line (Hexa lapwing) between Merta-II (HVDC) & Baripada (HVDC) (with Dedicated Metallic Return) Start Date of Connectivity under GNA: 01.04.2029(Interim). Final date shall be confirmed upon award of the system										
7.	2200000358 (Enhancement)	NTPC Renewable Energy Limited	Bikaner distt., Rajasthan	22.11.2023	Generator (Solar)	Land Route	250	30.06.2026	Bikaner-III/IV PS	• Conn BG1:50 lakh • Conn BG2: Nil (Bay in sharing) • Conn BG3: 2 lakh/MW
It was informed that M/s NTPC Renewable Energy Limited(NTPCREL) has applied for enhancement of connectivity for 250 MW at Bikaner-IV PS. As discussed earlier with the previous application of NTPC(150 MW, 700 MW & 400 MW), it was proposed to grant connectivity to NTPCREL at Bikaner-IV PS through 400 kV D/c line. For the Conn BG2 point of view, 400 MW & 250 MW applications are considered together with 400 MW application as lead. Further, it was also informed that M/s NTPC Renewable Energy Limited has requested Start Date of Connectivity under GNA from 30/06/2026. However,										

Sl. No.	Application ID	Applicant	Project Location	Submission Date	Nature of Applicant	Criterion for applying	Injection at ISTS point (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)	Conn BGs to be submitted
considering the same and expected commissioning date of transmission system(i.e. 31.08.2026), the start date of Connectivity under GNA shall be 01.09.2026 (interim). M/s NTPC Renewable noted the same. Accordingly, Transmission system for connectivity of NTPC Renewable Energy Limited for 250 MW under GNA is as below: <u>Details of Transmission system for Connectivity under GNA:</u> A. <u>Associated Transmission System (ATS): NIL</u> B. <u>Transmission System under applicant scope</u> (i). Common pooling station for NTPC Renewable Energy Limited Solar Power Projects (App No. 2200000244(150 MW) , 2200000308(700 MW), 2200000373(400 MW) & 2200000358(250 MW) – Bikaner-IV PS 400 kV D/c line (Suitable to carry minimum 900 MW per circuit at nominal voltage) C. <u>Transmission system for Connectivity under GNA</u> 1. Establishment of 6x1500 MVA, 765/400 kV Bikaner-IV Pooling Station along with 2x240 MVar (765kV) & 2x125 MVar (420kV) Bus Reactors at a suitable location near Bikaner 2. 400kV Sectionalizer bay (1 set) at Bikaner-IV PS 3. STATCOM (2x+300MVar) along with MSC (4x125 MVar) & MSR (2x125 MVar) along with 2 nos. 400kV bays at Bikaner-IV PS 4. LILO of both ckts of 400kV Bikaner II PS- Bikaner III PS (Quad) direct line at Bikaner-IV PS 5. Establishment of 765/400 kV, 6x1500 MVA S/s at suitable location near Siwani (Distt. Bhiwani) along with 2x240 MVar (765kV) Bus Reactor & 2x125 MVar (420kV) Bus Reactor 6. Bikaner-IV PS – Siwani 765 kV 2xD/c line along with 240 MVar switchable line reactor for each circuit at each end 7. Siwani – Jind (PG) 400 kV D/c line (Quad) 8. Siwani – Patran (Indi Grid) 400 kV D/c line (Quad) along with 80 MVar switchable line reactor for each circuit at Siwani S/s end 9. Siwani – Fatehabad (PG) 400 kV D/c line (Quad) 10. Siwani – Sonipat (PG) 400 kV D/c line (Quad) along with 63 MVar switchable line reactor for each circuit at Siwani S/s end 11. STATCOM (2x+300MVar) along with MSC (4x125 MVar) & MSR (2x125 MVar) along with 2 nos. 400kV bays at Siwani S/s 12. 400kV Sectionalizer bay (1 set) at at Siwani S/s D. <u>Additional Transmission system for Inter-regional power transfer:</u>										

Sl. No.	Application ID	Applicant	Project Location	Submission Date	Nature of Applicant	Criterion for applying	Injection at ISTS point (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)	Conn BGs to be submitted
1. Establishment of 765kV Mandsaur Pooling Station along with 2x330 MVAR (765 kV) Bus Reactors 2. Beawar- Mandsaur PS 765 kV D/c line along with 240 MVAR switchable line reactor for each circuit at each end 3a) Mandsaur PS – Indore(PG) 765 kV D/c Line along with 1x330 MVAR switchable line reactor on each ckt at Mandsaur end of Mandsaur PS – Indore(PG) 765 kV D/c Line OR 3b) Establishment of 765 kV Kurawar S/s with 2x330 MVAR 765 kV bus reactor 3c) Mandsaur – Kurawar 765 kV D/c line along with 240 MVAR switchable line reactors on each ckt at both ends of Mandsaur – Kurawar 765 kV D/c line 3d) LILO of Indore – Bhopal 765 kV S/c line at Kurawar Start Date of Connectivity under GNA: 01.09.2026(Interim). Final date shall be confirmed upon award of the system										
8.	2200000387	ACME Cleantech Solutions Private Limited	Jaisalmer distt., Rajasthan	29.11.2023	Generator (Solar)	Land BG route	600	31.03.2026	Fatehgarh-II PS	Shall be informed in the next meeting.
9.	2200000396	Acme Cleantech Solutions Private Limited	Jaisalmer distt., Rajasthan	30.11.2023	Generator (Hybrid)	LOA (SJVN)	250	31.03.2026	Fatehgarh-II PS	Shall be informed in the next meeting.
It was informed that M/s Acme Cleantech Solutions Private Limited has applied for connectivity for 600 MW & 250 MW at Fatehgarh-II PS at 400 kV. It was also informed that 1000 MW (500 MW at 220 kV & 500 MW at 400 kV) of connectivity margin was vacated at Fatehgarh-II PS due to surrender of connectivity by applicants under GNA transition process. In the reallocation meeting held on 20.06.23, issues regarding Fatehgarh-II PS falling under GIB additional important area, non clarity on clearance for overhead dedicated tr. Line (DTL) by GIB committee for connectivity at Fatehgarh-II PS as well as request of other RE developers for shifting from Fatehgarh-II PS to other nearest stations was discussed. Based on above, Fatehgarh-II PS was not considered for any reallocation till clarity of GIB clearance of RE developers DTL. In fact no developer showed their interest in occupying vacated margins at Fatehgarh-II PS, rather developers at Fatehgarh-II PS were willing to go out of Fatehgarh-II PS to alternate locations due to GIB issues. However, CTUIL has recently received a request from ABC RJ Land 01 Pvt. Ltd.(380 MW) granted connectivity at Fatehgarh-IV PS(Sec-I) to shift their connectivity from Fatehgarh-IV PS(Sec-II) to Fatehgarh-II PS. The matter is to be examined comprehensively. Accordingly, it was informed that the above applications from M/s ACME for grant of connectivity at Fatehgarh-II PS shall be discussed in the next CMETS NR meeting.										

Sl. No.	Application ID	Applicant	Project Location	Submission Date	Nature of Applicant	Criterion for applying	Injection at ISTS point (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)	Conn BGs to be submitted
It was also informed that, M/s ACME vide letter dated 09.01.24 had also raised its concerns on not taking up its applications (600 MW+250 MW) for grant at Fatehgarh-II PS in 27th CMETS-NR meeting. M/s ACME has also mentioned that ABC RJ Land 01 should have made another application at Fatehgarh-II PS after its application is disposed off by grant of connectivity at Fatehgarh-IV PS and it cannot claim connectivity at Fatehgarh-II PS at a later stage without having any live application pending with CTU. CTU reiterated that the matter is to be examined comprehensively. Accordingly, the above applications from M/s ACME for grant of connectivity at Fatehgarh-II PS shall be discussed in the subsequent CMETS NR meeting.										

C6. Application for Grant of GNA_{RE} to entities other than STU under Regulation 20.1, 20.3 and 20.4 to entities under Regulation 17.1(ii), (iii), (v) and (vi):

Sl No .	Application ID	Name of the Applicant	Submission Date	Region	Nature of applicant	GNA _{RE} within Region (MW)	GNA _{RE} outside Region (MW)	Total Quantum (MW) of GNA _{RE} Required	Start date of GNA _{RE}	End date of GNA _{RE}
1	2200000272	HHP Two Private Limited	06.10.2023	NR	Bulk consumer seeking to connect to ISTS at Sonipat PG)	0.0	110	110	01.05.2026	01.05.2051

It was informed that application for seeking GNA_{RE} as a bulk Consumer interconnected to ISTS (GNA_{RE} of 110MW -outside the region) at Sonipat (PG) was received from M/s HHP Two Private Limited in the month of Oct'23. CTUIL informed that for identification of necessary augmentation for grant of GNA, tentative location of the Bulk consumer is needed which was asked before the CMETS meeting, but same is not yet received. HHP Two Private Limited was again requested to share the tentative location of the project. Accordingly, the grant of GNA_{RE} to HHP Two shall be taken up in the next meeting after receipt of tentative location of the project. M/s HHP Two noted and agreed to share the same

C7. Discussion on Bhadla HVDC. Tr. System

As per queries of some of the RE developers regarding Bhadla HVDC schedule and margin availability on parallel EHVAC Corridor (Bhadla-III PS onwards), it was considered prudent to share transmission planning aspects in specific to Bhadla-III PS.

CTU informed that as part of Rajasthan REZ Ph-III (20GW) Transmission scheme, Bhadla-III & Ramgarh PS are being established for integration of 6.5GW & 2.9 GW RE potential respectively. Ramgarh PS is also being inter-connected with Bhadla-III PS for evacuation of RE power. In order to facilitate evacuation of 9.4GW RE power from Ramgarh/Bhadla-III PS (6.5GW+2.9GW) from Bhadla-III onwards, 765kV Bhadla-III - Sikar-II D/c line with implementation schedule of Mar'25 [for about 2.9GW power transfer requirement] as well as 6GW HVDC corridor (± 800 kV Bhadla (HVDC) -Fatehpur (HVDC)) with implementation schedule (tentative) of Feb'28 (Pole-1) & Aug'28 (Pole-2) shall be implemented as part of Ph-III scheme. The scheme is under advance stage of bidding.

Further, 765kV Bhadla-III - Bikaner-III D/c line was also agreed to meet some power transfer requirement from Bhadla-III PS onwards in 2025-26 timeframe as well as Optimal utilization of EHVAC transmission system beyond Bikaner-III PS while providing flexibility of power transfer from Bhadla/Bikaner RE clusters. The scheme is under bidding.

Considering both above EHVAC schemes, about 3.7 GW power can be transferred from both Ramgarh & Bhadla-III complex which has been allocated to RE developers based on application priority.

Some of the RE developer raised query that in case of some RE projects (with EHV AC system as per grant) getting delayed, whether their margins can be utilized by other RE developers having HVDC as their GNA system. It was clarified that in case of generation delays, for short term, such margins may be available but once the delayed generation commissions the project say in 3-4 months, the margin will exhaust. As Bhadla-III EHVAC & Bhadla HVDC system schedule have 2 years gap, RE developer should plan their project with indicated connectivity start date only at Bhadla-III & Ramgarh PS. However, all efforts shall be made to expedite HVDC transmission system as well.

-----XX-----

Annexure-A

List of Participants of 27th Consultation meeting for Evolving Transmission Schemes in NR held on 10.01.2024

CEA

Shri Nitin Deswal	Deputy Director
Shri Kanhaiya Singh Kushwaha	Assistant Director

NRPC

Shri D.K.Meena	SE(O)
----------------	-------

SECI

Shri R.K.Agarwal	Consultant
Shri Himanshu Gulati	Senior Engineer
Shri Prashant Kumar Upadhyay	Senior Manager

Grid India

Shri Gaurav Malviya	Manager
Shri Raj Kishan	Assistant Manager

CTU

Shri V Thiagarajan	Sr. GM (CTU)
Shri Kashish Bhambhani	GM (CTUIL)
Shri Sandeep Kumawat	DGM (CTU)
Smt. Ankita Singh	Ch. Manager (CTU)
Shri Tej Prakash Verma	Ch. Manager (CTU)
Shri R Narendra Sathvik	Manager (CTU)
Shri Madhusudan Meena	Engineer (CTU)
Shri Prakhar Pathak	Assistant Engineer (CTU)

POWERGRID

Shri Ravindra Nath Gupta	CGM
--------------------------	-----

DGM

Shri Sanjay Arora
Shri Deepak Sarit
Smt. Palak Sinha

CE/PD&C
XEN SS
AE SS

Shri Dr. Om Prakash Mahela

Executive Engineer (PP&D)

Shri Sujeet Kumar

Assistant Engineer

Sh. Anupam Sharma
Sh. ILLa Pant
Sh. Lalit Kumar
Sh. Neeraj Pathak
Sh. Himanshu Balyan

CE(P)
CE(C&R)
SE(SS&NP)
SE(C&R)
EE(C&R)

Shri Rajan Gupta
Shri K. K. Khapa

Shri Rohit Ahuja
Shri Ashish Shukla
Shri Sanjay
Shri Angshuman Rudra
Shri Ajay Pradhan

Anboto Solar Private Limited
Anboto Solar Private Limited
Anboto Solar Private Limited
Avaada Energy Pvt Ltd & Avaada RJBikaner Private Limited
Acme Cleantech Solutions Private Limited

Shri Yogesh Kumar Sanklecha	Acme Cleantech Solutions Private Limited
Shri Tushar Goyal	Acme Cleantech Solutions Private Limited
Shri Ajay Sunkari	AM Green Energy Private Limited
	AK Renewable Infra Private Limited & Shudh Solar Power Private Limited
Shri Rajendra Vyas	AM Green Energy Pvt Ltd
Shri Suchiket Sonar	Bundelkhand Saur Urja Limited
Shri Rajesh Kumar	Clean Max Gamma Private Limited
Shri Ashu Gupta	Clean Max Gamma Private Limited
Smt. Vidisha Dubey	Deshraj Solar Energy Pvt Ltd
Shri Shubham Roy	Deshraj Solar Energy Private Limited
Shri Kapil Pandey	First Energy Private Limited
Smt. Vasundhara Sen	Hazel Hybren Private Limited & Furies Solren Private Limited
Shri Abhinav Gupta	Hazel Hybren Private Limited & Furies Solren Private Limited
Shri Ankur Pathak	HHP Two Private Limited
Shri Ravindra Shekhawat	Iraax International (Opc) Private Limited
Shri Aniket Vyas	Juniper Green Energy Private Limited
Shri Pavan Kumar Gupta	Malur Renewables Private Limited
Shri Ketan mehta	Malur Renewables Private Limited
Shri Vaibhav Roongta	Malur Renewables Private Limited
Shri KC Narula	Malur Renewables Private Limited
Shri Lalit K Mehta	Malur Renewables Private Limited
Shri Animesh Manna	NTPC Renewable Energy Limited
Shri Sandeep Jain	Prerak Solar Urja Private Limited
Shri Kumar Vipul	Radiant Star Solar Park Private Limited
	ReNew Renew Solar (Shakti Six) Private Limited & Auxo Sunlight Private Limited
Shri Mohit Jain	Sunflag Power Ltd.
Shri Ranjan Chhibba	Sunflag Power Ltd.
Shri Rajendra Sharma	Sunflag Power Ltd.
Shri Rajeev Pandey	Solarcraft Power India 4 Private Limited
Shri Ayush Jain	Solarcraft Power India 4 Private Limited
Shri Ashish Agarwal	UPC Renewables
Shri Ashish Garg	

Mr. Fahim Alam
Shri Rohit Gera
Shri Pyasi Aditya
Shri Ajay S

UPC Renewables
Statkraft / Lanco Mandakini Hydro Energy Pvt.Ltd.
Statkraft / Lanco Mandakini Hydro Energy Pvt.Ltd.

**सेंद्रल ट्रान्समिशन यूटिलिटी ऑफ इंडिया लिमिटेड**

(पावर ग्रिड कॉर्पोरेशन ऑफ इंडिया लिमिटेड के स्वामित्व में)

(भारत सरकार का उद्यम)

CENTRAL TRANSMISSION UTILITY OF INDIA LTD.

(A wholly owned subsidiary of Power Grid Corporation of India Limited)

(A Government of India Enterprise)

Ref: CTU/N/ REZ Ph-IV (Part-3&4)

Date: 19th January, 2024

As per distribution list

Sub: Minutes of meeting for Joint study meetings held on 28.12.23 & 09.01.24 to deliberate & finalize the Transmission system for evacuation of power from Bikaner & Fatehgarh/Barmer Complex as part of Rajasthan REZ Ph-IV (Part-3a :2.4GW) and REZ Ph-IV (Part-4 :3.5GW) respectively

Dear Sir,

Please find enclosed the Minutes of meeting for Joint study meetings held on 28.12.23 & 09.01.24 to deliberate & finalize the Transmission system for evacuation of power from Bikaner & Fatehgarh/Barmer Complex as part of Rajasthan REZ Ph-IV (Part-3a :2.4GW) and REZ Ph-IV (Part-4 :3.5GW) respectively through virtual mode.

Thanking you,

Yours Faithfully,

(Sandeep Kumawat)
DGM (CTU)

Encl : Minutes of Meeting

Distribution List:

Chief Engineer (PSP&A – I) Central Electricity Authority Sewa Bhawan, R.K.Puram, New Delhi-110 066	Member Secretary Northern Regional Power Committee 18A, Shaheed Jeet Singh Sansanwal Marg, Katwaria Sarai, New Delhi – 110 016
Director (Power System) Solar Energy Corporation of India Ltd. D-3, 1 st Floor, A wing, Religare Building, District Centre, Saket, New Delhi-110017	Director Ministry of New and Renewable Energy, Block 14, CGO Complex, Lodhi Road, New Delhi-110003
Director (SO) Grid Controller of India Limited (erstwhile Power System Operation Corporation Ltd.) 9 th Floor, IFCI Towers, 61, Nehru Place, New Delhi-110 016	Executive Director Northern Regional Load Despatch Centre 18-A, Qutab Institutional Area, Shaheed Jeet Singh Sansanwal Marg, Katwaria Sarai, New Delhi– 110 016
Director (P&C), HPPTCL, Headoffice, Himfed Bhawan,Panjari, Shimla-171005, Himachal Pradesh	Director(W&P) UP Power Transmission Company Ltd. Shakti Bhawan Extn, 3rd floor, 14, Ashok Marg, Lucknow-226 001
Director (Technical) Punjab State Transmission Corporation Ltd. Head Office, The Mall, Patiala 147001, Punjab	Director (Projects) Power Transmission Corporation of Uttarakhand Ltd. Vidyut Bhawan, Near ISBT Crossing, Saharanpur Road, Majra, Dehradun.
Development Commissioner (Power) Power Development Department Grid Substation Complex, Janipur, Jammu	Director (Technical) Rajasthan Rajya Vidyut Prasaran Nigam Ltd. Vidyut Bhawan, Jaipur, Rajasthan-302005.
Member (Power) Bhakra Beas Management Board Sector-19 B, Madhya Marg, Chandigarh - 160019	Superintending Engineer (Operation) Electricity Circle, 5 th Floor, UT Secretariat, Sector-9 D, Chandigarh - 161009
Director (Operations) Delhi Transco Ltd. Shakti Sadan, Kotla Road, New Delhi-110 002	Director (Technical) Haryana Vidyut Prasaran Nigam Ltd. Shakti Bhawan, Sector-6, Panchkula-134109, Haryana

Minutes of meeting for Joint study meetings held on 28.12.23 & 09.01.24 to deliberate & finalize the Transmission system for evacuation of power from Bikaner & Fatehgarh/Barmer Complex as part of Rajasthan REZ Ph-IV (Part-3a :2.4GW) and REZ Ph-IV (Part-4 :3.5GW) respectively.

Joint study meetings were held on 28.12.23 & 09.01.24 to deliberate & finalize the Transmission system for evacuation of power from Bikaner & Fatehgarh/Barmer Complex as part of Rajasthan REZ Ph-IV (Part-3a :2.4GW) and REZ Ph-IV (Part-4 :3.5GW) respectively.

Details of Deliberations are as under

Gist of discussion in Joint study meeting held on 28.12.23

It was stated that Renewable Energy Zones (REZs) were identified by MNRE/SECI with a total capacity of 181.5 GW for likely benefits by the year 2030 in eight states, which includes 75 GW REZ potential in Rajasthan comprising of 15 GW Wind and 60 GW Solar. In this regard a Committee on Transmission Planning for RE was constituted by MOP for planning of the requisite Inter State Transmission System required for the targeted RE capacity by 2030 for which a Comprehensive transmission plan for evacuation of 75GW RE potential from Rajasthan was evolved. Details of schemes approved/Under Planning scheme as part of above is as under:

S.No	Transmission Scheme	RE Potential	Status
A	Under Bidding/ Approved		
1	Rajasthan REZ Ph-IV (Part-1 :7.7GW) (Bikaner Complex)	14 GW (Solar 14GW, BESS:6GW) Bikaner-II : 3.7GW Bikaner-III: 4GW	Awarded
2	Rajasthan REZ Ph-IV (Part-2 :5.5GW) (Jaisalmer/Barmer Complex)	5.5GW (Solar) Fatehgarh-IV: 4 GW Barmer-I: 1.5 GW	Under Bidding
3	Rajasthan REZ Ph-IV (Part-3 :3.6GW) (Bikaner Complex)	3.6 GW (Solar) Bikaner-IV:3.6GW	Under approval with NCT
B	Planned/Under Planning		
1	Rajasthan REZ Ph-IV (Part-3a :2.4GW) (Bikaner Complex)	2.4 GW (Solar) Bikaner-IV : 2.4GW	Scheme planned for 2026-27 time frame (EHVAC)- Present proposal
2	Rajasthan REZ Ph-IV (Part-4 :3.5GW) (Jaisalmer/Barmer Complex)	3.5 GW (Solar) Fatehgarh-IV: 1 GW Barmer-I: 2.5 GW	
3	Rajasthan REZ Ph-IV (Part-5 : 6GW) (Barmer Complex)	6 GW (Solar) Barmer-II : 6GW	Scheme planned for 2029 time frame (HVDC)
4	Rajasthan REZ Ph-IV (Part-6 : 6GW) (Bhadla/Bikaner Complex)	6 GW (Solar) Bhadla-IV: 2 GW Bikaner-V: 4 GW*	

***SECI to confirm total RE potential at Bikaner-V**

A. Rajasthan REZ Ph-IV (Part-3a :2.4GW) (Bikaner Complex)

It was stated that transmission scheme is evolved for about 7.7GW (Solar) in Bikaner complex (14 GW potential along with 6 GW BESS) in Rajasthan for RE potential identified at Bikaner complex as part of committee report. However, no application of BESS (linked with RE) against envisaged 6GW was received. Accordingly, RE potential of about 7.7GW (in place of 14GW) can be evacuated from planned system (Ph-IV scheme) from Bikaner complex (Bikaner-II(3.7 GW) & Bikaner-III(4 GW)).

Further with development of RE transmission scheme planned as part of Ph-IV transmission schemes in Bikaner and Fatehgarh/Barmer complex, as well as other strengthening schemes, additional margin is created at Bikaner-III PS, which will be evacuated with Bikaner-IV tr. System and allocated as per priority of applications at Bikaner-IV.

Subsequently, Transmission scheme was evolved and approved in 25th CMETS-NR meeting for about 3.6 GW (Solar) as part of Rajasthan REZ Ph-IV (Part-3 :3.6GW) (Bikaner Complex) for injection at Bikaner-IV in Rajasthan. At present connectivity of about 5.4 GW is already agreed/received against its total potential (6GW) and more applications are expected due to land availability and being outside of GIB area. Therefore, HVDC from Bikaner complex is planned.

As part of EHVAC system, evacuation system is planned for 2.4GW capacity (thus makes total capacity of 6 GW at Bikaner-IV PS) comprising Bikaner-IV PS – Churu 765 kV D/c line (2nd line and LILO of 2nd ckt of 765 kV Sikar-II (PG) -Khetri (PG) D/c line at Churu S/s. For onward dispersal of power, Siwani – Sonipat (PG) 400 kV D/c line (Quad) is also planned.

For EHVAC system, studies were carried out in 2026-27 time frame in solar maximized scenario and Study files for solar maximized scenario was shared on 20.12.23 with all constituents. Considering grant of connectivity to new RE generators in Bikaner complex as well as for evacuation of power beyond Bikaner complex, following transmission scheme is proposed for evacuation of power from Rajasthan REZ Ph-IV (Part-3a :2.4GW) [Bikaner complex]. Details of proposed EHVAC scheme is as under

A. Transmission system for evacuation of power from Bikaner Complex as part of Rajasthan REZ Ph-IV (Part-3a :2.4GW)

In the above meeting following transmission system was proposed for evacuation of power from Rajasthan REZ Ph-IV (Part-3a :2.4GW) for evacuation of power from Bikaner complex

Bikaner-IV: 2.4GW (Solar)

- Augmentation of 1x1500 MVA(5th), 765/400 kV ICT at Bikaner-IV Pooling Station
- Augmentation of 3x500 MVA (5th to 7th), 400/220 kV ICT at Bikaner-IV Pooling Station[#]
- 220kV line bays (3 nos.) for RE connectivity at Bikaner-IV PS[#]
- Bikaner-IV PS – Churu 765 kV D/c line (2nd) along with 240 MVA^r switchable line reactor for each circuit at Bikaner-IV PS end (~175 km)
- LILO of other ckt of 765 kV Sikar-II (PG) -Khetri (PG) D/c line at Churu S/s* (~80 km)
- Augmentation of 765/400 kV, 1x1500 MVA ICT (4th) at Siwani S/s
- Siwani – Sonipat (PG) 400 kV D/c line (Quad Moose) (~150 km) along with 63 MVA^r switchable line reactor for each circuit at Siwani S/s end

****one ckt is already LILOed as part of Rajasthan REZ Ph-IV (Part-3 :3.6GW) scheme)***

To be finalized as per application received

In the meeting, CEA stated that with Phase-IV Part 3a (2.4GW) proposed scheme, there will be no requirement of 765/400kV Churu S/s & associated interconnections, which were earlier agreed in 25th CMETS-NR meeting as part of Phase-IV Part 3 (3.6GW) scheme.

CEA stated that in the earlier agreed Phase-IV Part 3 (3.6 GW) scheme, Churu S/s & associated transmission scheme (LILO of one ckt of 765 kV Sikar-II (PG) -Khetri (PG) D/c line at Churu S/s and Churu- Fatehabad (PG) 400 kV D/c line) was considered to address issue on synchronization of line after corridor outage (765kV Bikaner-IV-Siwani D/c line) in N-1-1/N-2 in peak solar hours as emphasized by Grid-India. CEA stated that in place of Bikaner-IV – Churu D/c line, a parallel corridor towards Siwani S/s i.e. 765kV Bikaner-IV- Siwani D/c line may be considered for additional 2.4GW evacuation of power as part of scheme. With the above parallel 765kV corridor i.e. 765kV Bikaner-IV – Siwani D/c line, angular stability issue under N-1-1 contingency will be resolved as 4 nos. of 765kV circuits are available towards Siwani S/s. Further in future, if there is requirement of establishment of Churu S/s, Bikaner-IV – Siwani D/c line may be LILOed at Churu S/s.

Further, it was deliberated that Transmission system for evacuation of power from Bikaner Complex as part of Rajasthan REZ Ph-IV (Part-3 :3.6GW) is being taken up for approval in next NCT. In view of that both the schemes viz Ph-IV: Part 3 (3.6GW) & Ph-IV: Part 3a (2.4GW)) may be considered comprehensively for 6GW evacuation.

CTU stated that they will review the proposal based on CEA observations and deliberations in meeting and evaluate various alternatives (with and wo Churu S/s) in a comprehensive manner. Accordingly, the scheme will be further deliberated and finalized in next Joint study meeting.

Gist of discussion in Joint study meeting held on 09.01.24

As per deliberation held in Joint study meeting held on 28.12.23, CTU examined both the proposals as below with two alternatives

- **Alternative-1** : Bikaner-IV- Churu- Siwani 765kV D/c & Bikaner-IV- Siwani 765kV D/c & associated transmission interconnection at Churu S/s i.e. LILO of one ckt of 765 kV Sikar-II (PG) -Khetri (PG) D/c line at Churu S/s and Churu- Fatehabad (PG) 400 kV D/c line & onwards 400kV transmission system from Siwani S/s (Towards Jind, Patran, Sonipat)
- **Alternative-2**: Bikaner-IV- Siwani 765kV 2xD/c line & onwards 400kV transmission system from Siwani S/s (Towards Jind, Patran, Fatehabad, Sonipat)

Details of both the alternatives are enclosed in **Annexure-1**.

It was stated that recently additional margin of about 600MW (over 4GW planned capacity) was released at Bikaner-III PS, with Bikaner-IV transmission system. In the earlier agreed system (LILLO of one ckt of 765 kV Bikaner-III -Neemrana-II D/c line at Bikaner-IV PS), 765/400kV ICTs at Bikaner-III PS and Bikaner (PG) is higher in N-1 contingency. Space is not available for augmentation of new at 765/400kV ICT (5th) at Bikaner (PG) and 765/400kV ICT (7th) at Bikaner-III PS. Therefore, in place of LILLO of one ckt of 765 kV Bikaner-III -Neemrana-II D/c line at Bikaner-IV PS, LILLO of both ckts of 400kV Bikaner II- Bikaner-III D/c line (Quad) at Bikaner-IV is proposed which will relieve the above ICT loadings and increase the operational margin on above ICTs. Additionally STATCOM at Siwani (± 300 MVar) is proposed in Alternate-2 to mitigate low voltage issue.

Further, Observations were received on 04.01.24 from Grid-India and CEA on both the alternatives for transmission system for evacuation of power from Bikaner complex as part of Rajasthan REZ Ph-IV (Part-3). Major observations are as under

A. Grid-India Observations/Comments

- With a dispatch of 1470 MW of the IC of 1840 MW for RAPP-C, N-1 violations of 400 kV RAPP-C – Shujalpur D/c is being observed for both the cases. However, with the establishment of 765 kV Churu S/S (Alternative - 1), relief of 10-11 MW is seen in each circuit.
- STATCOMs at Bikaner – IV & other stations kept in switched off conditions. All STATCOMs may be modelled and made in service as per approved plan.
- During peak solar period, voltages at Siwani (730kV) and Bikaner-IV (740kV) are very low even after taking B/Rs out. If line reactors are to be taken out of service for maintaining voltages, the same needs to be mentioned along with the scheme. Further, line reactor ratings for 765kV Bikaner-IV- Siwani ckts may be corrected in June base case (330MVAR reactor shown)
- Considering the identified solar potential at Churu by SECI, establishment of Churu substation will help in tapping the solar generation in that complex. CTUIL may clarify its plan and the timelines for establishing the ISTS Churu substation, if the same is not being taken up right now.

Additionally, Grid-India in 25th CMETS-NR meeting opined that in case of direct interconnection of Bikaner-IV to Siwani i.e. not considering 765/400kV Churu in between and onward transmission system, the Transmission system is kind of radial system connected with RE generation pocket and poses stability issues in various operational scenarios in future. Further Grid-India also requested to share Study cases for off peak scenarios. Other study related comments received from Grid-India was also deliberated in above Joint study meeting

B. CEA Observations/Comments

- Power flow in both the alternatives seems to be in order.
- Transmission system requirement is less in Alt 2. This would result in saving of hundreds of crore.
- In Alt 2, with Bikaner-IV- Siwani 765 kV 2xD/c line, issue of synchronization of line after corridor outage (765 kV Bikaner-IV-Siwani D/c line) in N-1-1/N-2 in peak solar hours as highlighted by Grid-India gets resolved.
- As intimated by CTUIL, connectivity applications of 5.4 GW RE has already been received against potential of 6 GW. Therefore, a comprehensive scheme for 6 GW RE evacuation could be implemented.
- In view of the above, Alt 2 seems to be preferable at present. Churu substation could be implemented in future, if required, with LILO of Bikaner-IV- Siwani 765 kV line to cater to the potential in Churu district.
- Inputs regarding firm schedule of requirement of Churu S/s (for RE potential) may be taken from SECI.
- In case Grid-India anticipates or highlights major operational issues other than N-1-1/N-2 in Alt 2, both the alternatives may be reviewed accordingly.

CTU analyzed both the alternatives with above changes and observations are as under :

S.No.	Alternative-1 : Bikaner-IV- Churu- Siwani 765kV D/c & Bikaner-IV- Siwani 765kV D/c & associated transmission interconnection at Churu S/s	Alternative-2: Bikaner-IV- Siwani 765kV 2xD/c line)
1	Rajasthan REZ Ph-IV (Part-3 :3.6GW) is under approval in NCT and change in earlier agreed scheme (3.6GW) will delay the implementation timeline for 3-4 months and may impact the evacuation of RE generation from Bikaner-IV PS. Tentative timeline for scheme (3.6GW) : Jul'26 Tentative timeline for scheme (2.4GW) : Sep'26	Comprehensive scheme for 6GW will be implemented with tentative timeline of Sep'26
2	SECI vide mail 07.12.23 informed that Southwest Churu has solar potential & therefore considering solar potential at Churu as indicated by SECI, 220kV future scope was retained at Churu S/s as earlier agreed in 25th CMETS-NR meeting.	Churu Substation may be planned with future transmission proposals i.e. Bikaner-V, other strengthening schemes based on requirement. Therefore, RE evacuation from Churu S/s will only be possible at later stage (beyond 2027) in Alt-2.
3	Due to shorter line length (Bikaner-IV-Churu & Churu-Siwani section), under N-1 contingency of 765kV Bikaner-IV-Churu-Siwani corridor, voltage dipping at Bikaner-IV and Siwani in Alt-1 will be lesser (about 10kV than Alt-2) and in the range of 740 -750kV considering STATCOM at Bikaner-IV under various operating scenario.	Due to longer line length, under N-1 contingency of 765kV Bikaner-IV-Siwani , voltage dipping in Alt-2 will be higher (about 10kV than Alt-1) and in the range of 730-740kV considering STATCOM at Bikaner-IV under various operating scenario. In above scenario STATCOM (+300MVar) at Siwani may also be required (in case of operational difficulty of line reactors switching on EHVAC lines in RE pockets). Grid-India may comment.
4		Estimated Cost of scheme is about 12% lower (about Rs 1200 Cr) is lower than Alt-1*

* incl. cost of Siwani STATCOM (+300MVar)

Revised study files with proposed changes (LILO of both ckts of 400kV Bikaner II- Bikaner-III D/c line at Bikaner-IV PS in place of LILO of one ckt of 765 kV Bikaner-III -Neemrana-II D/c line at Bikaner-IV PS & Siwani STATCOM (in Alt-2)) was circulated on 06.05.24 along with study plots.

CTU enquired SECI that with Alt-2, Churu substation is not being considered for evacuation of RE potential indicated by SECI in earlier meetings. SECI stated that Churu complex was not identified in solar potential zone as part of 500GW report and thereafter potential was indicated only because the substation was planned as part of agreed Bikaner-IV transmission scheme (3.6GW), however no response from RE developers is yet received to assess & harness RE potential at Churu complex. CTU stated that in above case till no confirmation received from SECI, RE potential at Churu will not be considered and may be evolved based on SECI inputs at later stage.

Grid-India stated that in June solar scenario studies, pf of Punjab/Haryana load is considered near to 0.99 pf which is very optimistic. However, in real time they have faced voltage dipping issues in Punjab and Haryana intra state network and interstate network (incident to intra state) due to high reactive power drawl in paddy season. This Intra state reactive support may

further reduce in paddy season due to envisaged hydro generation from J&K/HP and solar generation from Rajasthan.

CTU enquired to Grid-India about capacitive support by STUs/DISCOMs in real time to improve load pf and mitigate voltage dipping issues in peak loading season. Grid-India stated that in real time capacitor bank support is available in Punjab, however in Haryana low voltage issue is still persist in June-July peak loading season.

CTU stated that in the integrated planning of transmission system, reactive power (MVar) drawl/absorption requirement at load end to be compensated at intra state network level only and ISTS system is not supposed to be planned to compensate MVar injection/ drawl from intra state network under various operating scenario as it will also increase losses due to MVar flow to downstream network from upstream network which is not desirable. In view of that load pf (>0.98 pf) may be maintained by STUs/DISCOMs specially in Punjab & Haryana through new capacitive banks, if required and also provide progress of installing new capacitors in monthly OCC/NRPC meetings.

Considering above, it was decided that STUs specially Haryana may assess the reactive compensation requirement for peak load season in next 2-3 years and may take necessary actions to maintain pf and mitigating low voltage issues in intra state network. HVPN told that they have planned capacitor banks installation at downstream network.

Grid-India stated that in present proposal (in both alternatives), with LILO of both ckts of 400kV Bikaner II- Bikaner-III D/c line (Quad) at Bikaner-IV in place of LILO of one ckt of 765 kV Bikaner-III -Neemrana-II D/c line (2nd) at Bikaner-IV PS, more than 7GW power evacuated through Bikaner-IV system. Grid-India highlighted the voltage dipping issue at Siwani substation (~ 730 kV) in N-1 contingency. The voltage will further decreases to 725kV, with no STATCOM reactive support (STATCOM support reserved for dynamic support only)

CEA stated that line reactors on EHVAC lines can be switched off during low voltage as it will give direct relief to improve voltage to 15-20kV, however operational difficulties in switching on/off of line reactors may be deliberated. Further STATCOM at Siwani S/s may be considered ± 600 MVar (earlier ± 300 MVar proposed in scheme) along with MSC/MSR to provide voltage support.

Grid-India stated that as per the earlier experience, line reactor switching (switched off) is avoided in RE pockets connected through radial system (Fatehgarh/bhadla complex) due to tripping of RE plants in HVRT.

CEA stated the proposed Bikaner-IV system is not radial and proposed to be well connected to Bikaner-II/Bikaner-III system through 400kV D/c lines. CTU stated this at present, only Ph-I and part Ph-II system is commissioned, however in next 2-3 years transmission scheme of about 40-45GW will be commissioned through various interlinking lines (within various RE complexes) as well as inter regional links (NR-WR). With the increase in SCMVA and high SCR of various RE substations, impact of line reactor switching on HVRT tripping may be not as that severe.

Grid-India stated that earlier, voltage in Fatehgarh complex was observed up to 720kV in peak solar hours and under such condition, sequential line reactor switching was carried out in for volage control, however they have faced resistance from TSPs regarding line reactor switching and line opening for voltage control. In view of that provision of switching of line reactors may be clearly stipulated in the RfP of proposed scheme.

CEA stated that line reactor switching in RE pockets is in general operational requirement and need not to be specific for one scheme and should be carried out in real time as per Grid requirement.

CTU stated that presently as per IEC standard followed for breaker duty cycle, class M2 circuit breakers i.e frequently operated CBs for special service requirements and designed so as to require only limited maintenance, mechanical type tested for 10,000 operations.

CEA stated that there is any difference in bus reactors switching and line reactor switching. CTU stated that as per information available, the position of breakers w.r.t reactors is the only difference in bus or line reactors, however its impact to be confirmed by the TSP. however whenever such line reactor switching issue raised by TSP in future, it should come with proper reasoning & justification from TSP side.

Grid-India stated that in new STATCOMs, MSC/MSR may only be used for dynamic range due to controller design features in which MSC/MSR support is only available once certain dynamic reserve is already used. It was also stated that MSC has its own switching time and with fault recovery, MSC may not come in picture Therefore, in steady state study file, STATCOMs along with MSC/MSR may be kept off for worst scenario. To mitigate low voltage issues without STATCOM support, line reactors to be switched off as per requirement.

CTU stated that in base case voltage at Siwani S/s is about 733kV and in N-1 contingency of one ckt of Bikaner-IV-Siwani 2xD/c lines, voltage may further reduce to 728kV. In above scenario, voltage will be improved to 740kV considering STATCOM (at Bikaner-IV and Siwani) and 748kV with STATCOM+MSC at Bikaner-IV & Siwani S/s. However, in case STATCOM along with MSCs is kept off as suggested by Grid-India and line reactor switching is carried out, it will further improve the voltage of Siwani S/s by 750kV (line reactors on other 3 ckts of 765kV Bikaner-IV-Siwani kept off). In view of that it is recommended that steady state voltage to be maintained over 750kV in planning files considering line reactor switching (760kV at Siwani S/s in present case with line reactor on 765kV Bikaner-IV -Siwani kept off) . Accordingly, it was decided by CEA, CTU & Grid-India that line reactor switching may be carried out in a sequential manner in RE pockets for better voltage control.

Grid-India enquired that studies may be rechecked in nine scenarios for any additional requirement. CTU stated in RE related planning studies, studies were carried out in maximum stressed scenarios considering NR peak/off peak scenarios, NR import/export, reactive compensation etc. studies for All India nine scenarios will be shared in March month as part of Rolling plan report and based on which any additional requirement, if any may be discussed separately.

CTU stated that in Feb solar maximized scenario, loading of 400kV RAPP- Shujalpur D/c line is critical (about 940MW) in N-1 contingency. To relieve loading of above line, EHVAC corridor from RAPP generating station or other transmission corridor in parallel to above line to be planned. In 500GW report also, 765kV Ajmer-Kota-Shujalpur transmission corridor was planned to integrate RE complexes of Ajmer. In view of that, it was decided that loading of above line will be reviewed with progress of RE generation projects at Rajasthan and planning to be carried out in next phase to relieve loading on above line. Grid-India & CEA agreed for the same

Grid-India enquired that timeline for implementation of two nos. of Bikaner-IV-Siwani D/c lines will be same or different. CEA stated that scheme will be segregated in various packages and efforts to be made that all the packages are implemented simultaneously with implementation timeline decided in NCT meeting.

Grid-India stated that as per past experiences, there will be always some mismatch (up to 6 months) in implementation time of two parallel double ckt corridors from same RE pooling stations. In such situation, in the event of only one corridor i.e. 765kV Bikaner-IV-Siwani D/c

is commissioned and 2nd 765kV D/c gets delayed, RE generation of 2-3GW may only be evacuated in base case with one D/c and in N-1 contingency, however in N-1-1 contingency, restoration of line becomes delayed in solar peak hours and shall impact the RE generation.

CEA enquired about connectivity timelines of RE generators applying for connectivity at Bikaner-IV. CTU stated that timelines for RE generators is progressively from Jun'25.

CTU stated that considering HVPNL existing & proposed interconnections, interconnections as well as GNA granted to bulk consumer for 100MW GNA at 220kV Jind(PG), 400/220kV ICTs (3x500 MVA) at Jind(PG) are critically loaded in N-1 contingency. However, decision on implementation timeframe of ICT augmentation at Jind (PG) may be taken based on timeline of HVPNL proposed interconnections and future load growth (on 220kV Nain, Jind and IOCL S/s of HVPNL).

Grid-India stated that at present 400/220kV Jind(PG) S/s has 2x500MVA ICTs & 4 nos. of 220kV interconnections whereas 3rd ICT (500MVA) in ISTS by POWERGRID and 4 nos. of STU feeders are under implementation by HVPNL. At present loading on 2x500MVA ICT is about 500MW (peak) and in view of that ICT augmentation at Jind may be considered at later stage with commissioning of HVPNL interconnections & 400/220kV ICT (3rd) at Jind(PG). HVPNL stated that 400kV Munakh S/s for power evacuation of YTPP (800MW) is under planning. Additionally, 640MVA load will be envisaged progressively from 2027 by IOCL. In view of that it was decided that ICT loading will be monitored in real time with implementation of HVPNL feeders and load growth in Jind area. Accordingly, requirements will be assessed based on Grid-India operational feedback report and loading diversion in planning studies considering HVPNL proposal for evacuation of power from YTPP generation (800MW). CEA agreed for the same.

Grid-India also stated to provide off peak file with proposed scheme. CTU stated that off peak file is under preparation and same will be provided within this week. Based on which reactive compensation requirement will be checked and any additional requirement will be included in final proposal. Subsequently, off peak file was circulated on 13.01.24 to all stakeholders. Result of system studies incorporating observations received from stakeholders is enclosed in **Exhibit-1**.

In view of above deliberations, following transmission scheme was agreed in the above Joint study meetings (28.12.23 & 09.01.24)

Transmission system for evacuation of power from Rajasthan REZ Ph-IV (Part-3 : 6GW)

Bikaner-IV: 6 GW (Solar)

- Establishment of 6x1500 MVA, 765/400 kV & 6x500 MVA, 400/220 kV Bikaner-IV Pooling Station along with 2x240 MVar (765kV) & 2x125 MVar (420kV) Bus Reactors at a suitable location near Bikaner *

Future provisions at Bikaner-IV PS (excl. scope for present scheme): Space for

- 765 kV line bays along with switchable line reactors – 6 nos.
- 765kV Bus Reactor along with bay: 1 no.
- 400 kV line bays along with switchable line reactor –4 nos.
- 400 kV line bays–2 nos.
- 400/220kV ICT along with bays –4 nos.
- 400 kV Bus Reactor along with bay: 1 no.
- 400kV Sectionalizer bay: 2 sets
- 220 kV line bays for connectivity of RE Applications -5 nos.
- 220kV Sectionalizer bay: 2 sets
- 220 kV BC (2 nos.) bays and 220 kV TBC (2 nos.) bays

****along with provision of 80MVar spare reactor (Single phase) & 500MVA spare transformer unit (Single phase)***

- 220kV line bays (6 nos.) for RE connectivity at Bikaner-IV PS
- 400kV line bays (3 no.) for RE connectivity at Bikaner-IV PS
- 220kV Sectionalizer bay (1 set) along with BC (2 nos.) bays and 220 kV TBC (2 nos.) bays at Bikaner-IV PS
- 400kV Sectionalizer bay (1 set) at Bikaner-IV PS
- STATCOM (2x±300MVar) along with MSC (4x125 MVar) & MSR (2x125 MVar) along with 2 nos. 400kV bays at Bikaner-IV PS
- LILO of both ckts of 400kV Bikaner II PS- Bikaner III PS (Quad) direct line at Bikaner-IV PS (~20kms)
- Establishment of 765/400 kV, 6x1500 MVA S/s at suitable location near Siwani (Distt. Bhiwani) along with 2x240 MVar (765kV) Bus Reactor & 2x125 MVar (420kV) Bus Reactor*

Future provisions at Siwani S/s(excl. scope for present scheme): Space for

- 765 kV line bays along with switchable line reactors – 6
- 765kV Bus Reactor along with bay: 1 nos.
- 400 kV line bays along with switchable line reactor –4
- 400 kV Bus Reactor along with bays: 1 no.
- 400kV Sectionalizer bay: 1 set

****along with provision of 80MVar spare reactor (Single phase) & 500MVA spare transformer unit (Single phase)***

- Bikaner-IV PS – Siwani 765 kV 2xD/c line along with 240 MVar switchable line reactor for each circuit at each end (~260 km)
- Siwani – Jind (PG) 400 kV D/c line (Quad) (~110 km)
- Siwani – Patran (Indi Grid) 400 kV D/c line (Quad) (~160 km) (400kV GIS duct :700m) along with 80 MVar switchable line reactor for each circuit at Siwani S/s end
- Siwani – Fatehabad (PG) 400 kV D/c line (Quad) (~80 km)
- Siwani – Sonipat (PG) 400 kV D/c line (Quad) (~150 km) along with 63 MVar switchable line reactor for each circuit at Siwani S/s end
- STATCOM (2x±300MVar) along with MSC (4x125 MVar) & MSR (2x125 MVar) along with 2 nos. 400kV bays at Siwani S/s
- 400kV Sectionalizer bay (1 set) at at Siwani S/s

As discussed above, for evacuation of power beyond 6GW from Bikaner-IV PS, HVDC from Bikaner complex is planned. For additional power transfer from Bikaner complex, a new substation at suitable location in Bikaner i.e. Bikaner-V is planned. CTU enquired SECI to confirm RE potential for Bikaner-V PS as 14 GW evacuation system is already planned from Bikaner complex (Bikaner-II/III:8GW, Bikaner-IV: 6GW) against 14 GW RE potential (along with 6 GW BESS) in Rajasthan for RE potential identified at Bikaner complex as part of committee report. SECI informed that they will revert on the same in consultation with MNRE at the earliest.

Further as part of 75GW REZ in Rajasthan (as part of 500GW report), 5 GW RE potential (3GW solar, 2 GW wind) along with net dispatch of 2GW considering 2GW BESS was considered at Bhadla complex (Bhadla-IV: 2GW). A comprehensive HVDC scheme is planned for 6GW evacuation from Bikaner-V (4GW) and Bhadla-IV (2GW) PS towards Baripada in Orissa (ER) through HVDC system, for which studies is undergoing for 2029 timeframe and will be taken up in other Joint study meeting. However grant of connectivity shall depend on

application received at Bikaner-V & Bhadla-IV which may differ than Bikaner-V (4GW) & Bhadla-IV (2GW). Details of broad scheme is attached in **Annexure-2**

B. Transmission system for evacuation of power from Fatehgarh/Barmer Complex as part of Rajasthan REZ Ph-IV (Part-4 :3.5GW)

Gist of discussion in Joint study meeting held on 28.12.23

It was informed that as part of Renewable Energy Zones (REZs) identified by MNRE/SECI with a total capacity of 75 GW REZs in Rajasthan, evacuation system for 13GW was planned at Fatehgarh (5GW) & Barmer (8GW) complex. Transmission scheme for Rajasthan REZ Ph-IV (Part-2:5.5GW) (Jaisalmer/Barmer Complex) was approved in 14th NCT meeting for injection at Fatehgarh-IV PS (4GW) & Barmer-I PS (1.5GW) in Rajasthan and under bidding.

At present connectivity of about 8 GW capacity is already granted/received. For evacuation of power beyond 5.5GW from Fatehgarh-IV PS/Barmer-I PS, hybrid transmission system (EHVAC(3.5GW)+HVDC(6GW)) from Fatehgarh/Barmer complex is planned. Studies were carried out in 2026-27 time frame in solar maximized scenario for EHVAC system and Study files for solar maximized scenario was shared on 20.12.23 with all constituents. Study results were deliberated in above meeting

It was stated in meeting that the EHVAC evacuation system comprises 3.5 GW capacity transmission system from Fatehgarh-IV/Barmer-I (thus making total planned capacity of 9GW (5.5GW+3.5GW)) and 6GW HVDC system from Barmer-II PS (new pooling station is being established as part of HVDC scheme). The 6GW HVDC scheme comprises Barmer-II to Suitable locations in WR/SR through HVDC system.

In the joint study meeting, CTU stated that as per committee report Barmer-II PS was planned for 6GW RE potential (2GW BESS capacity) with 4GW evacuation capacity. In 25th CMETS-NR meeting, SECI informed that at present there is no clear visibility for RE projects with BESS before 2027 as award process will take time (1-2 years). In view of that 7.5 GW RE potential (solar) remains untapped due to non-materialisation of BESS capacity (Fatehgarh-IV:4GW+Barmer-I : 1.5GW+Barmer-II: 2GW). Accordingly, it was suggested that 2 GW additional RE potential may be considered for planning of transmission scheme from Barmer-II PS. SECI informed that they will provide their inputs on enhancement of potential at Barmer-II PS (4GW to 6GW) consultation with MNRE and RE developers.

Details of RE potential considered in Fatehgarh/Barmer complex is under:

S.No	Transmission Scheme	RE Potential	Cumulative RE Potential
1	Rajasthan REZ Ph-IV (Part-2:5.5GW) (Jaisalmer/Barmer Complex) EHVAC system -Under Bidding	5.5GW (Solar) Fatehgarh-IV: 4 GW Barmer-I: 1.5 GW	5.5GW (Solar) Fatehgarh-IV: 4 GW Barmer-I: 1.5 GW
2	Rajasthan REZ Ph-IV (Part-4:3.5GW) (Jaisalmer/Barmer Complex) EHVAC system-Present proposal	3.5 GW (Solar) Fatehgarh-IV: 1 GW Barmer-I: 2.5 GW	9GW (Solar) Fatehgarh-IV: 5 GW Barmer-I: 4 GW
3	Rajasthan REZ Ph-IV (Part-5 : 6GW) (Barmer Complex)	6 GW (Solar) Barmer-II : 6 GW*	15 GW (Solar) Fatehgarh-IV: 5 GW

S.No	Transmission Scheme	RE Potential	Cumulative RE Potential
	HVDC system towards WR/WR-SR		Barmer-I: 4 GW Barmer-II: 6 GW*

**Subject to inputs from MNRE/SECI for enhancement of RE potential at Barmer-II PS*

In the Joint study meeting held on 28.12.2023, following transmission system for evacuation of power from Fatehgarh/Barmer Complex as part of Rajasthan REZ Ph-IV (Part-4 :3.5GW) was proposed

Fatehgarh-IV: 1 GW (Solar), Barmer-I :2.5GW (Solar)

- Establishment of 765/400kV, 2x1500 MVA S/s at suitable location near Merta (Merta-II Substation) along with 2x125 MVar & 2x240 MVar bus reactor at Merta-II* S/s
- STATCOM (2x+300MVar) along with MSC (4x125 MVar) & MSR (2x125 MVar) along with 2 nos. 400kV line bays at Barmer-I PS
- Augmentation with 765/400 kV, 2x1500 MVA Transformer (4th & 5th) at Barmer-I PS
- Augmentation of 3x500 MVA (3rd to 5th), 400/220 kV ICT at Barmer-I PS
- 220kV line bays (4 nos.) for RE connectivity at Barmer-I PS
- 220kV Sectionalization bay (1 set) along with BC (1 nos.) and 220 kV TBC (1 nos.) at Barmer-I PS
- Fatehgarh-IV PS – Barmer-I PS 400kV D/c line (Quad) (~40km)
- Barmer-I PS – Merta-II 765 kV D/c line along with 240 MVar switchable line reactor for each circuit at each end of Barmer-I PS – Merta-II 765 kV D/c line (~320 km)
- Merta-II – Beawar 400 kV D/c line (Quad) (~55 km)
- Merta-II – Dausa 765 kV D/c line along with 240 MVar switchable line reactor for each circuit at each end of Merta-II – Dausa 765 kV D/c line line (~250 km)
- Establishment of 765/400kV, 2x1500 MVA S/s at suitable location near Ghiror (Distt. Mainpuri) along with 2x125 MVar & 2x240 MVar bus reactor at Ghiror S/s (UP)
- Dausa - Ghiror 765 kV D/c line along with 240 MVar switchable line reactor for each circuit at each end of Dausa - Ghiror 765 kV D/c line (~285 km)
- LILO of both ckt of 765 kV Aligarh (PG) -Orai (PG) D/c line (~40 km) at Ghiror S/s along with 240 MVar switchable line reactor for each circuit at Ghiror S/s end of 765 kV Ghiror -Orai D/c line
- LILO of one ckt of 765kV Agra (PG) – Fatehpur(PG) 765kV D/c line at Ghiror along with 240 MVar switchable line reactor at Ghiror S/s end of 765 kV Ghiror -Fatehpur line (~40 kms)

In above meeting, Grid-India enquired about installation of STATCOMs at new pooling stations in Bikaner and Fatehgarh/Barmer complex. CTU stated that at present 3 nos. STATCOMs are already operational at Fatehgarh-II, Bhadla-II & Bikaner-II and 2 nos. STATCOMs at Ramgarh and Fatehgarh-III are under implementation. Additionally, 3 nos. STATCOMs at Bikaner-IV, Siwani and Barmer-I is also planned in proposed schemes (Ph-IV: Part 3 & 4). Additionally, space provision to be kept STATCOM at Merta-II, & Ghiror S/s as part of future scope.

Grid-India stated that proposal is in order, however 400/220kV ICTs (2x315MVA) at Agra (PG) is critically loaded in N-1 contingency. CTU stated that with proposed scheme there is marginally increment (~5MW) in 400/220kV ICT loading in Agra. CTU stated that space is not available for installation of 3rd ICT at Agra(PG) S/s, therefore replacement of ICTs may be considered in future based on real time loadings.

CTU stated that in the present proposal they have optimally utilized the Aligarh-Orai and Fatehpur -Agra high capacity 765kV corridors as the loadings of lines are less in solar maximized scenario. 765/400kV Orai Substation is well connected with WR Grid and Power

from 765/400kV Orai shall be dispersed to load centres of WR through various high capacity lines. CTU stated that they are planning to interconnect proposed Ghiror S/s to nearby UPPTCL substation i.e. 400/220kV Firozabad at 400kV level as part of present scheme. CTU requested UPPTCL to provide space availability for 2 nos. of 400kv line bays at Firozabad S/s. This will also relieve the loading of Agra ICTs and increase resiliency of system. UPPTCL agreed for the same.

UPPTCL requested that studies may be performed with Bhadla-Fatehpur HVDC system to check loadings with proposed system. CTU stated that Bhadla-Fatehpur system is under bidding with scheduled implementation Feb'28 (Pole-1) and therefore not considered in studies, however they will carry out sensitivity case considering HVDC system and commensurate RE generation and will provide to UPPTCL. Same was provided to UPPTCL on 02.01.24.

CEA stated that proposal in order however, studies may be reviewed with UP intra state transmission system and associated RE generation in GEC-II. CTU agreed for the same. CTU enquired SECI and UPPTCL for envisaged RE potential in Bundelkhand region. UPPTCL stated that UPNEDA communicated 10GW RE potential in Bundelkhand region (UP), out of which UPPTCL has planned transmission scheme for evacuation of 4GW solar in intra state and transmission scheme is already approved in GEC-II. CTU stated that they have also received 1200MW solar application in Bundelkhand region for injection at Orai S/s which will be discussed in ensuing CMETS-NR meeting. It was stated that the above 4GW potential (out of 10GW potential) is part of 500GW report, however balance 6GW potential as indicated by UPNEDA is not part of 500GW report, therefore UPPTCL may provide communication to CEA/CTU in 2-3 days, so same can be forwarded to SECI for evaluation of developable RE potential in this region. UPPTCL agreed for the same

Further, UPPTCL informed in the meeting that space is available for 2 nos. of 400kV line bays at Firozabad S/s however, they shall confirm the same after verification. Prima facie loading of 400kV Ghiror- Firozabad D/c line is in order. In view of that CTU included 400kV Ghiror-Firozabad as part of proposed Transmission system for evacuation of power from Rajasthan REZ Ph-IV (Part-4 :3.5 GW) [Fatehgarh/Barmer] Complex. This will also relieve the loading at Agra ICTs and increase resiliency of the system.

Revised study files with proposed changes (considering 400kV Ghiror- Firozabad D/c line & Barmer-I STATCOM as part of proposed scheme) and considering UP intra state solar generation and associated transmission scheme (GEC-II) was circulated on 01.05.24 along with study plots.

Gist of discussion in Joint study meeting held on 09.01.24

CTU stated that in Joint study meeting held on 28.12.23, Transmission system for evacuation of power from Fatehgarh/Barmer Complex as part of Rajasthan REZ Ph-IV (Part-4 :3.5GW) was in-principally agreed.

Grid-India also stated that the surplus power in NR and WR during solar peak period is flowing towards ER and high loading is observed in 765 kV Ranchi - Dharamjaygarh line and other EHV lines towards ER with higher angular separation after N-1. Due to high loading, low voltages observed in ER region on various EHVAC buses.

CTU stated that based on Grid-India observations, in revised studies, Guzuwaka HVDC is already reversed (1000MW) towards ER and Talchar-Kolar capacity is also kept minimum in forward direction (600-700MW). This will relieve loading of various WR-ER corridors incl. 765kV Ranchi – Dharamjaygarh D/c up to a certain extent.. The files were already sent to

other planning groups (CTU) to review loadings (ER-WR) commensurate to envisaged thermal generation in both the regions.

CTU stated the loading of some of WR-ER corridors are in general high and with proposed scheme (Rajasthan Ph-IV (Part-4 :3.5 GW)), there is marginal increment on above Inter regional links in winter solar maximized scenario. In the next phase of studies (2028-29) with proposed HVDC corridors from NR to ER and NR to WR, loadings will be reviewed, and measures will be taken if required. Grid-India & CEA agreed for the same

CTU enquired about RE potential at Ghiror complex for provision of 220kV scope at Ghiror S/s. SECI replied that solar irradiation level at Ghiror complex is less and at present they are not anticipating RE applications for injection at Ghiror. In view of that it was decided that 220kV future scope shall not be considered at Ghiror for RE injection.

UPPTCL informed that space is available at 400kV Firozabad S/s for interconnection of proposed 400kV Ghiror – Firozabad D/c line and power flow is in order on the above line. CTU requested UPPTCL to provide the above information in the mail at the earliest. Regarding RE potential in Bundelkhand region, UPPTCL stated that they have received inputs from UPNEDA however UPPTCL sought some clarifications as well as some additional inputs from UPNEDA. CTU requested UPPTCL to provide the same at the earliest.

Further CTU enquired for drawl requirement at Merta-II and Ghiror S/s. UPPTCL stated that in view of growing load of UP, space provision may be kept for future drawl requirement at Ghiror S/s, however they will confirm the same on mail. RVPN stated that they will revert with the drawl requirement at Merta-II S/s. No other comments received from any Stakeholders. Revised study files with proposed changes in Ph-IV (Part-3: 6GW) and incorporating stakeholder comments on all India LGB, study files were circulated on 06.05.24 along with study plots.

As discussed earlier, off peak file were circulated on 13.01.24 to all stakeholders and based on voltages specially in Ghiror complex, reactive compensation was reviewed. Result of system studies incorporating observations received from stakeholders is enclosed in **Exhibit-1**

In view of above deliberations, following transmission scheme was agreed in above Joint study meetings (28.12.23 & 09.01.24)

Transmission system for evacuation of power from Rajasthan REZ Ph-IV (Part-4 :3.5 GW) [Fatehgarh/Barmer] Complex

Fatehgarh-IV: 1 GW (Solar), Barmer-I :2.5GW (Solar)

- Establishment of 765/400kV, 2x1500 MVA S/s at suitable location near Merta (Merta-II Substation) along with 2x125 MVAR & 2x240 MVAR bus reactor at Merta-II* S/s

Future provisions at Merta-II S/s(excl. scope for present scheme): Space for

- 765/400kV ICTs along with bays- 4
- 765 kV line bays along with switchable line reactors – 8
- 765kV Bus Reactor along with bay: 1 nos.
- 400 kV line bays along with switchable line reactor –8
- 400 kV line bays – 2 nos.
- 400 kV Bus Reactor along with bays: 1 no.
- 400kV Sectionalizer bay: 2 sets
- 400/220kV ICT along with bays -6 nos.
- 220 kV line bays for RE injection -8 nos.
- 220kV Sectionalizer bay: 2 set

- 220 kV BC (3 nos.) bays and 220 kV TBC (3 nos.) bays
- 6000 MW, ± 800 kV Merta (HVDC) [LCC] terminal station (4x1500 MW) along with bays
- STATCOM (2x $\pm$ 300MVA, 4x125MVA MSC, 2x125MVA MSR) along with 400kV bays (2 nos.)

**along with provision of 80-MVA & 110MVA spare reactors (Single phase) & 500MVA spare transformer unit (Single phase)*

- STATCOM (2x $\pm$ 300MVA) along with MSC (4x125 MVA) & MSR (2x125 MVA) along with 2 nos. 400kV bays at Barmer-I PS
- Augmentation with 765/400 kV, 2x1500 MVA Transformer (4th & 5th) at Barmer-I PS
- Augmentation of 5x500 MVA (5th to 9th), 400/220 kV ICTs at Barmer-I PS
- 220kV line bays (6 nos.) for RE connectivity at Barmer-I PS
- 220kV Sectionalizer bay (1 set) along with 220 kV BC (1 nos.) bay and 220 kV TBC (1 nos.) bay at Barmer-I PS
- 400kV Sectionalizer bay (1 set) at Barmer-I S/s
- Fatehgarh-IV PS (Sec-2) – Barmer-I PS 400kV D/c line (Quad) (~25km)
- Barmer-I PS – Merta-II 765 kV D/c line along with 240 MVA switchable line reactor at Barmer-I PS end and 330MVA switchable line reactor at Merta-II PS end for each circuit of Barmer-I PS – Merta-II 765 kV D/c line (~320 km)
- Merta-II – Beawar 400 kV D/c line (Quad) (~55 km)
- Merta-II – Dausa 765 kV D/c line along with 240 MVA switchable line reactor for each circuit at each end of Merta-II – Dausa 765 kV D/c line line (~250 km)
- Establishment of 765/400kV, 2x1500 MVA S/s at suitable location near Ghiror (Distt. Mainpuri) along with 2x125 MVA & 2x240 MVA bus reactor at Ghiror S/s (UP)

Future provisions at Ghiror S/s (excl. scope for present scheme): Space for**

- 765/400kV ICTs along with bays- 4
- 765 kV line bays along with switchable line reactors – 6
- 765kV Bus Reactor along with bay: 1 nos.
- 400 kV line bays along with switchable line reactor –6
- 400 kV Bus Reactor along with bays: 1 no.
- 400kV Sectionalizer bay: 1 set
- 400/220kV ICT along with bays -4 nos.
- 220 kV line bays for drawl -6 nos.
- 220kV Sectionalizer bay: 1 set
- 220 kV BC (2 nos.) bay and 220 kV TBC (2 nos.) bay
- STATCOM (2x $\pm$ 300MVA, 4x125MVA MSC, 2x125MVA MSR) along with 400kV bays (2 nos.)

**along with provision of 80MVA spare reactor (Single phase) & 500MVA spare transformer unit (Single phase)*

***Drawl requirement at Ghiror S/s to be confirmed by UPPTCL, accordingly future scope will be finalized*

- Dausa - Ghiror 765 kV D/c line along with 240 MVA switchable line reactor for each circuit at each end of Dausa - Ghiror 765 kV D/c line (~285 km)
- LILO of both ckt of 765 kV Aligarh (PG) -Orai (PG) D/c line (~40 km) at Ghiror S/s along with 240 MVA switchable line reactor for each circuit at Ghiror S/s end of 765 kV Ghiror -Orai D/c line
- LILO of one ckt of 765kV Agra (PG) – Fatehpur(PG) 765kV D/c line at Ghiror along with 240 MVA switchable line reactor at Ghiror S/s end of 765 kV Ghiror -Fatehpur line (~40 kms)
- 400kV Ghiror-Firozabad (UPPTCL) D/c line (Quad) (~40 kms)

It was also deliberated that HVDC system is to be planned for additional 6GW at Barmer complex with 2029 implementation schedule. Studies for HVDC scheme is undergoing and will be taken up in next Joint study meeting. Details of broad HVDC scheme is enclosed in

Annexure-2

Alternative-I

A. Ph-IV (Part-3 : 3.6GW) at Bikaner-IV PS (Already agreed in 25th CMETS-NR meeting)

- Establishment of 4x1500 MVA, 765/400 kV & 4x500 MVA, 400/220 kV Bikaner-IV Pooling Station along with 2x240 MVA (765kV) & 2x125 MVA (420kV) Bus Reactors at a suitable location near Bikaner
- STATCOM (2x±300MVA) along with MSC (4x125 MVA) & MSR (2x125 MVA) along with 2 nos. 400kV line bays at Bikaner-IV PS
- 220kV line bays (5 nos.) for RE connectivity at Bikaner-IV PS
- 400kV line bays (1 no.) for RE connectivity at Bikaner-IV PS
- 220kV Sectionalization bay (1 set) along with BC (2 nos.) and 220 kV TBC (2 nos.) at Bikaner-IV PS
- 400kV Sectionalization bay (1 set) at Bikaner-IV PS
- LILO of both ckts of 400kV Bikaner II PS- Bikaner III PS (Quad) at Bikaner-IV PS
- Establishment of 765/400 kV, 2x1500 MVA S/s at suitable location near Churu along with 2x240 MVA (765kV) Bus Reactor & 2x125 MVA (420kV) Bus Reactor
- Bikaner-IV PS – Churu 765 kV D/c line along with 240 MVA switchable line reactor for each circuit at Bikaner-IV PS end (~180 km)
- LILO of one ckt of 765 kV Sikar-II (PG) -Khetri (PG) D/c line at Churu S/s (~90 km)
- Churu – Fatehabad (PG) 400 kV D/c line (Quad) along with 63 MVA switchable line reactor for each circuit at Churu S/s end (~150 km)
- Establishment of 765/400 kV, 3x1500 MVA S/s at suitable location near Siwani (Distt. Bhiwani) along with 2x240 MVA (765kV) Bus Reactor & 2x125 MVA (420kV) Bus Reactor
- Siwani – Churu 765 kV D/c line (~95 km)
- Siwani – Jind (PG) 400 kV D/c line (Quad Moose) (~110 km)
- Siwani – Patran (Indi Grid) 400 kV D/c line (Quad) (~160 km) (400kV GIS duct :700m) along with 80 MVA switchable line reactor for each circuit at Siwani S/s end

B. Ph-IV (Part-3a : 2.4GW) at Bikaner-IV PS (New Proposal)

- Augmentation of 1x1500 MVA(5th), 765/400 kV ICT at Bikaner-IV Pooling Station
- Augmentation of 3x500 MVA (5th to 7th), 400/220 kV ICT at Bikaner-IV Pooling Station[#]
- 220kV line bays (3 nos.) for RE connectivity at Bikaner-IV PS[#]
- Bikaner-IV PS – Siwani 765 kV D/c line along with 240 MVA switchable line reactor for each circuit at each end (~260 km)
- Augmentation of 765/400 kV, 2x1500 MVA ICT (4th & 5th) at Siwani S/s
- Siwani – Sonipat (PG) 400 kV D/c line (Quad Moose) (~150 km) along with 63 MVA switchable line reactor for each circuit at Siwani S/s end

[#] To be finalized as per application received

Alternative-II

Bikaner-IV: 6 GW (Solar)

- Establishment of 5x1500 MVA, 765/400 kV & 7x500 MVA[#]. 400/220 kV Bikaner-IV Pooling Station along with 2x240 MVA (765kV) & 2x125 MVA (420kV) Bus Reactors at a suitable location near Bikaner
- STATCOM (2x±300MVA) along with MSC (4x125 MVA) & MSR (2x125 MVA) along with 2 nos. 400kV line bays at Bikaner-IV PS
- 220kV line bays (8 nos.) for RE connectivity at Bikaner-IV PS [#]
- 400kV line bays (3 no.) for RE connectivity at Bikaner-IV PS

- 220kV Sectionalization bay (1 set) along with BC (2 nos.) and 220 kV TBC (2 nos.) at Bikaner-IV PS
- 400kV Sectionalization bay (1 set) at Bikaner-IV PS
- LILO of both ckts of Bikaner II PS- Bikaner III PS (Quad) at Bikaner-IV PS
- Establishment of 765/400 kV, 6x1500 MVA S/s at suitable location near Siwani (Distt. Bhiwani) along with 2x240 MVA (765kV) Bus Reactor & 2x125 MVA (420kV) Bus Reactor
- Bikaner-IV PS – Siwani 765 kV 2xD/c line along with 240 MVA switchable line reactor for each circuit at each end (~260 km)
- Siwani – Jind (PG) 400 kV D/c line (Quad Moose) (~110 km)
- Siwani – Patran (Indi Grid) 400 kV D/c line (Quad) (~160 km) (400kV GIS duct :700m) along with 80 MVA switchable line reactor for each circuit at Siwani S/s end
- Siwani – Fatehabad (PG) 400 kV D/c line (Quad) (~80 km)
- Siwani – Sonipat (PG) 400 kV D/c line (Quad Moose) (~150 km) along with 63 MVA switchable line reactor for each circuit at Siwani S/s end
- STATCOM (± 300 MVA) along with MSC (2x125 MVA) & MSR (1x125 MVA) along with 2 nos. 400kV line bays at Siwani S/s

To be finalized as per application received

A. Proposed Transmission system for evacuation of power from Rajasthan REZ Ph-IV (Part-5 :6 GW) [Barmer Complex] Barmer-II : 6GW (Solar)

- Establishment of 400/220kV, 4x500MVA S/s at suitable location near Barmer (Barmer-II Substation) along with 2x125 MVar bus reactor

Future provisions at Barmer-II S/s:

Space for

- 400 kV line bays along with switchable line reactor –6
- 400 kV line bays –4
- 400 kV Bus Reactor along with bays: 1 no.
- 400/220 kV ICT along with bays -10 Nos.
- 400 kV Sectionalization bays: 2 sets
- 220 kV line bays for connectivity of RE Applications -15 Nos.
- 220kV Sectionalization bay: 3 sets
- 220 kV BC (4 Nos.) & TBC (4 Nos.)
- 220kV line bays (4 nos.) for RE connectivity at Barmer-II PS
- Barmer-I PS – Barmer-II PS 400kV D/c line (Quad)
- Establishment of 6000 MW, $\pm$ 800 kV Barmer-II (HVDC) [LCC] terminal station (4x1500 MW) at a suitable location near Barmer-II substation
 - 400/33 kV, 2x50 MVA transformers for exclusively supplying auxiliary power to HVDC terminal.
 - 400kV bus sectionaliser -2 nos (1 Set) at Barmer-II (HVDC) station
- Establishment of 6000 MW, $\pm$ 800 kV (HVDC) [LCC] terminal station (4x1500 MW) at suitable location in WR/SR or 2 nos. 3000MW, $\pm$ 800 kV (HVDC) [LCC] terminal stations (Multi terminal) at suitable location(s) in WR/SR
- $\pm$ 800 kV HVDC line (Hexa lapwing) between Barmer-II (HVDC) & WR /SR (HVDC) (with Dedicated Metallic Return)
- Associated EHVAC system strengthening in WR/SR

Exact HVDC terminal points in WR/WR-SR boundary are proposed to be identified based on detailed system studies involving stakeholders from NR, WR & SR. Decision on HVDC technology i.e. LCC/ VSC will be carried out at later stage during joint study meeting

B. Proposed Transmission system for evacuation of power from Rajasthan REZ Ph-IV (Part-6 :6 GW) [Bhadla/Bikaner Complex]

Bhadla-IV: 2 GW (Solar), Bikaner-V : 4GW (Solar)

- Establishment of 6000 MW, $\pm$ 800 kV Merta (HVDC) [LCC] terminal station (4x1500 MW) in Merta-II substation
 - 400/33 kV, 2x50 MVA transformers for exclusively supplying auxiliary power to HVDC terminal.
 - 400kV bus sectionaliser -2 nos (1 Set) at Merta-II (HVDC) station
- Establishment of 765/400kV, 4x1500 MVA S/s & 400/220kV 4x500MVA pooling station at suitable location near Bikaner (Bikaner-V PS) along with 2x125 MVar & 2x240 MVar bus reactor

Future provisions

Space for

- 765/400kV ICTs along with bays- 2
- 765 kV line bays along with switchable line reactors – 10
- 765kV Bus Reactor along with bay: 1 nos.
- 400 kV line bays along with switchable line reactor –10
- 400kv line bays : 4 nos.
- 400 kV Bus Reactor along with bays: 1 no.
- 400kV Sectionalization bay: 2 sets
- 400/220kV ICT along with bays -6 nos.
- 220 kV line bays - 10 nos.
- 220kV Sectionalization bay: 3 sets

****along with provision of 80-MVAr spare reactor (Single phase) & 500MVA spare transformer unit (Single phase)***

- 220kV line bays (4 nos.) for RE connectivity at Bikaner-V PS
- Bikaner-V PS – Merta-2 765KV D/c line (~ 200kms) along with 240 MVAr switchable line reactor for each circuit at each end of Bikaner-V PS – Merta-2 765KV D/c line
- Establishment of 765/400kV, 3x1500 MVA S/s & 400/220kV 3x500MVA pooling station at suitable location near Bhadla (Bhadla-IV PS) along with 2x125 MVAr & 2x240 MVAr bus reactor

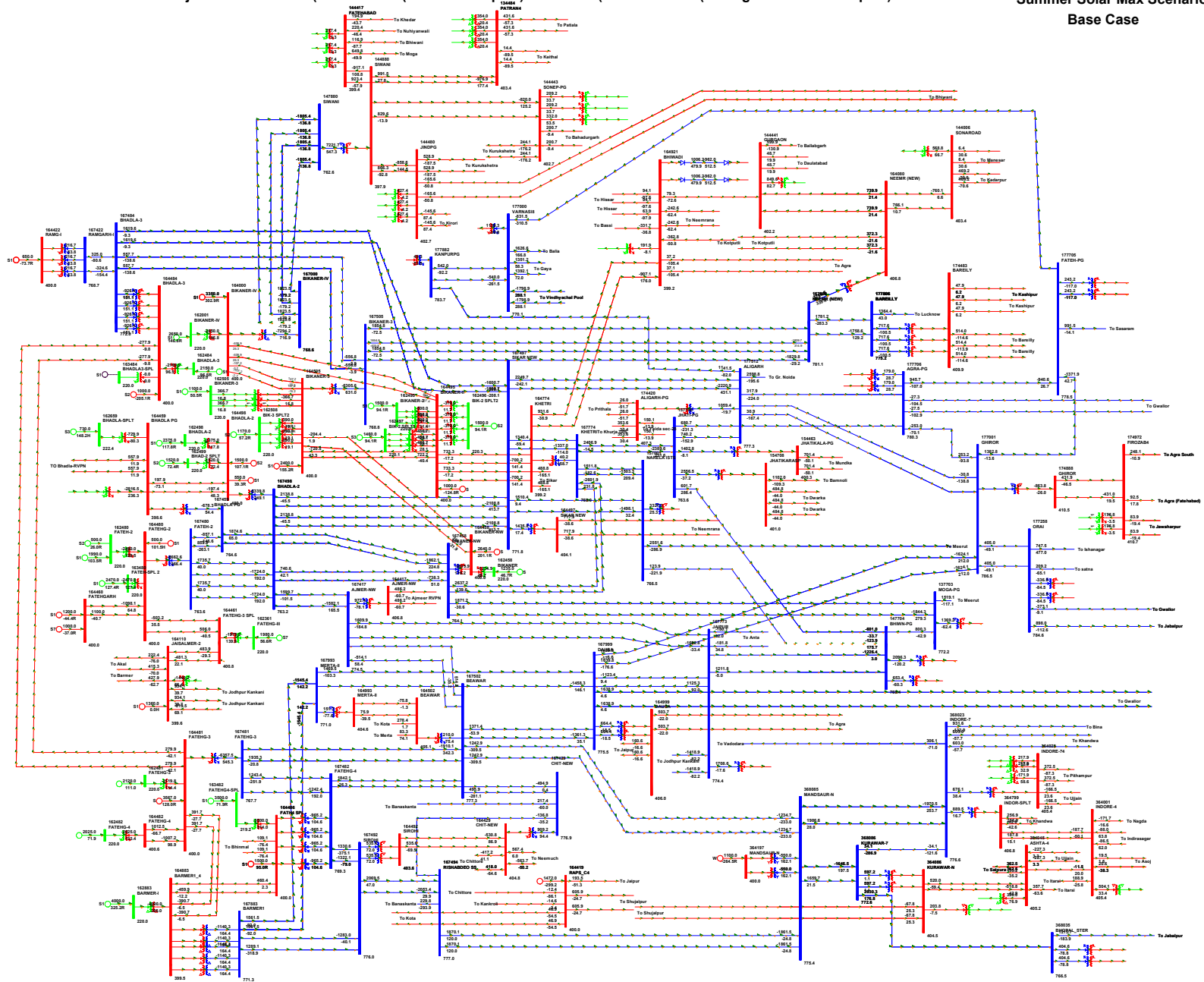
Future provisions

Space for

- 765/400kV ICTs along with bays- 3
 - 765 kV line bays along with switchable line reactors – 10
 - 765kV Bus Reactor along with bay: 1 nos.
 - 400 kV line bays along with switchable line reactor –10
 - 400kv line bays : 4 nos.
 - 400 kV Bus Reactor along with bays: 1 no.
 - 400kV Sectionalization bay: 2 sets
 - 400/220kV ICT along with bays -7 nos.
 - 220 kV line bays - 10 nos.
 - 220kV Sectionalization bay: 3 sets
-
- Bhadla-IV PS – Merta-II 765KV D/c line (~ 250kms) along with 240 MVAr switchable line reactor for each circuit at each end of Bhadla-IV PS – Merta-2 765KV D/c line
 - Augmentation of 765/400 kV, 3x1500 MVA ICTs at Merta-II S/s
 - 6000 MW, ± 800 kV Merta (HVDC) [LCC] terminal station (4x1500 MW)
 - Establishment of 6000 MW, ± 800 kV Baripada (HVDC) [LCC] terminal station (4x1500 MW) at suitable location near Baripada (Distt. Mayurbhanj)
 - ± 800 kV HVDC line (Hexa lapwing) between Merta-II (HVDC) & Baripada (HVDC) (with Dedicated Metallic Return) - 1600 km

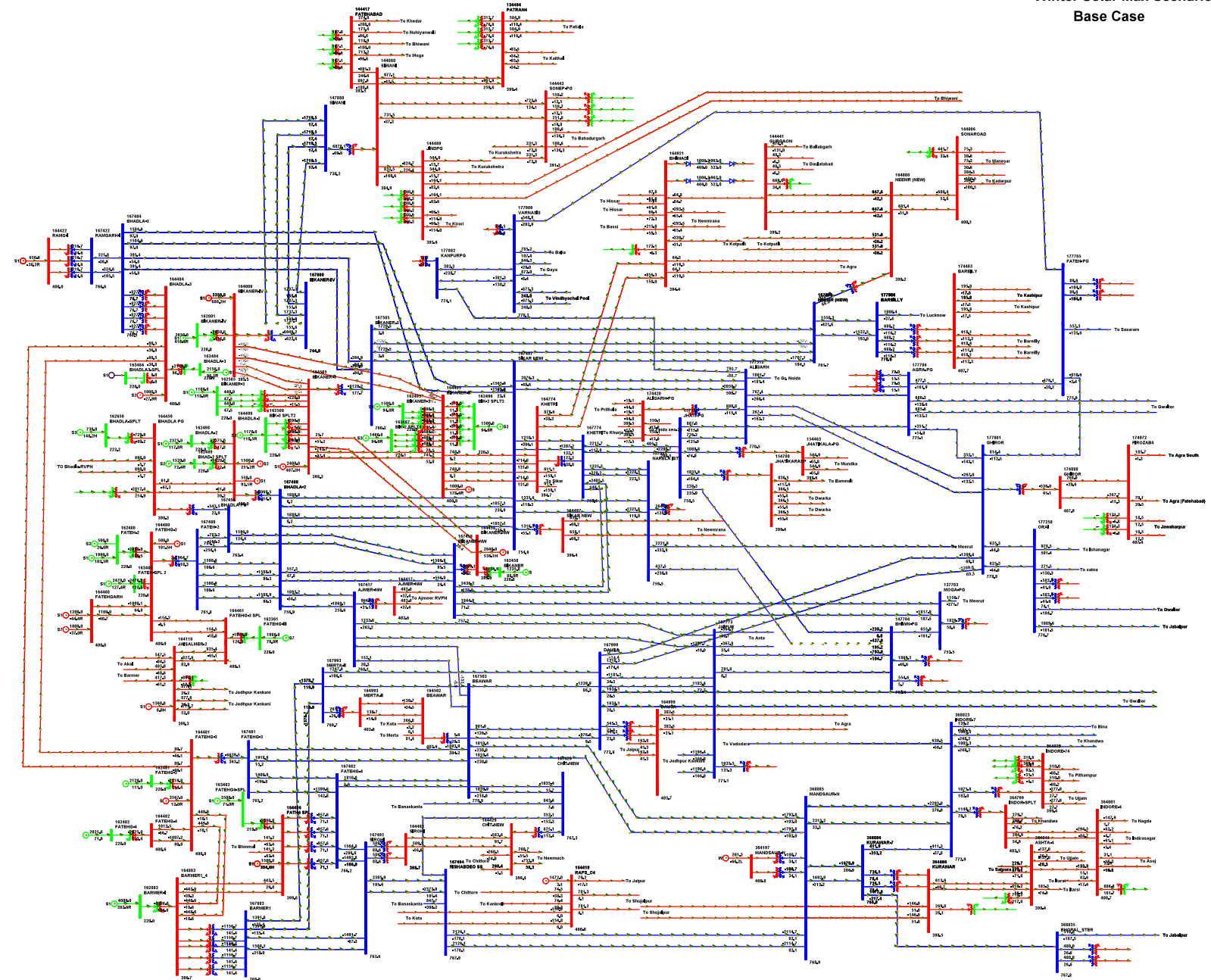
EHVAC system beyond Baripada (HVDC) S/s is under planning.

Transmission scheme for Rajasthan REZ Ph-IV (Part-3 :6GW (Bikaner complex) and Ph-IV (Part-4 :3.5GW (Fatehgarh/Barmer complex)

Summer Solar Max Scenario
Base Case

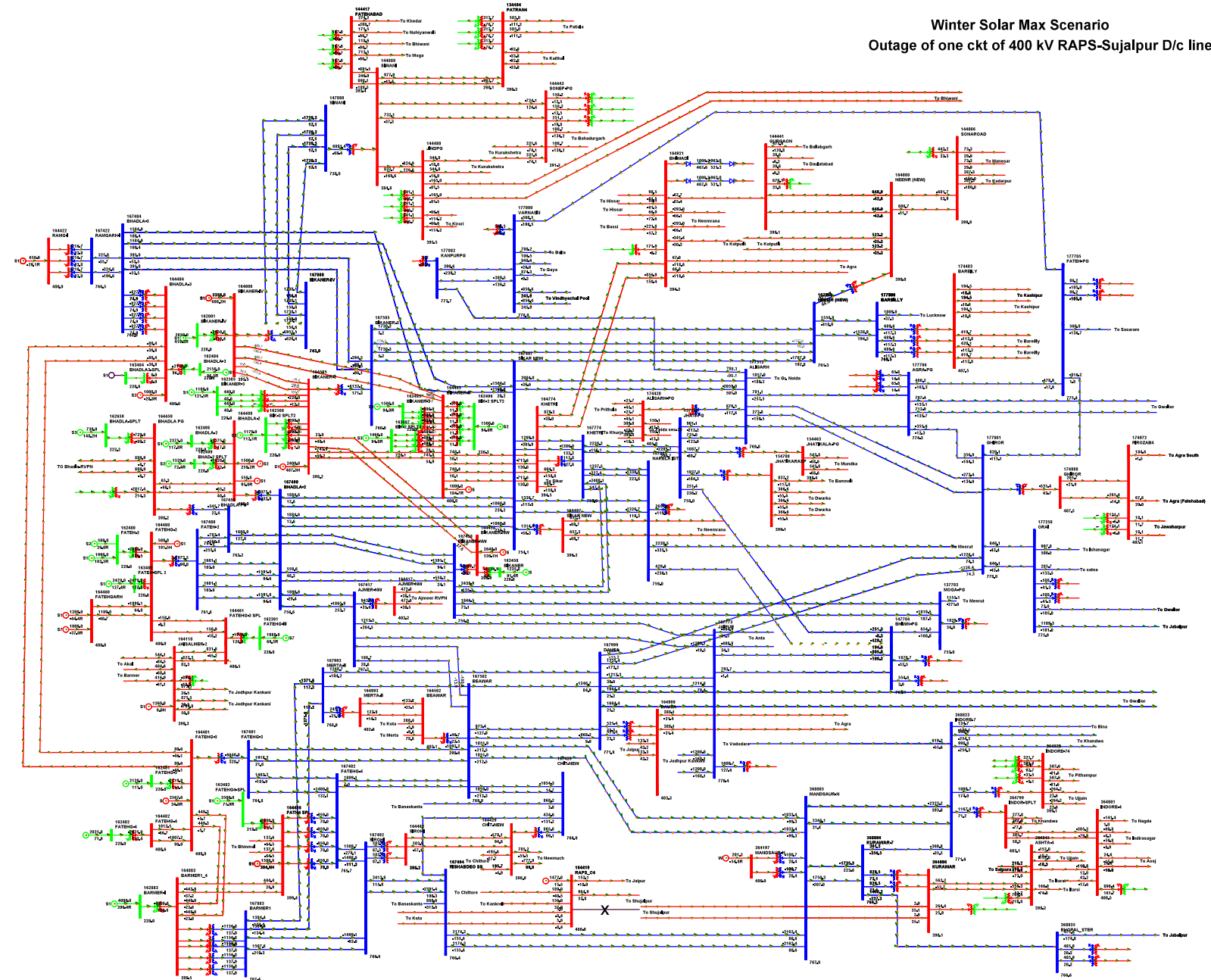
Transmission scheme for Rajasthan REZ Ph-IV (Part-3 :6GW (Bikaner complex) and Ph-IV (Part-4 :3.5GW (Fatehgarh/Barmer complex)

Winter Solar Max Scenario
Base Case



Transmission scheme for Rajasthan REZ Ph-IV (Part-3 :6GW (Bikaner complex) and Ph-IV (Part-4 :3.5GW (Fatehgarh/Barmer complex)

Winter Solar Max Scenario
Outage of one ckt of 400 kV RAPS-Sujalpur D/c line



Tej Prakash Verma {तेजप्रकाश वर्मा}

From: Ankur Gulati (अंकुर गुलाटी) <ankurgulati@grid-india.in>
Sent: Thursday, December 7, 2023 4:09 PM
To: Tej Prakash Verma {तेजप्रकाश वर्मा}
Cc: M M Hassan (एम एम हसन); H S Kaushal {एच.एस. कौशल}; Kashish Bhambhani {कशिश भम्भानी}; Sandeep Kumawat {संदीप कुमावत}; V Thiagarajan {वी. त्यागराजन}; Gaurav Malviya (गौरव मालवीय)
Subject: RE: Comments on the OPGW installation on 2nd E/W peak of Kishenpur- Dulhasti line

Sir,

Proposal of Installation of OPGW on 2nd E/w Peak of Kishanpur-Dulhasti can be considered as it would provide redundancy for Kishtwar & Dulhasti Data.

Regards
 Ankur Gulati
 DGM NRLDC

From: Tej Prakash Verma {तेजप्रकाश वर्मा} <tejprakash@powergrid.in>
Sent: 01 December 2023 16:21
To: Ankur Gulati (अंकुर गुलाटी) <ankurgulati@grid-india.in>
Cc: M M Hassan (एम एम हसन) <mm.hassan@grid-india.in>; H S Kaushal {एच.एस. कौशल} <hsk@powergrid.in>; Kashish Bhambhani {कशिश भम्भानी} <kashish@powergrid.in>; Sandeep Kumawat {संदीप कुमावत} <sandeepk@powergrid.in>; V Thiagarajan {वी. त्यागराजन} <vthiagarajan@powergrid.in>; Gaurav Malviya (गौरव मालवीय) <gauravmalviya@grid-india.in>
Subject: Comments on the OPGW installation on 2nd E/W peak of Kishenpur- Dulhasti line

****Warning****

This email has not originated from Grid-India. Do not click on attachment or links unless sender is reliable.
 Malware/ Viruses can be easily transmitted via email.

Dear Sir,

As discussed in the meeting of 26th Consultation Meeting for Evolving Transmission Schemes in Northern Region, following communication scheme needs to be reviewed by NRLDC communication deptt. for their observation:

- (i) OPGW installation on 2nd E/W peak of 400kV Kishenpur-Dulhasti line

This is with reference to Transmission System Scheme for Ratle HEP (850MW), Where LILO of 400 kV Kishenpur-Dulhasti line (Twin Zebra) at Kishtwar S/s (one ckt already LILOed as part of Pakaldul transmission system) is planned. Along with the scheme we are proposing OPGW installation on 2nd E/W peak of Kishenpur-Dulhasti line, this will create additional redundancy to Dulhasti & Kishtwar S/s. Redundant communication for Dulhasti was also deliberated in the 23rd meeting of TeST.

Please provide your comments at the earliest so that scheme may be prepared for approval in NCT.

Thanks & Regards,
T P Verma,
Chief Manager (Comm),
Central Transmission Utility of India Ltd.,

1st Floor, Suadamini, Plot No. 2, Sector-29,
Gurugram (Haryana) – 122001.
Phone No.- +91-124-2822154
Mo. +91-9650598191

From: Sandeep Kumawat {संदीप कुमावत} <sandeepk@powergrid.in>

Sent: Wednesday, November 29, 2023 2:52 PM

To: cea-pspa1@gov.in; manjari.cea@gmail.com; ms-nrpc@nic.in; directorso@grid-india.in; Gaurav Malviya (गौरव मालवीय) <gauravmalviya@grid-india.in>; ceplg@hvpn.org.in; ceplng@upptcl.org; dir.opr@dtl.gov.in; dgmplanning.dtl2016@gmail.com; dir-tech@pstcl.org; dir.tech@rvpn.co.in; dcpijkd@yahoo.com; pmupddjk@gmail.com; cesocomml@hvpn.org.in; secoml@hvpn.org.in; cepdc@hvpn.org.in; ce-chd@nic.in; md@seci.co.in; Member Power System <memberscea@nic.in>; ce_candr@ptcul.org; alok.kumar@grid-india.in; sharan_ishan@yahoo.com; i.sharan@nic.in; directorpc.tcl@hmail.in; harmanjeet.tcl@hmail.in; dmplg@hpptcl.in; Santosh Kumar <seo-nrpc@nic.in>; Reeturaj Pandey <pandeyr.cea@gov.in>; xen2.pp@rvpn.co.in; director_op@upptcl.org; ceo.rattle@nhpc.nic.in; vindosharma@nhpc.nic.in; Rajesh Kumar Gupta <rajesh.gupta@adani.com>; mehendrasingh.dabi@adani.com; mehul.rupera@adani.com; Rohit Kumar Jain {रोहित कुमार जैन} <rohit.jain@powergrid.in>; Vibhay Kumar {विभय कुमार} <vibhay@powergrid.in>; Nitin Deswal <nitindeswal@nic.in>; Kanhaiya Singh Kushwaha <kanhaiya.cea@gov.in>; Sr.Xen Planning I <srxen-plann1@pstcl.org>; priyam.jain@grid-india.in

Cc: P C Garg {पी.सी. गर्ग} <pcgarg@powergrid.in>; Ashok Pal {अशोक पाल} <ashok@powergrid.in>; V Thiagarajan {वी. त्यागराजन} <vthiagarajan@powergrid.in>; Shiv Kumar Gupta {एस.के. गुप्ता} <shivkumar@powergrid.in>; Tej Prakash Verma {तेजप्रकाश वर्मा} <tejprakash@powergrid.in>; Kashish Bhambhani {कशिश भम्भानी} <kashish@powergrid.in>; Yatin Sharma {यतिन शर्मा} <yatinsharma@powergrid.in>; Roushan Kumar {रौशन कुमार} <roushan.k@powergrid.in>; MADHUSUDAN MEENA {मधुसूदन मीना} <madhusudanmeena@powergrid.in>; Ankita Singh {अंकिता सिंह} <ankita@powergrid.in>; NARENDRA SATHVIK RANGANATH . {नरेंद्र सात्विक आर} <rnsathvik@powergrid.in>

Subject: Meeting Rescheduling : 26th Consultation Meeting for Evolving Transmission Schemes in Northern Region to 01.12.2023 (postponement from earlier meeting date 30.11.23)

Dear Sir/Ma'am,

In view of NCT meeting schedule to be held on 30.11.23 (2 PM), the 26th Consultation Meeting for Evolving Transmission Schemes in Northern Region (monthly meeting) **has been rescheduled and would be held on 01st December 2023 (Friday) at 3PM through virtual mode.**

Revised Details in this regard are as below:

Topic	:	26 th Consultation Meeting for Evolving Transmission Schemes in NR
Date & Time	:	01 st December, 2023 (Friday), 03:00 PM
Meeting Link	:	Microsoft Teams meeting Join on your computer, mobile app or room device Click here to join the meeting

In this regard, agenda (same as per trailing mail) along with **revised meeting notice** of the meeting is enclosed which is also available on CTU website www.ctuil.in

Regards

Sandeep Kumawat

DGM (CTUIL)

Power Grid Corporation of India Limited

From: Sandeep Kumawat {संदीप कुमावत}

Sent: 24 November 2023 18:12

To: cea-pspa1@gov.in <cea-pspa1@gov.in>; manjari.cea@gmail.com <manjari.cea@gmail.com>; ms-nrpc@nic.in <ms-nrpc@nic.in>; directorso@grid-india.in <directorso@grid-india.in>; Gaurav Malviya (गौरव मालवीय) <gauravmalviya@grid-india.in>; ceplg@hvpn.org.in <ceplg@hvpn.org.in>; ceplng@upptcl.org <ceplng@upptcl.org>; dir.opr@dtl.gov.in <dir.opr@dtl.gov.in>; dgmplanning.dtl2016@gmail.com <dgmplanning.dtl2016@gmail.com>; dir-tech@pstcl.org <dir-tech@pstcl.org>; dir.tech@rvpn.co.in <dir.tech@rvpn.co.in>; dcpijkd@yahoo.com <dcpijkd@yahoo.com>; pmupddjk@gmail.com <pmupddjk@gmail.com>; cesocomml@hvpn.org.in <cesocomml@hvpn.org.in>; secoml@hvpn.org.in <secoml@hvpn.org.in>; cepdh@hvpn.org.in <cepdh@hvpn.org.in>; ce-chd@nic.in <ce-chd@nic.in>; md@seci.co.in <md@seci.co.in>; Member Power System <memberspscea@nic.in>; ce_candr@ptcul.org <ce_candr@ptcul.org>; alok.kumar@grid-india.in <alok.kumar@grid-india.in>; sharan_ishan@yahoo.com <sharan_ishan@yahoo.com>; i.sharan@nic.in <i.sharan@nic.in>; directorpc.tcl@hpmail.in <directorpc.tcl@hpmail.in>; harmanjeet.tcl@hpmail.in <harmanjeet.tcl@hpmail.in>; dmpplg@hpptcl.in <dmpplg@hpptcl.in>; Santosh Kumar <seo-nrpc@nic.in>; Reeturaj Pandey <pandeyr.cea@gov.in>; xen2.pp@rvpn.co.in <XEN2.PP@RVPN.CO.IN>; director_op@upptcl.org <director_op@upptcl.org>; ceo.ratle@nhpc.nic.in <ceo.ratle@nhpc.nic.in>; vindosharma@nhpc.nic.in <vindosharma@nhpc.nic.in>; Rajesh Kumar Gupta <rajesh.gupta@adani.com>; mehendrasingh.dabi@adani.com <mehendrasingh.dabi@adani.com>; mehul.rupera@adani.com <mehul.rupera@adani.com>; Rohit Kumar Jain {रोहित कुमार जैन} <rohit.jain@powergrid.in>; Vibhay Kumar {विभय कुमार} <vibhay@powergrid.in>; Nitin Deswal <nitindeswal@nic.in>; Kanhaiya Singh Kushwaha <kanhaiya.cea@gov.in>; Sr.Xen Planning I <srxen-plann1@pstcl.org>; priyam.jain@grid-india.in <priyam.jain@grid-india.in>

Cc: P C Garg {पी.सी. गर्ग} <pcgarg@powergrid.in>; Ashok Pal {अशोक पाल} <ashok@powergrid.in>; V Thiagarajan {वी. त्यागराजन} <vthiagarajan@powergrid.in>; Shiv Kumar Gupta {एस.के. गुप्ता} <shivkumar@powergrid.in>; Tej Prakash Verma {तेजप्रकाश वर्मा} <tejprakash@powergrid.in>; Kashish Bhambhani {कशिश भम्भानी} <kashish@powergrid.in>; Yatin Sharma {यतिन शर्मा} <yatinsharma@powergrid.in>; Roushan Kumar {रौशन कुमार} <roushan.k@powergrid.in>; MADHUSUDAN MEENA {मधुसूदन मीना} <madhusudanmeena@powergrid.in>; Ankita Singh {अंकिता सिंह} <ankita@powergrid.in>; NARENDRA SATHVIK RANGANATH . {नरेंद्र सात्विक आर} <rnsathvik@powergrid.in>

Subject: Meeting Notice & Agenda for 26th Consultation Meeting for Evolving Transmission Schemes in Northern Region scheduled to be held on 30.11.2023

Dear Sir/Ma'am,

The 26th Consultation Meeting for Evolving Transmission Schemes in Northern Region is scheduled to be held on 30th November, 2023 (Thursday) through virtual mode.

Details in this regard are as below:

Topic : 26th Consultation Meeting for Evolving Transmission Schemes in NR

Date & Time : 30th November, 2023 (Thursday), 03:00 PM

Meeting Link : Microsoft Teams meeting

Join on your computer, mobile app or room device
[Click here to join the meeting](#)

In this regard, please find enclosed agenda of the meeting which is also available on CTU website www.ctuil.in

It is requested to join the meeting through above link and send confirmation of participation in this regard through return mail.

Regards

Sandeep Kumawat

DGM (CTUIL)

Power Grid Corporation of India Limited

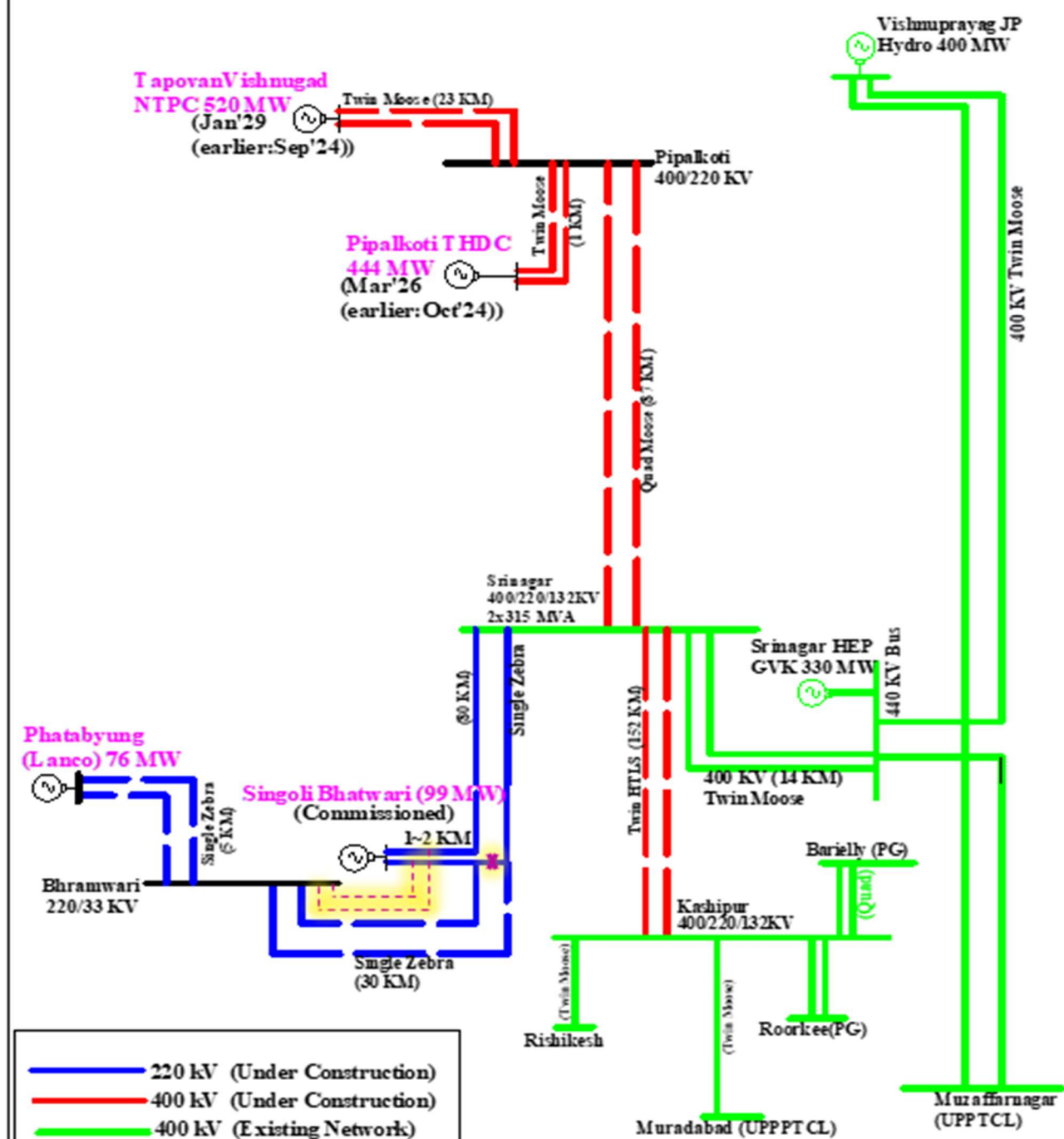
दावात्याग : यह ईमेल पावरग्रिड के दावात्याग नियम व शर्तों द्वारा शासित है जिसे <http://apps.powergrid.in/Disclaimer.htm> पर देखा जा सकता है। Disclaimer: This e-mail is governed by the Disclaimer Terms & Conditions of POWERGRID which may be viewed at <http://apps.powergrid.in/Disclaimer.htm>

Follow Grid-India on:



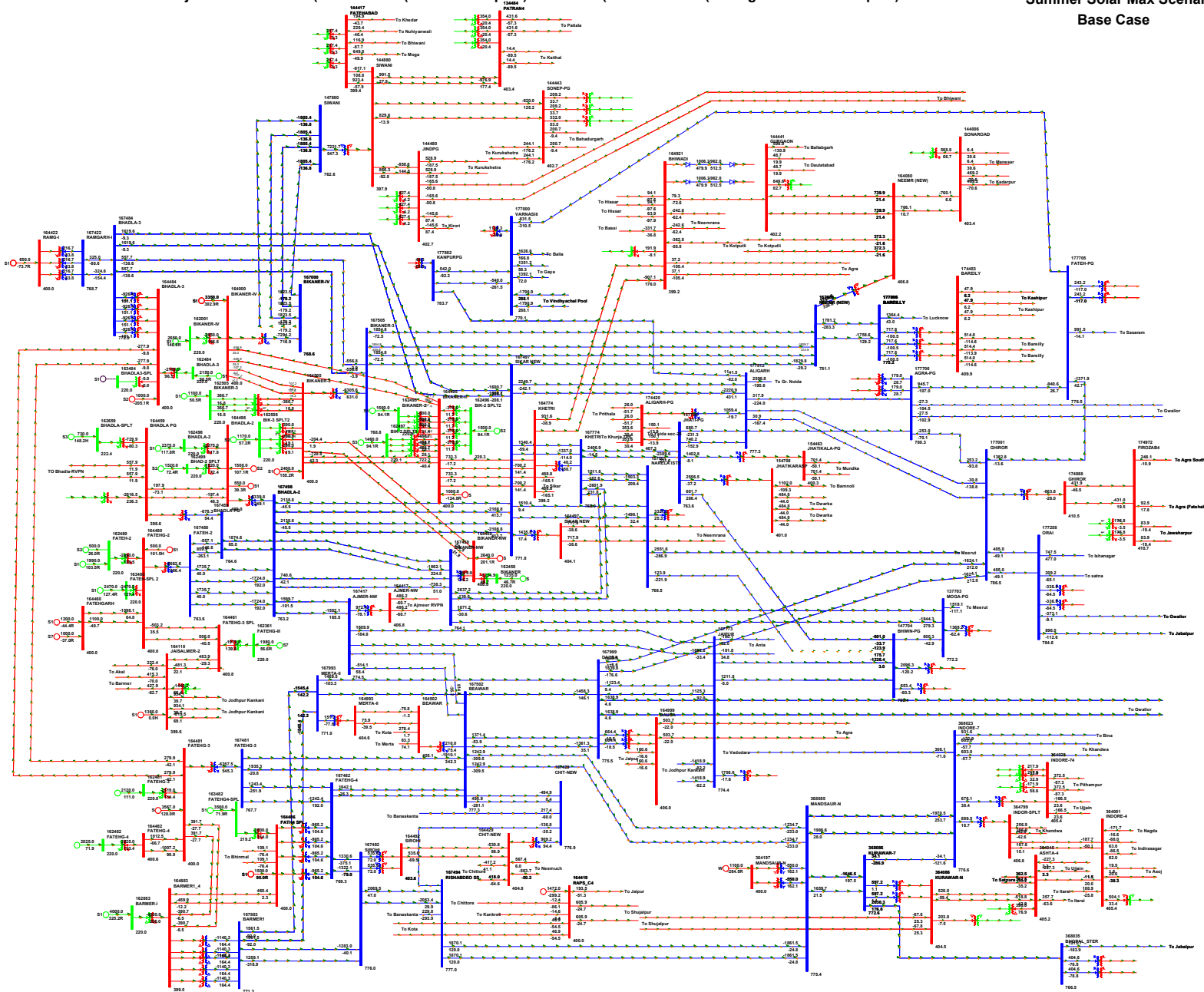
* This e-mail is an official email of Grid Controller of India Limited (Grid-India), is confidential and intended to use by the addressee only. If the message is received by anyone other than the addressee, please return the message to the sender by replying to it and then delete the message from your computer. Internet e-mails are not necessarily secure. The Grid Controller of India Limited (Grid-India) does not accept responsibility for changes made to this message after it was sent. Whilst all reasonable care has been taken to avoid the transmission of viruses, it is the responsibility of the recipient to ensure that the onward transmission, opening or use of this message and any attachments will not adversely affect its systems or data. No responsibility is accepted by the Grid-India in this regard and the recipient should carry out such virus and other checks as it considers appropriate. Visit our website at www.grid-india.in *

Proposed Transmission system of Alakanda Basin

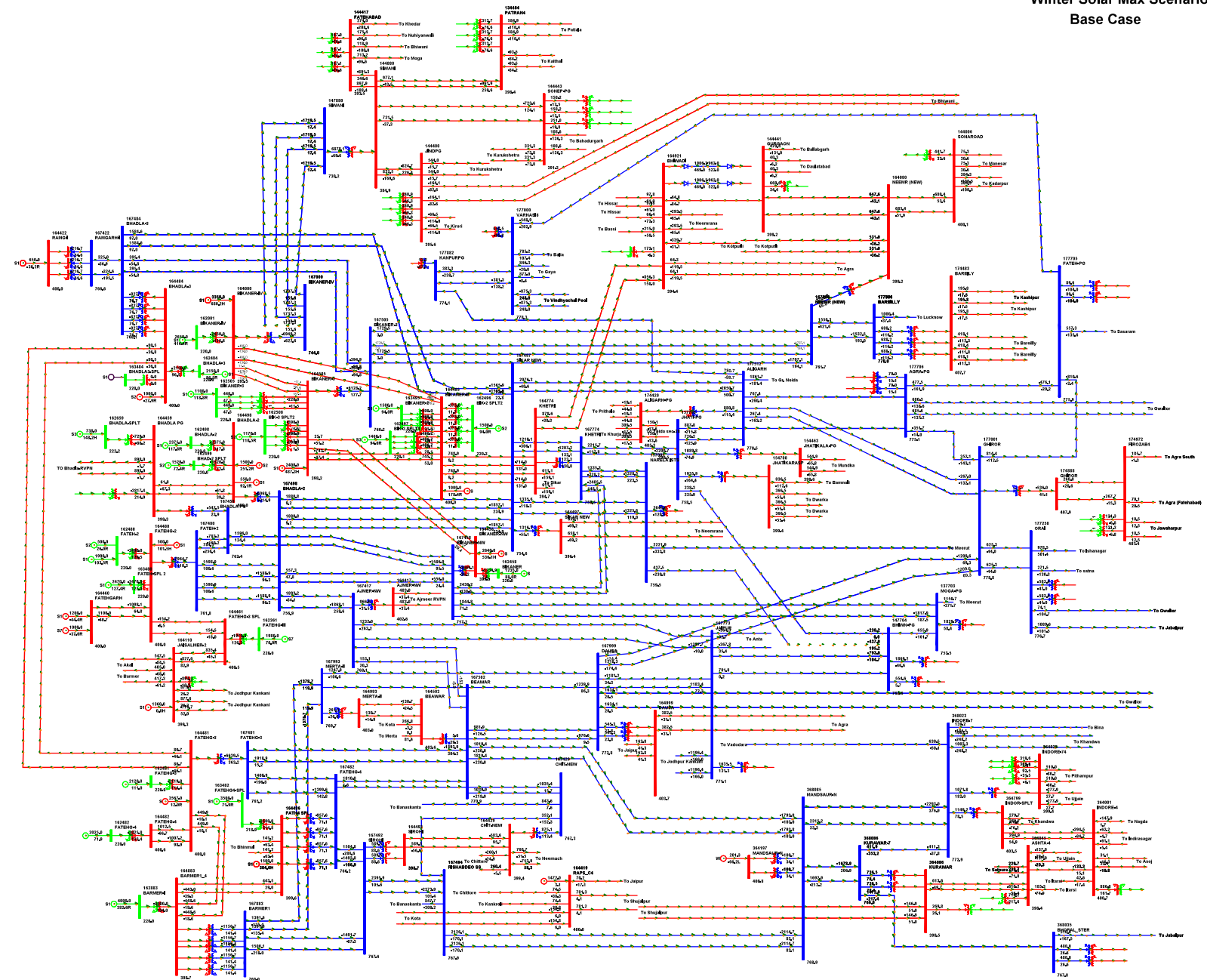


Transmission scheme for Rajasthan REZ Ph-IV (Part-3 :6GW (Bikaner complex) and Ph-IV (Part-4 :3.5GW (Fatehgarh/Barmer complex)

Summer Solar Max Scenario Base Case



Winter Solar Max Scenario Base Case



Winter Solar Max Scenario
Outage of one ckt of 400 kV RAPS-Sujalpur D/c line

